

Aula 10

lmsdmflmslmf

Roberta Marcondes Machado





Microbiota humana



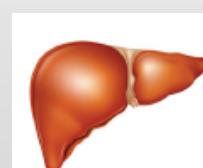
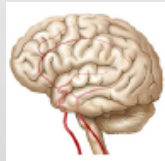
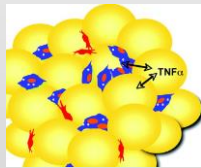
Microrganismos que habitam o corpo humano



Microbiota intestinal: bactérias, vírus, fungos, protozoários e parasitas



**Podem modular a permeabilidade intestinal e
exerce várias ações em diferentes órgãos**



Caracterização da Microbiota Intestinal

➤ Microbiota intestinal:

➤ Cerca de 40 trilhões de bactérias

- 150 grupos filogenéticos

- 1800 gêneros

- 15.000 – 36.000 espécies.

- 90% Firmicutes, Bacteroidetes e Actinobacteria, em menor proporção Proteobacteria e Verrucomicrobia.

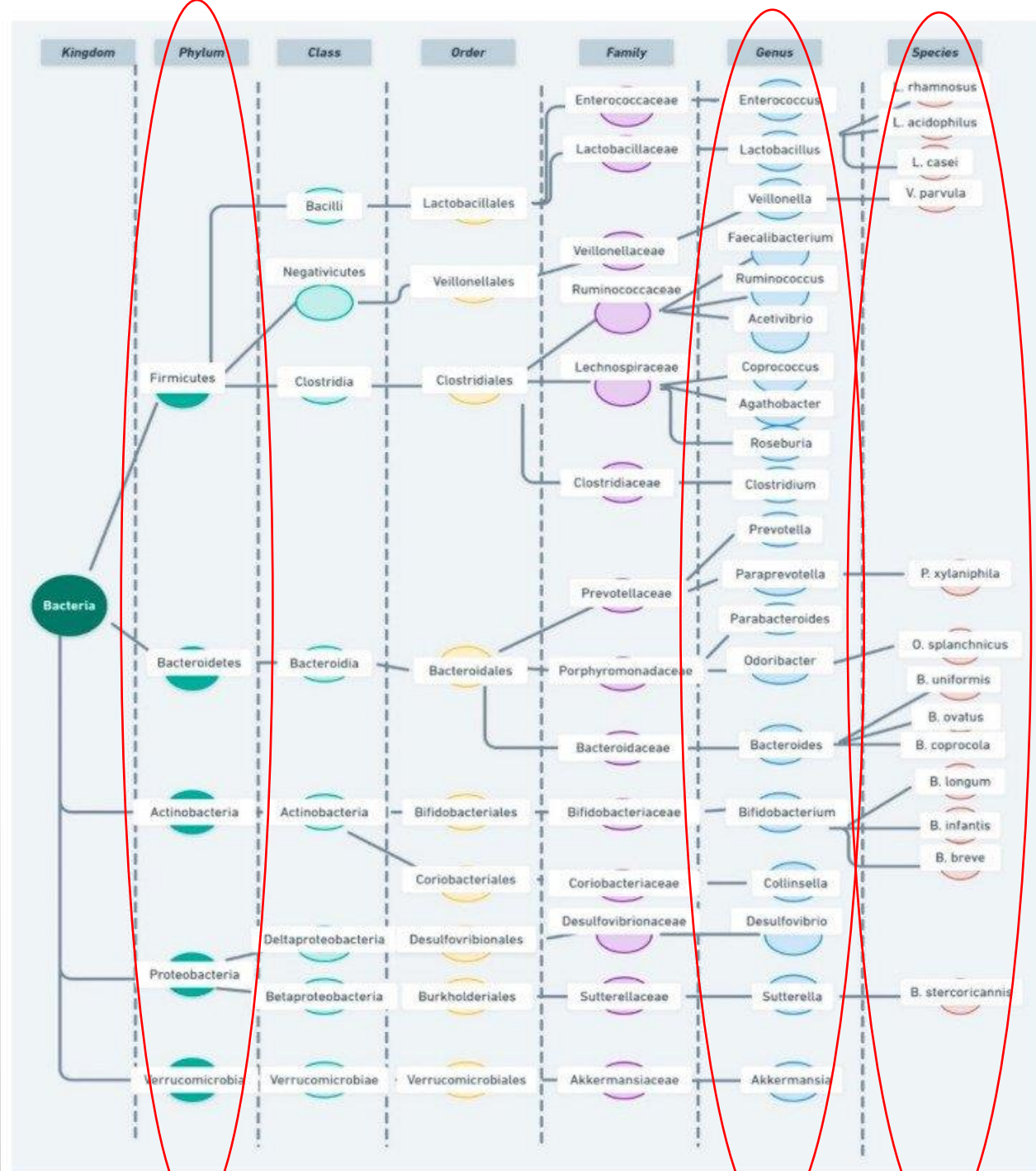
- Representa grande potencial regulador do metabolismo e da imunidade (Manson, et al., 2008; Qin et al, 2010).

- Grande variação ao longo do trato gastrointestinal (lumen, mucosa e superfície epitelial).

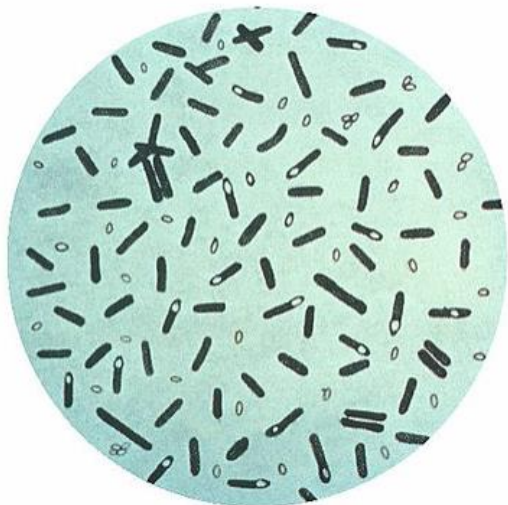


Classificação

- 150 grupos filogenéticos
- 1800 gêneros
- 15.000 – 36.000 espécies.



Firmicutes



Clostridium botulinum

Classificação científica

Domínio: Bacteria

Filo: Firmicutes

Classes e Ordens

Classe Bacilli

- Bacillales
- Lactobacillales

Classe Clostridia

- Clostridiales
- Halanaerobiales
- Thermoanaerobacteriales

Classe Mollicutes

Bacilli

Bacillales

1. *Alicyclobacillus*
2. *Ammoniphilus*
3. *Amphibacillus*
4. *Aneurinibacillus*
5. *Anoxybacillus*
6. *Bacillus*
7. *Brevibacillus*
8. *Brochothrix*
9. *Caryophanon*
10. *Exiguobacterium*
11. *Filibacter*
12. *Filobacillus*
13. *Gemella*
14. *Geobacillus*
15. *Gracilibacillus*
16. *Halobacillus*
17. *Jeotgalibacillus*
18. *Jeotgalicoccus*
19. *Kurthia*
20. *Lentibacillus*
21. *Listeria*

Lactobacillales

1. *Abiotrophia*
2. *Acetoanaerobium*
3. *Aerococcus*
4. *Agitococcus*
5. *Alkalibacterium*
6. *Allofustis*
7. *Alloiococcus*
8. *Atopobacter*
9. *Carnobacterium*
10. *Desemzia*
11. *Dolosicoccus*
12. *Dolosigranulum*
13. *Enterococcus*
14. *Eremococcus*
15. *Facklamia*
16. *Globicatella*
17. *Granulicatella*
18. *Ignavigranum*
19. *Isobaculum*
20. *Lactobacillus*
21. *Lactococcus*
22. *Lactosphaera*

22. *Lactosphaera*
23. *Leuconostoc*
24. *Marinilactibacillus*
25. *Melissococcus*
26. *Oenococcus*
27. *Oscillospira*
28. *Paralactobacillus*
29. *Pediococcus*
30. *Streptococcus*
31. *Syntrophococcus*
32. *Tetragenococcus*
33. *Trichococcus*
34. *Vagococcus*
35. *Weissella*

Ex. *L.casei*

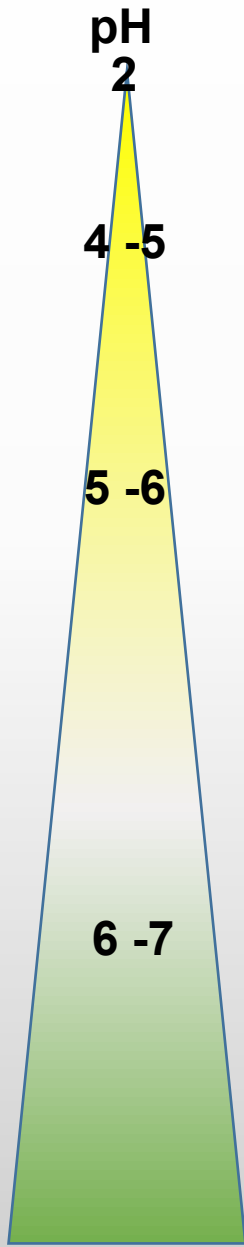
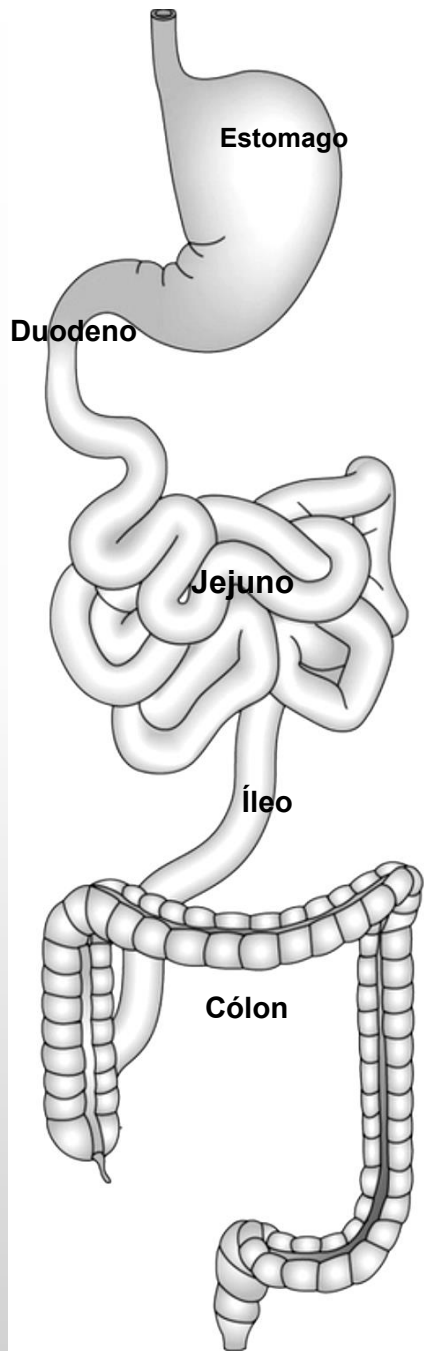
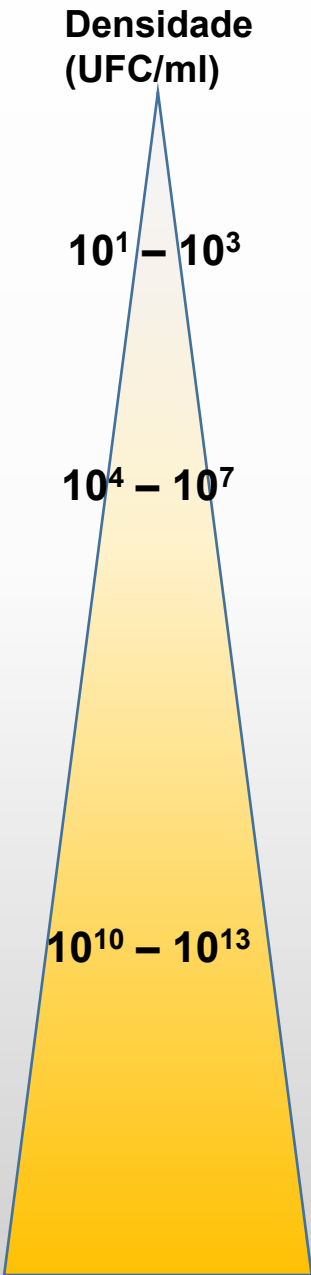


Composição da Microbiota

Lactobacillus
Streptococcus

Enterobacteriaceae

Bacteroides
Bifidobacterium
Clostridium
Eubacterium
Ruminococcus



Função por segmento

Digestão
e absorção de nutrientes
Imunomodulação

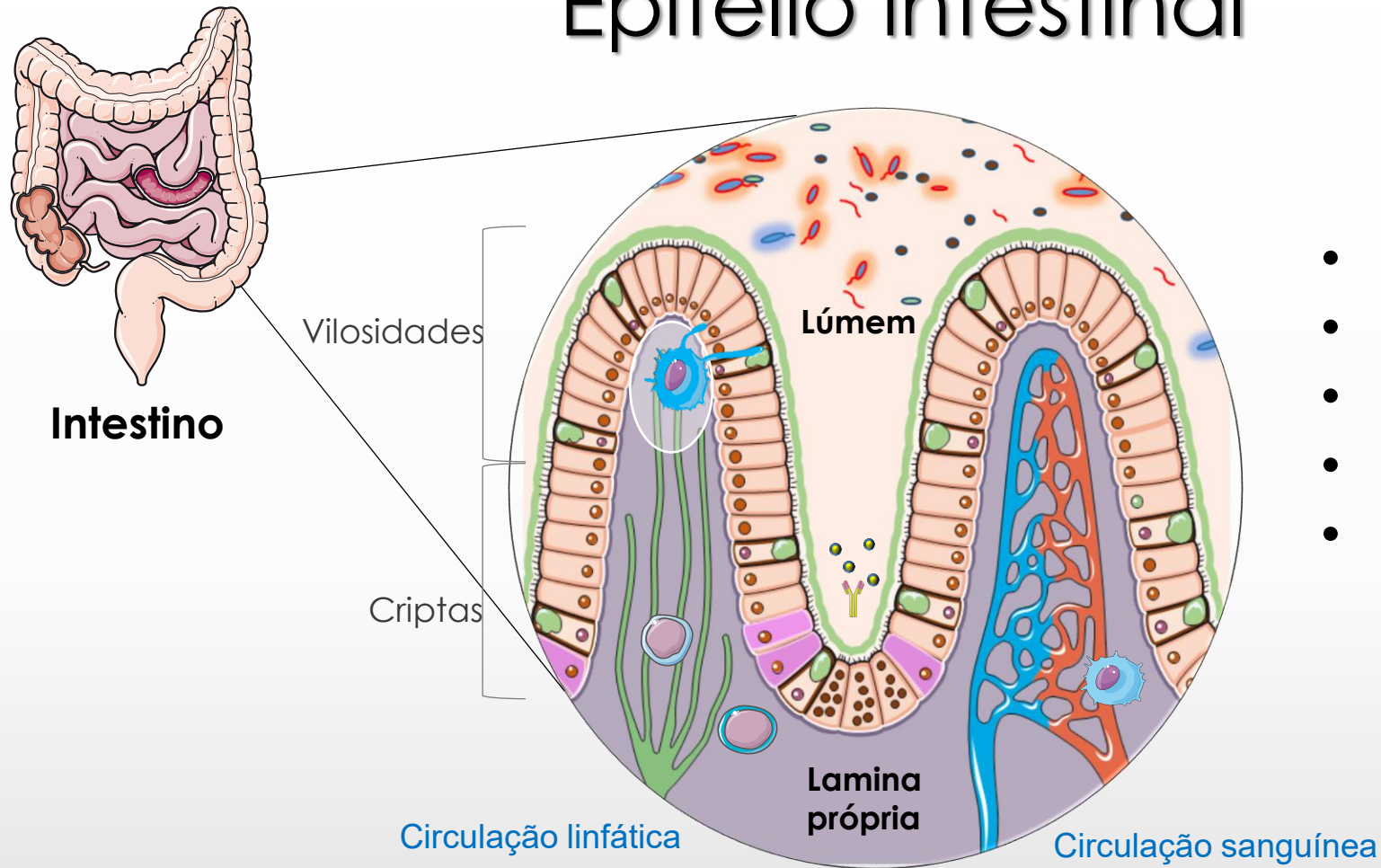
Cálcio e Vit. D
Vit. A,D,E,K

Ácido Biliar
Vit. B12

Água
AGCC



Epitélio intestinal

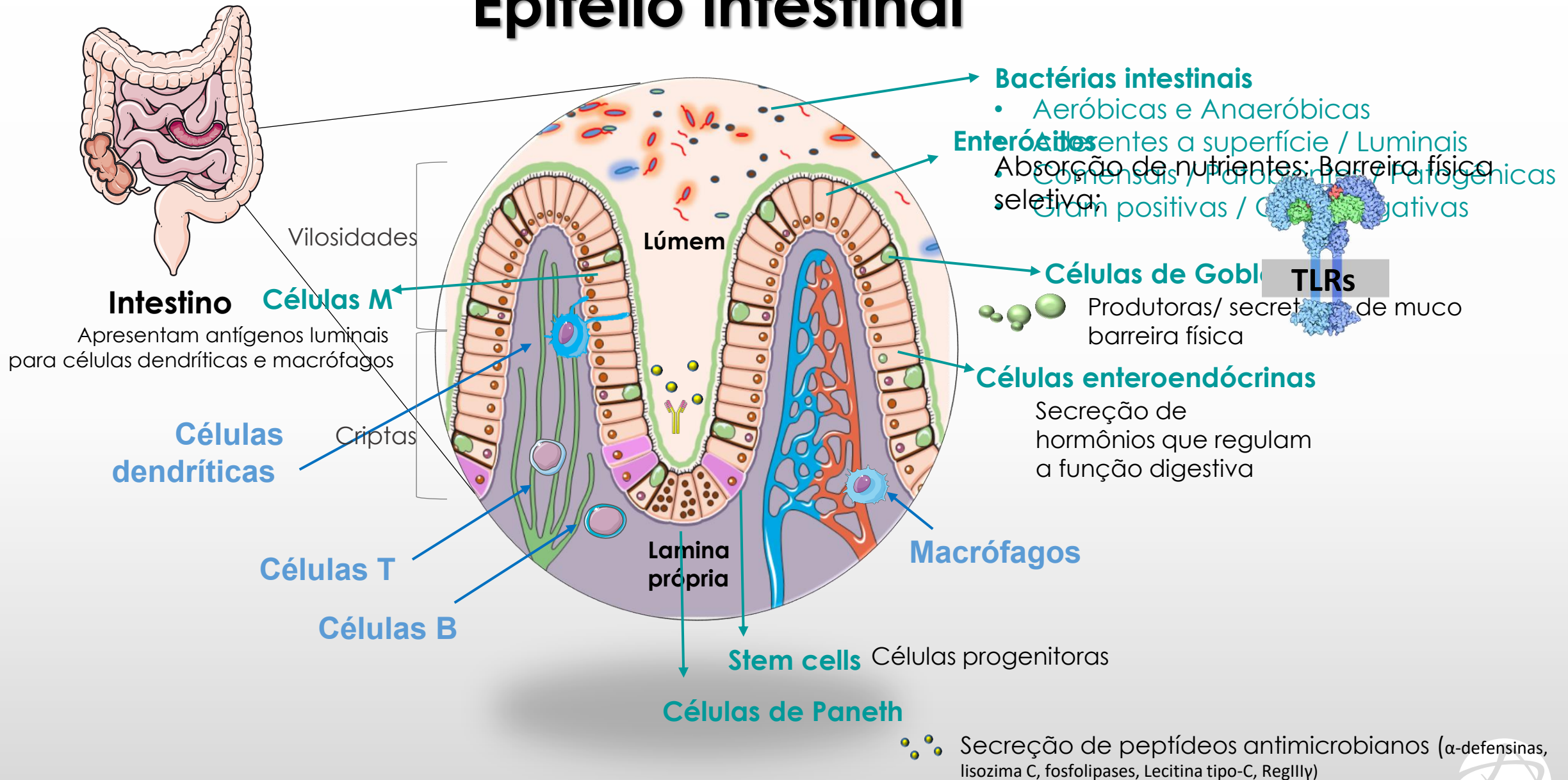


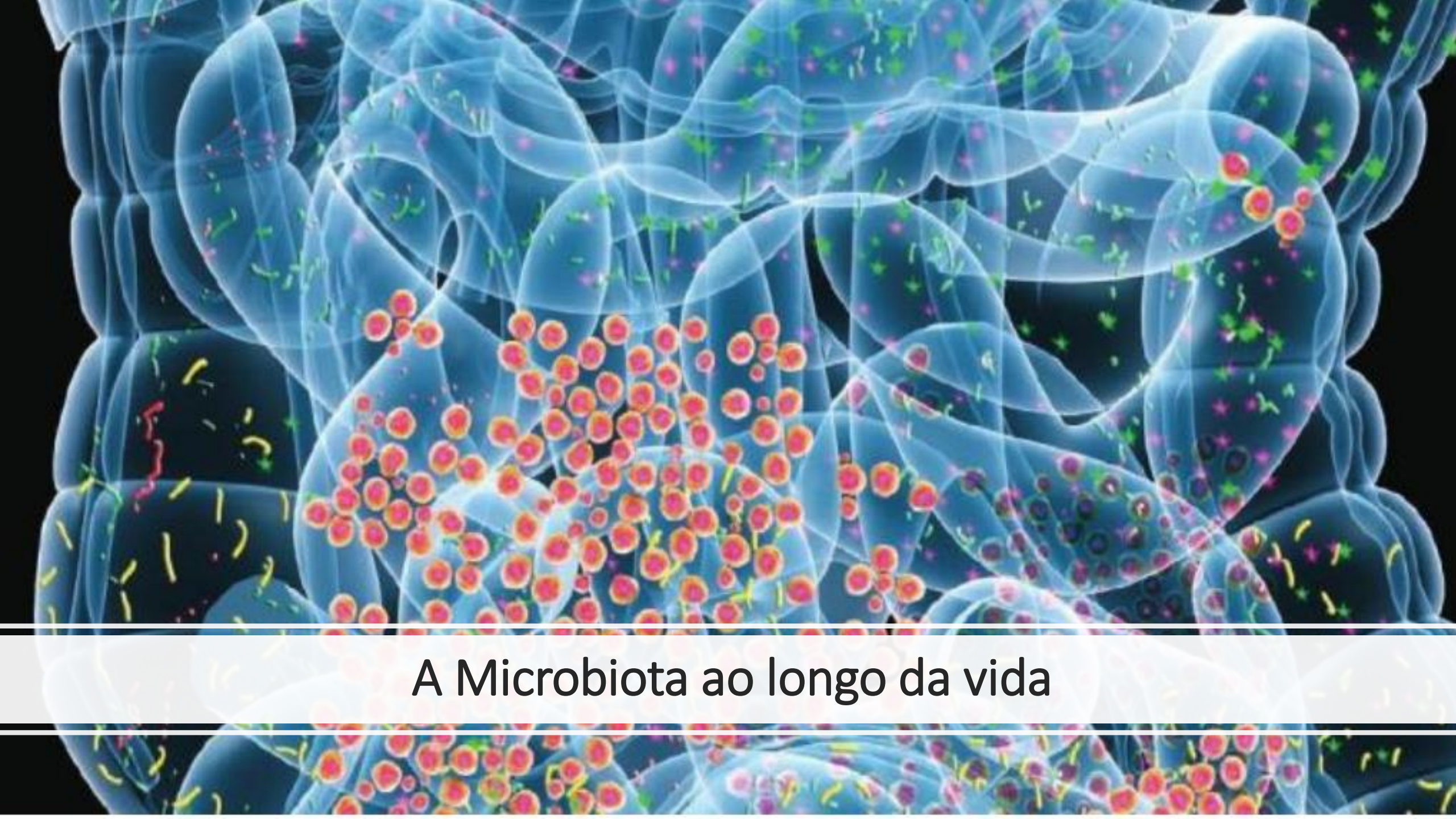
Funções

- Digestão
- Absorção
- Enteroendócrinas
- Barreira seletiva
- Linha de defesa



Epitélio intestinal

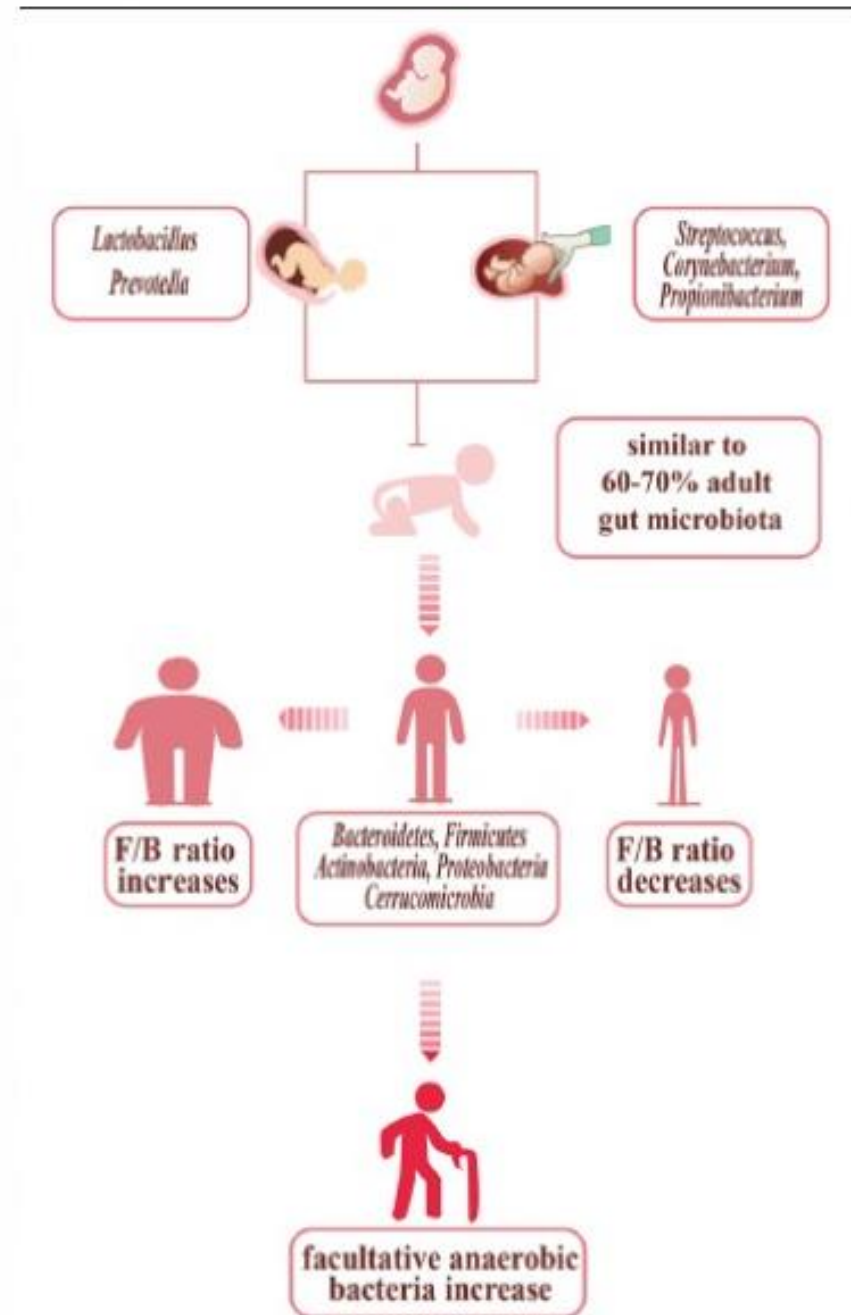




A Microbiota ao longo da vida

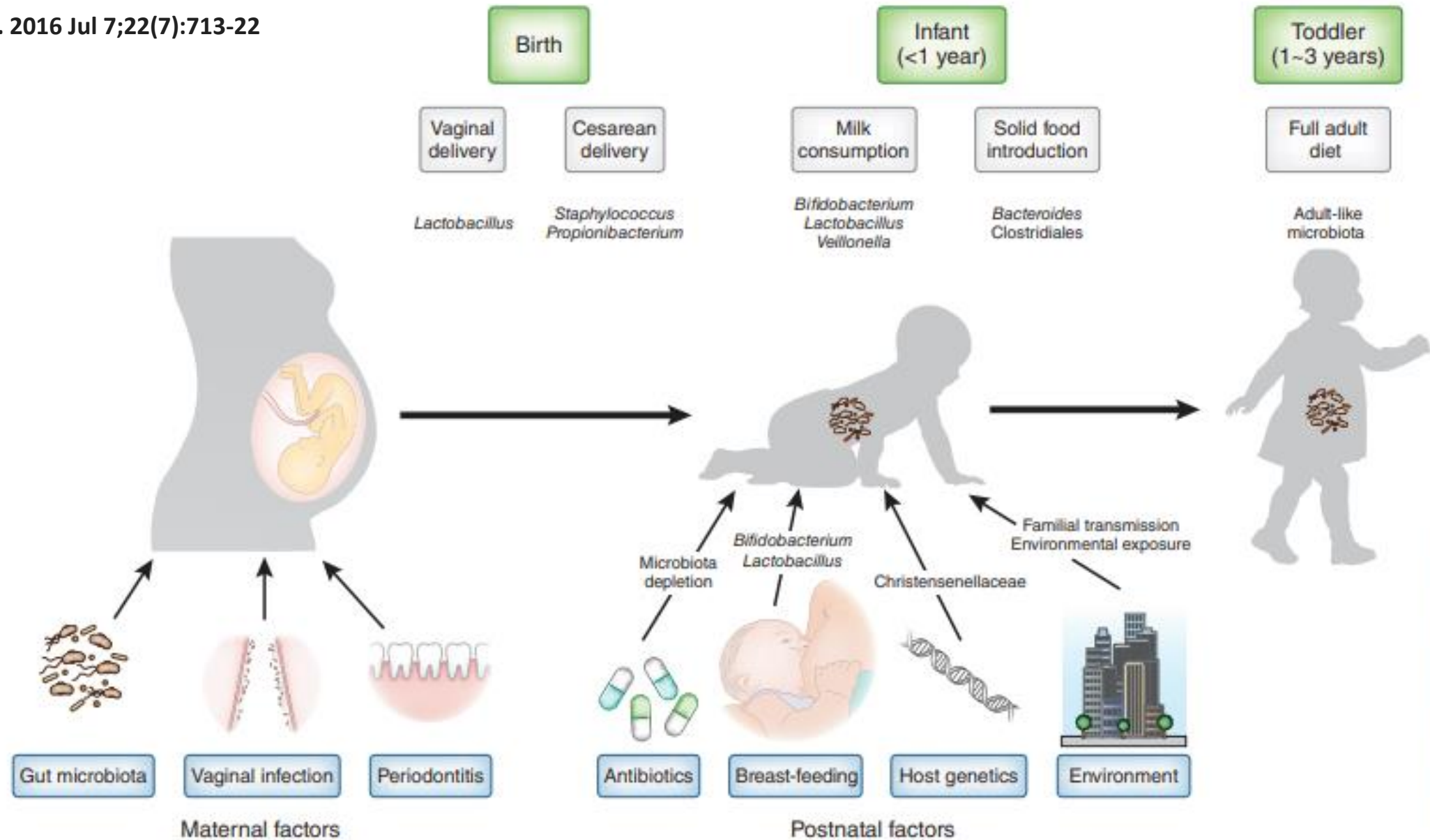


- A microbiota sofre alterações ao longo da vida.
- Feto é exposto ao microbioma materno desde o útero.
- Transmissão microbiana materno-fetal: parto vaginal x parto cesária
- Tipo de aleitamento: Aleitamento materno x fórmulas lácteas
- Introdução de alimentos
- Em importante estudo, o aleitamento materno foi o principal fator estrutural do microbioma.

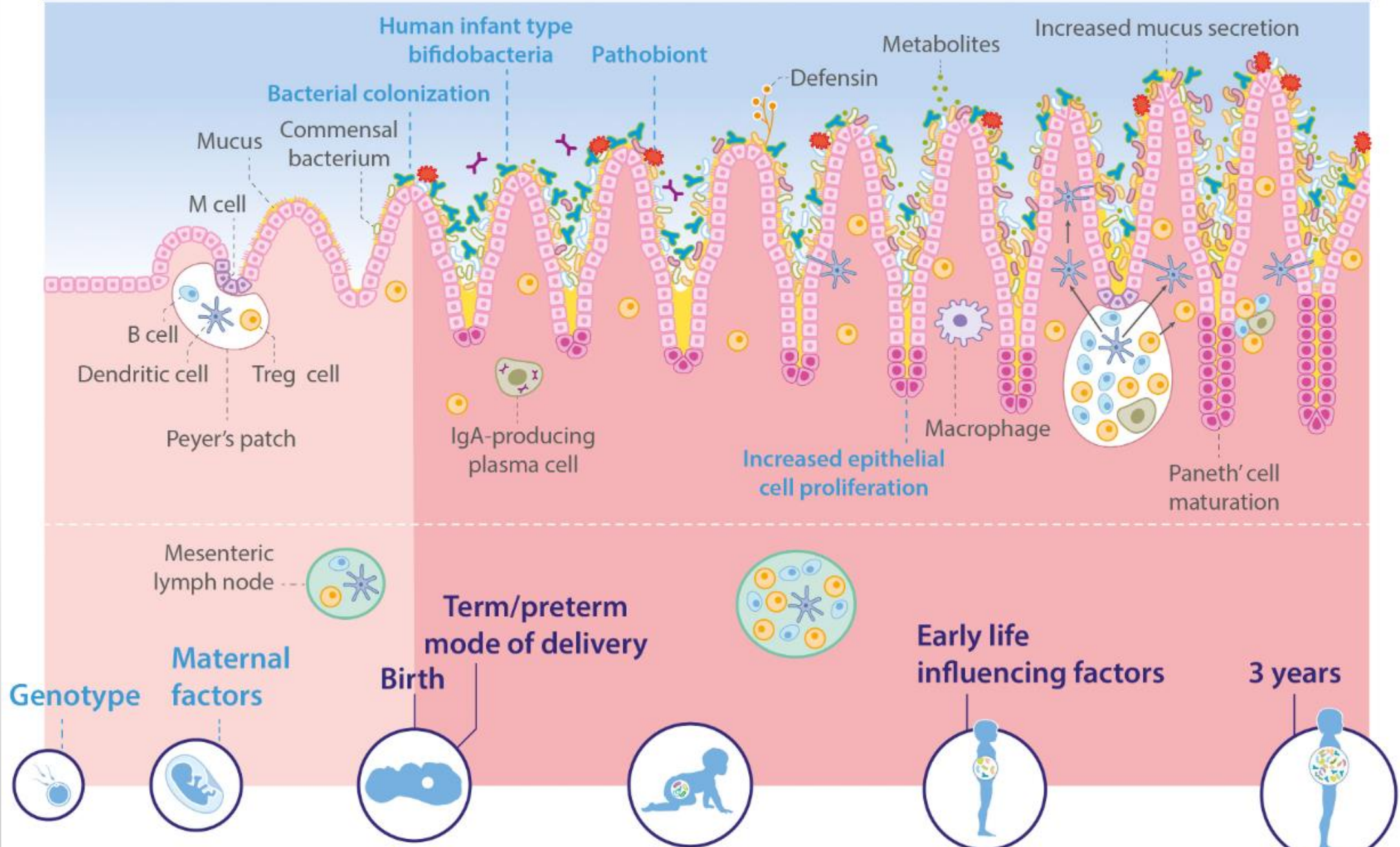


The microbiome in early life: implications for health outcomes

Tamburini S, et al., Nat Med. 2016 Jul 7;22(7):713-22



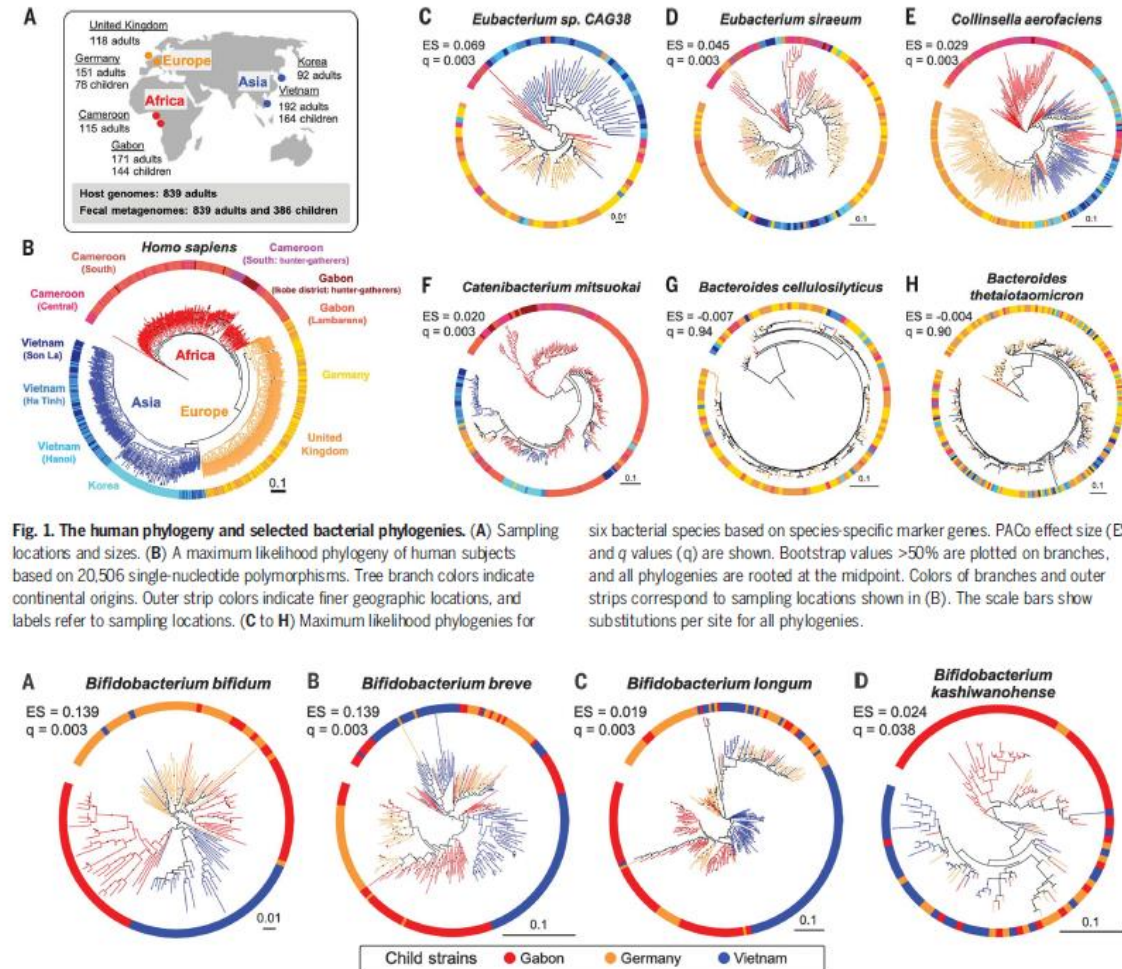
Normal gut, microbiota and immune development



Codiversification of gut microbiota with humans

Taichi A. Suzuki^{1†}, J. Liam Fitzstevens^{1†}, Victor T. Schmidt¹, Hagay Enav¹, Kelsey E. Huus¹, Mirabeau Mbong Ngwese¹, Anne Griebhammer², Anne Pfeleiderer³, Bayode R. Adegbite^{3,4}, Jeannot F. Zinsou^{3,4}, Meral Esen^{3,5,6}, Thirumalaisamy P. Velavan^{3,7}, Ayola A. Adegnika^{3,4,5,8}, Le Huu Song^{7,9}, Timothy D. Spector¹⁰, Amanda L. Muehlbauer¹¹, Nina Marchi¹², Hyena Kang¹³, Lisa Maier^{2,6}, Ran Blekhan¹⁴, Laure Ségurel^{12,15}, Gwangpyo Ko¹³, Nicholas D. Youngblut¹, Peter Kremsner^{3,4,5,6}, Ruth E. Ley^{16*}

RESEARCH | REPORT



Role of the microbiome in human development

Maria Gloria Dominguez-Bello,¹ Filipa Godoy-Vitorino,² Rob Knight,³ Martin J Blaser⁴

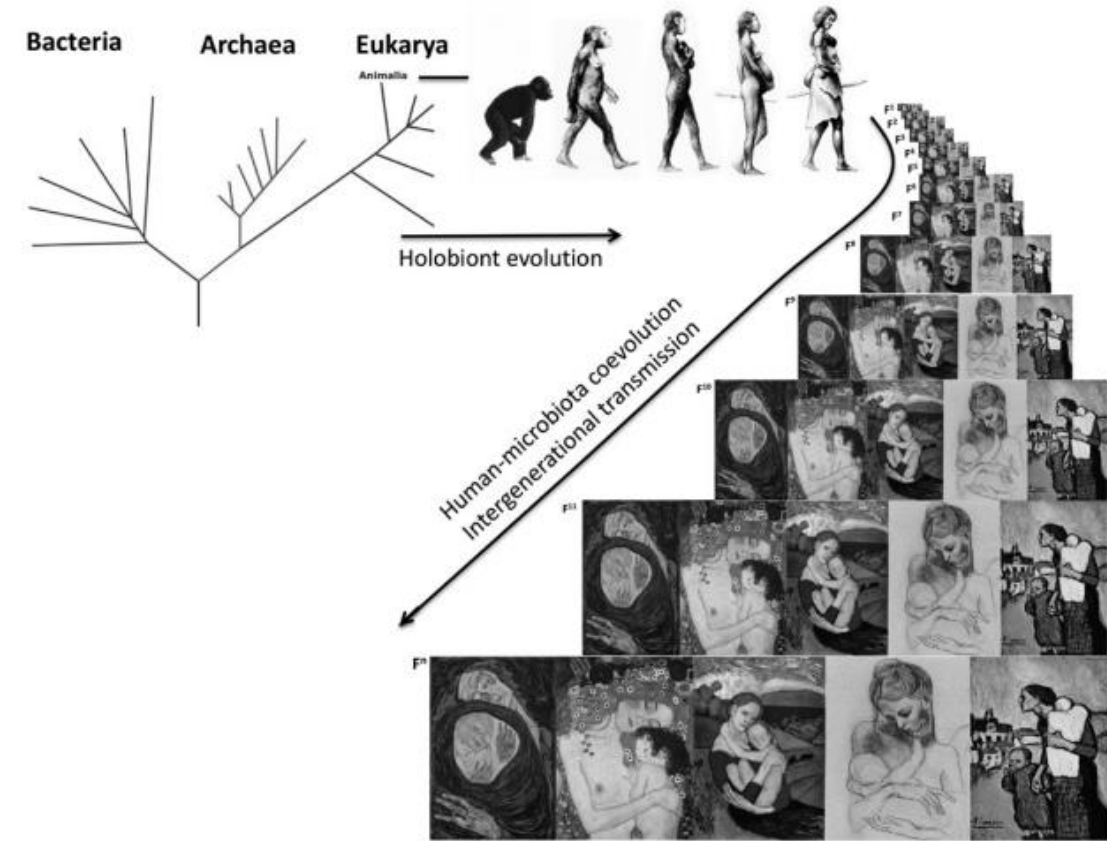
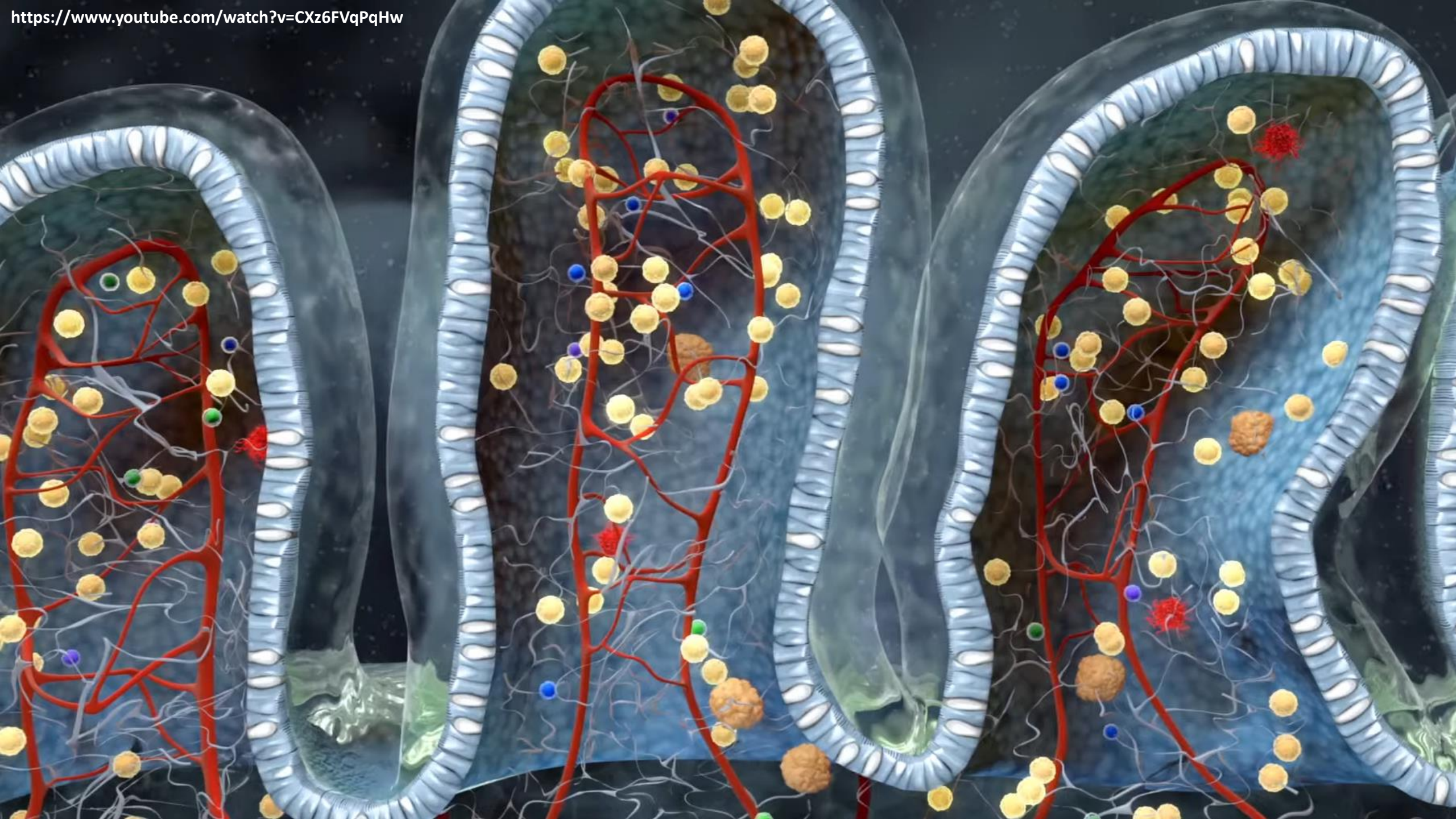
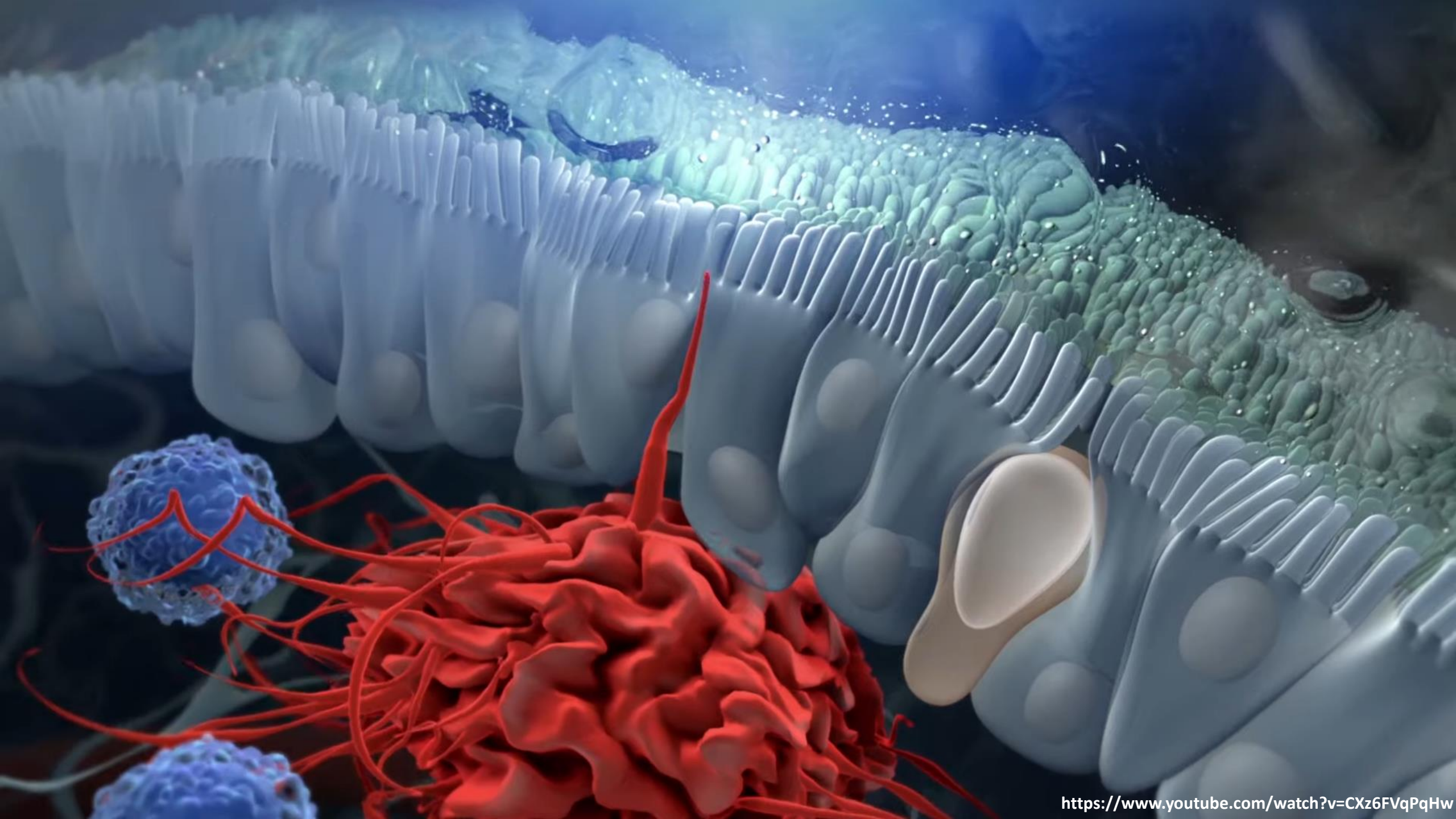
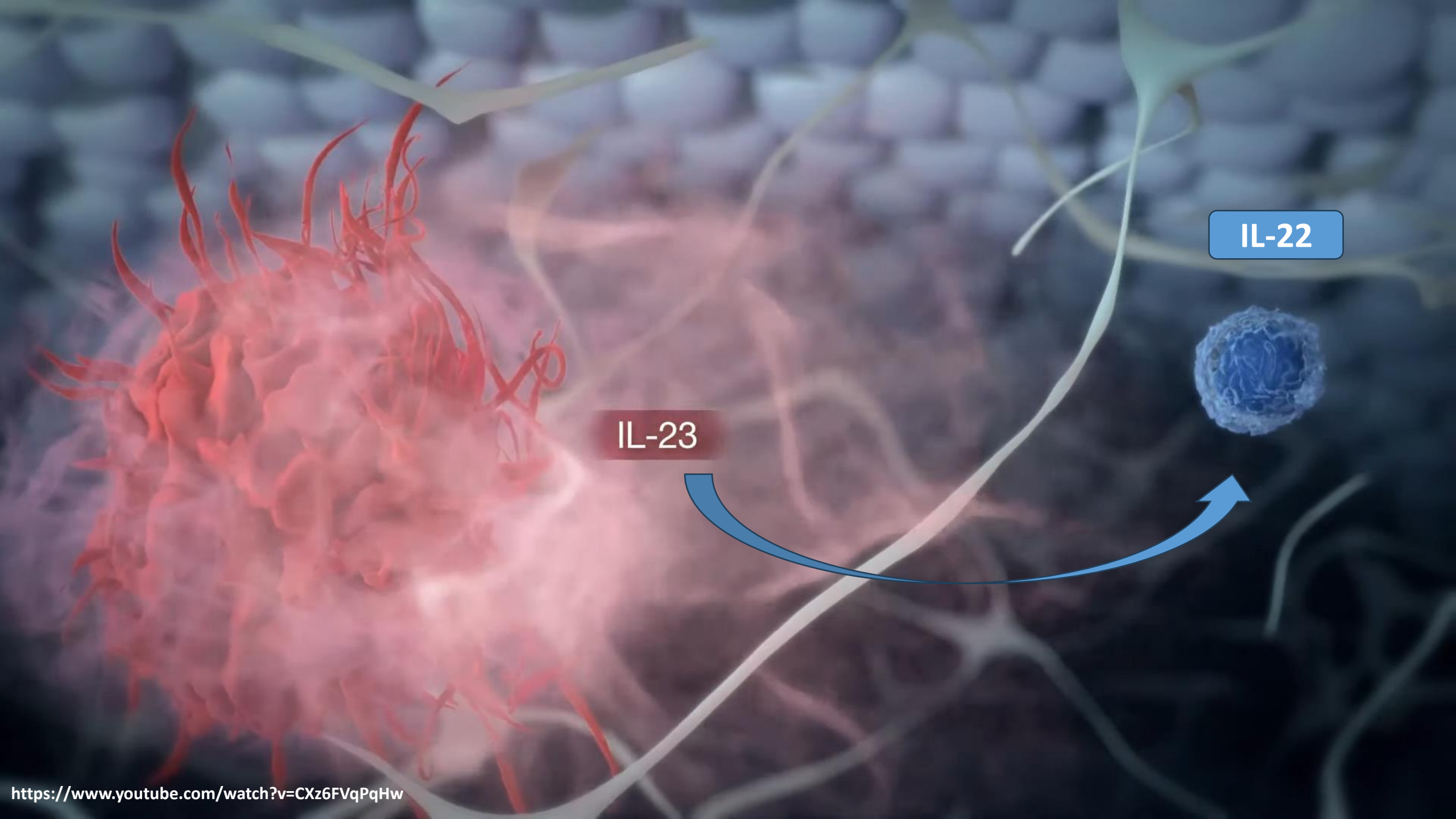


Figure 1 Evolution of the holobiont and vertical transmission through human generations.

Dominguez-Bello MG et al., Gut. 2019 Jun;68(6):1108-1114

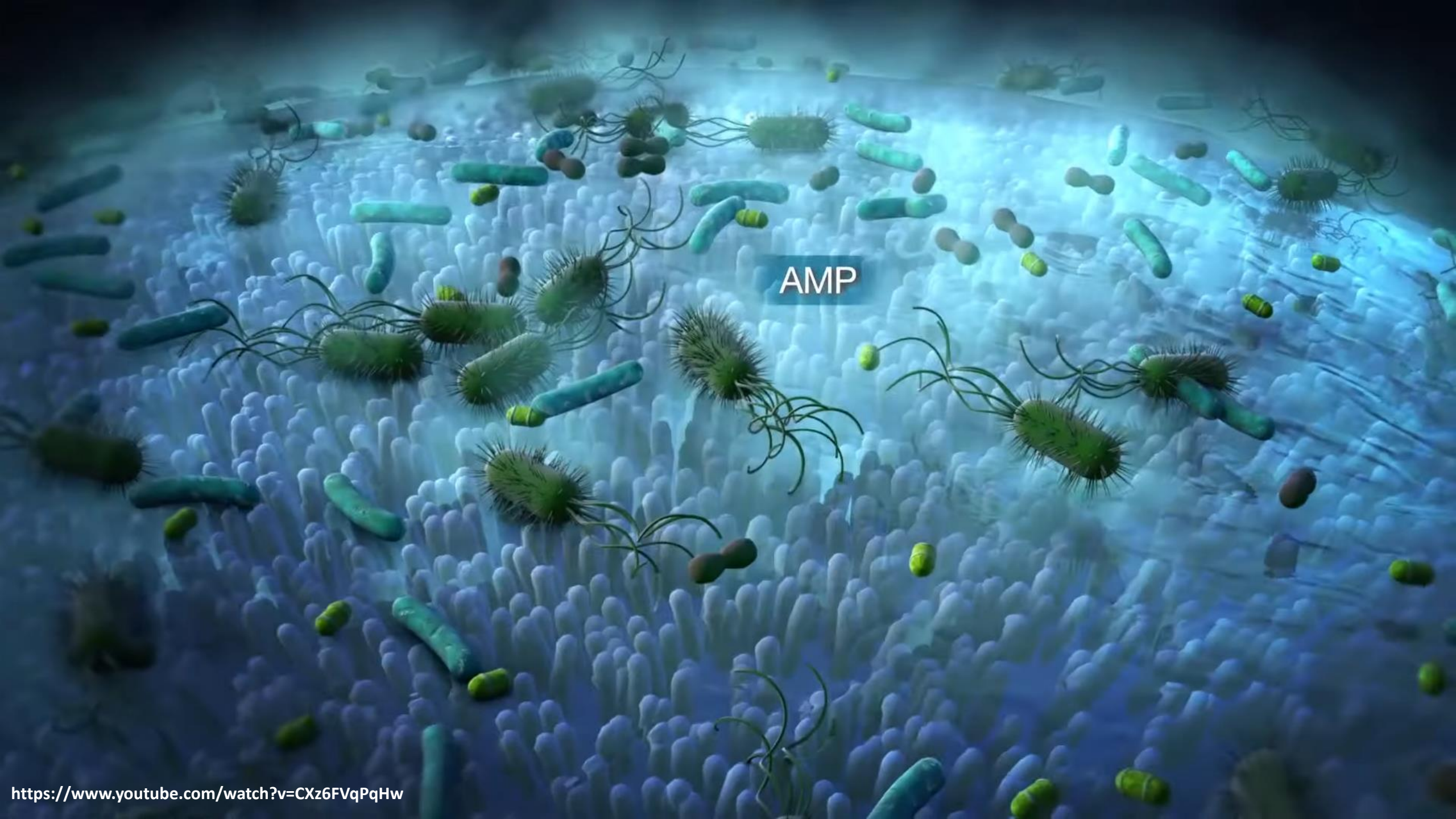




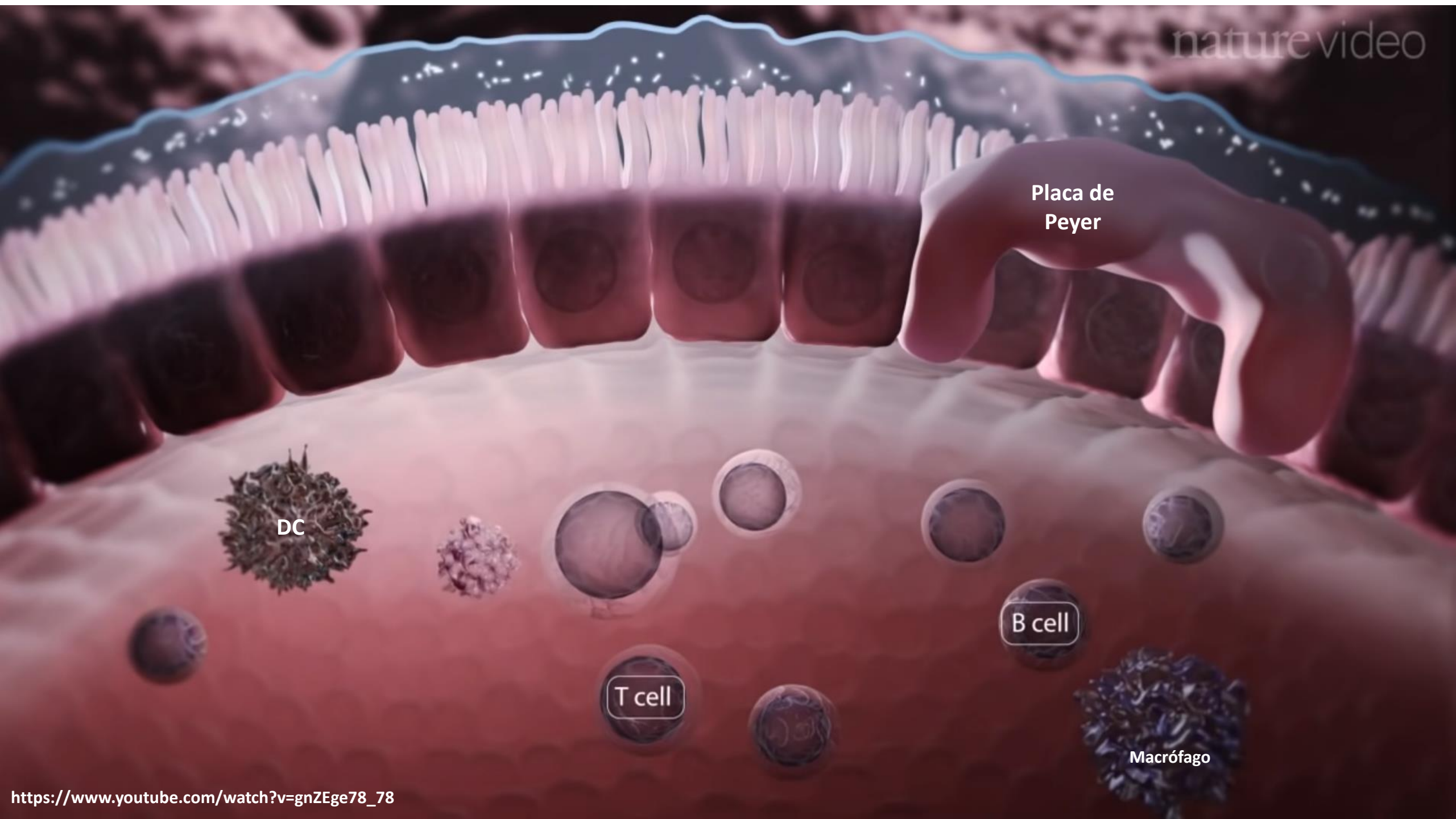


IL-22

IL-23



AMP



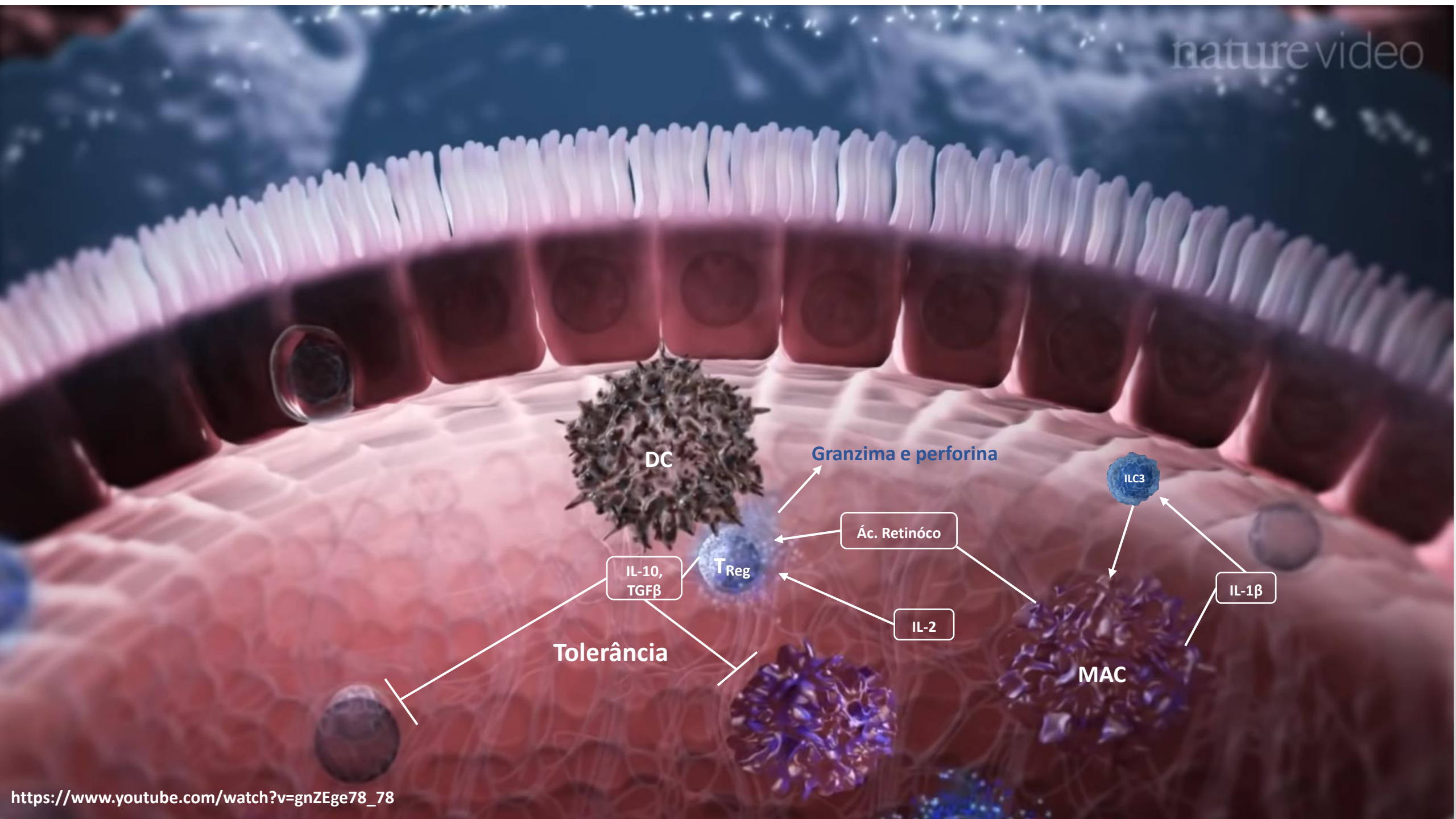
Placa de
Peyer

DC

T cell

B cell

Macrófago



DC

Granzima e perforina

ILC3

Ác. Retinóico

IL-10,
TGFβ

T_{Reg}

IL-2

IL-1β

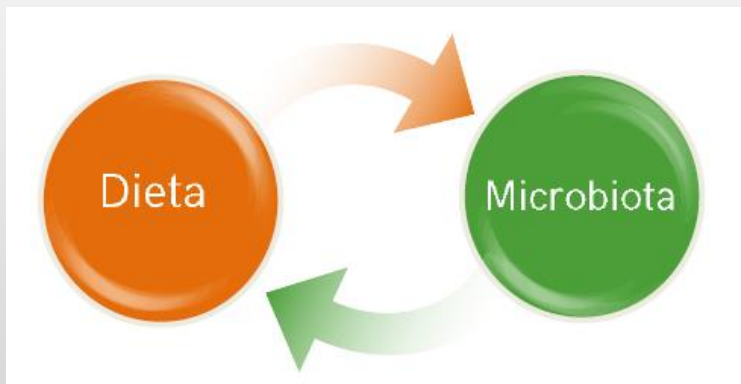
Tolerância

MAC

Modulação da microbiota



- Interações metabólicas
- Competição
- Sensibilidade de presença



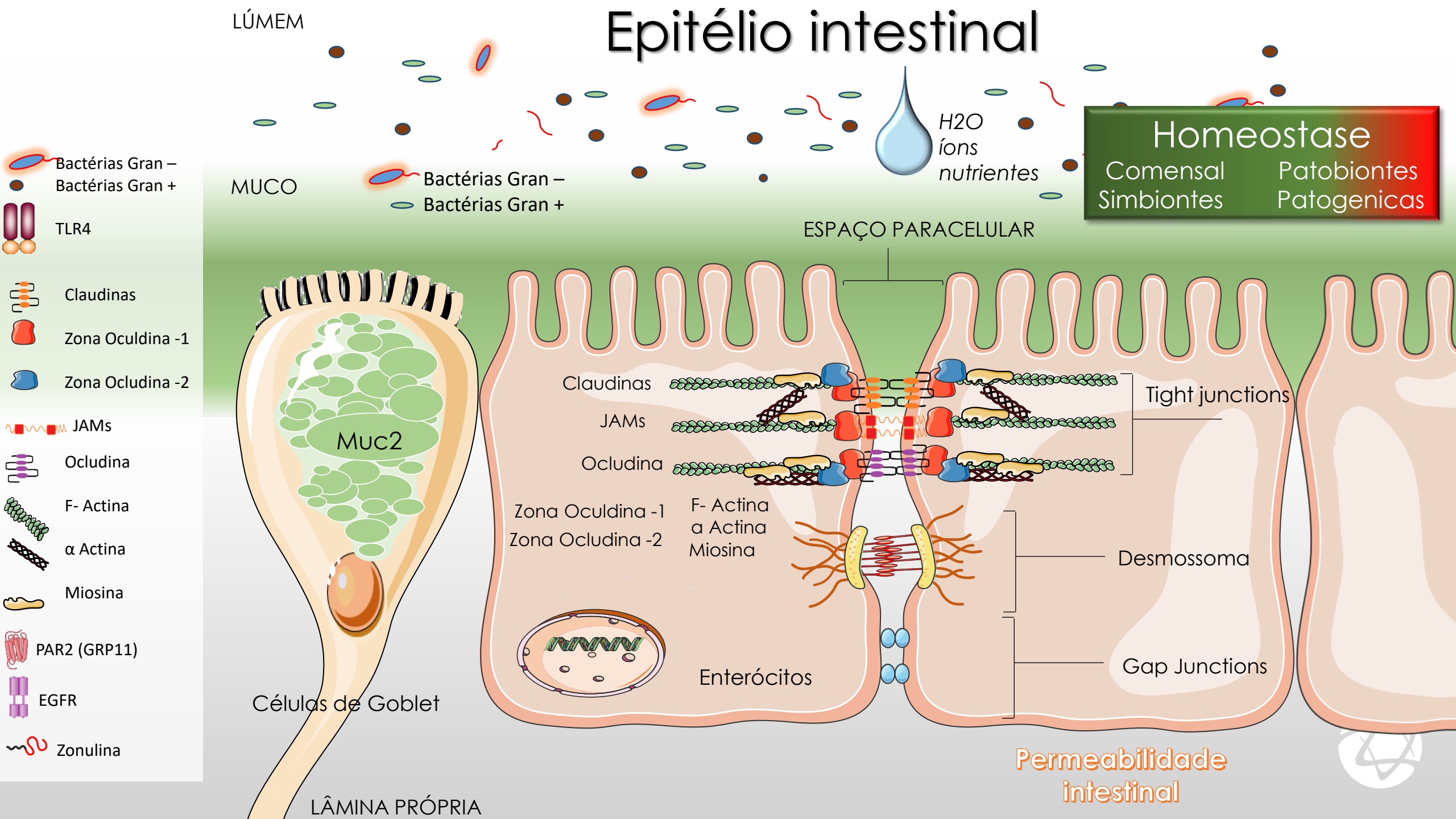
- **Genética e sistema imune do hospedeiro afetam a composição da microbiota**
 - **Desenvolvimento do tecido linfoide intestinal**
 - **Balanco entre as respostas imune Th1, Th2, Th17, Treg**
 - **Desenvolvimento do sistema enteroendócrino e sistema nervoso entérico**
-
- **Quantidade e qualidade da dieta afetam a composição da microbiota**
 - **Confere modulação mais determinante que a genética**

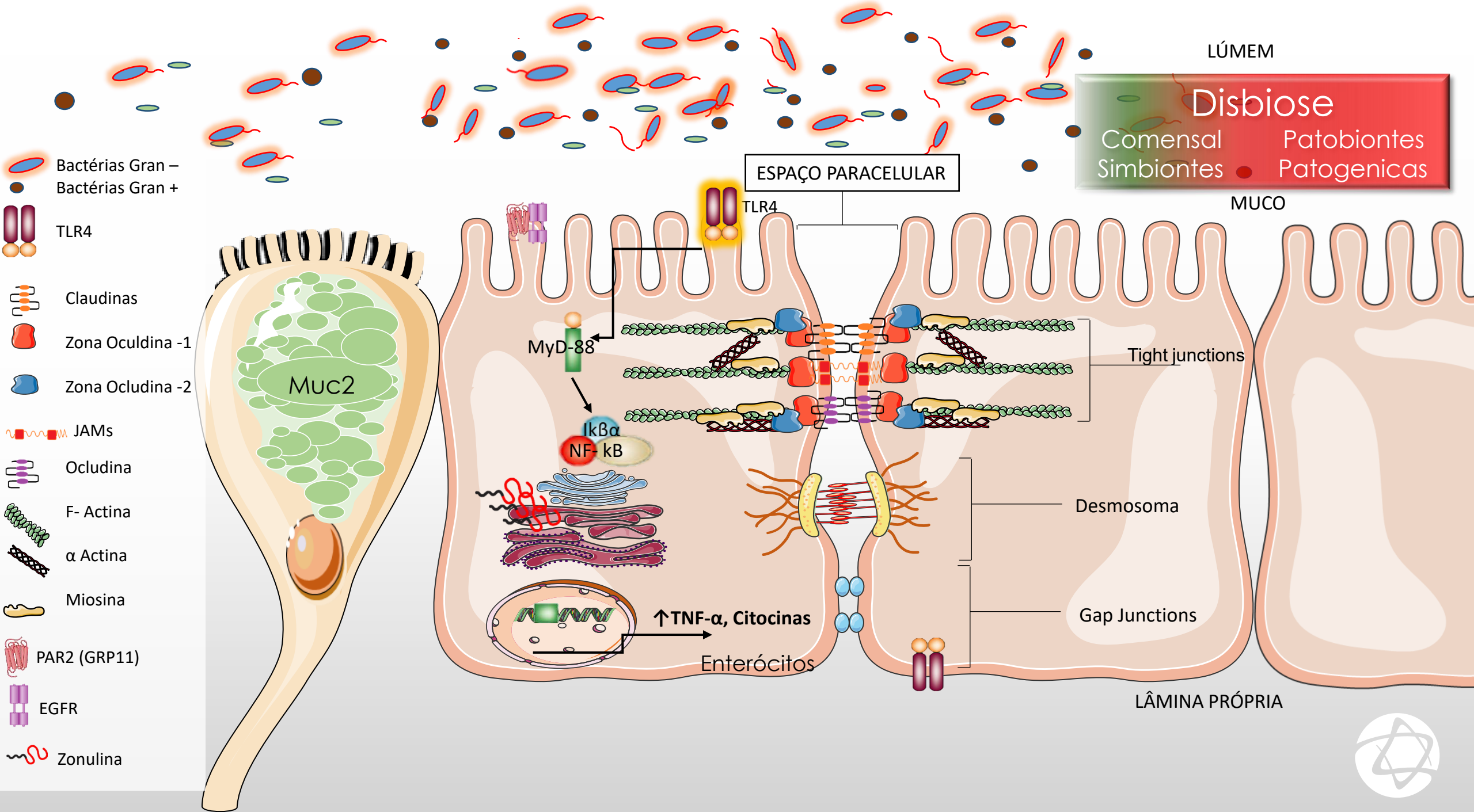


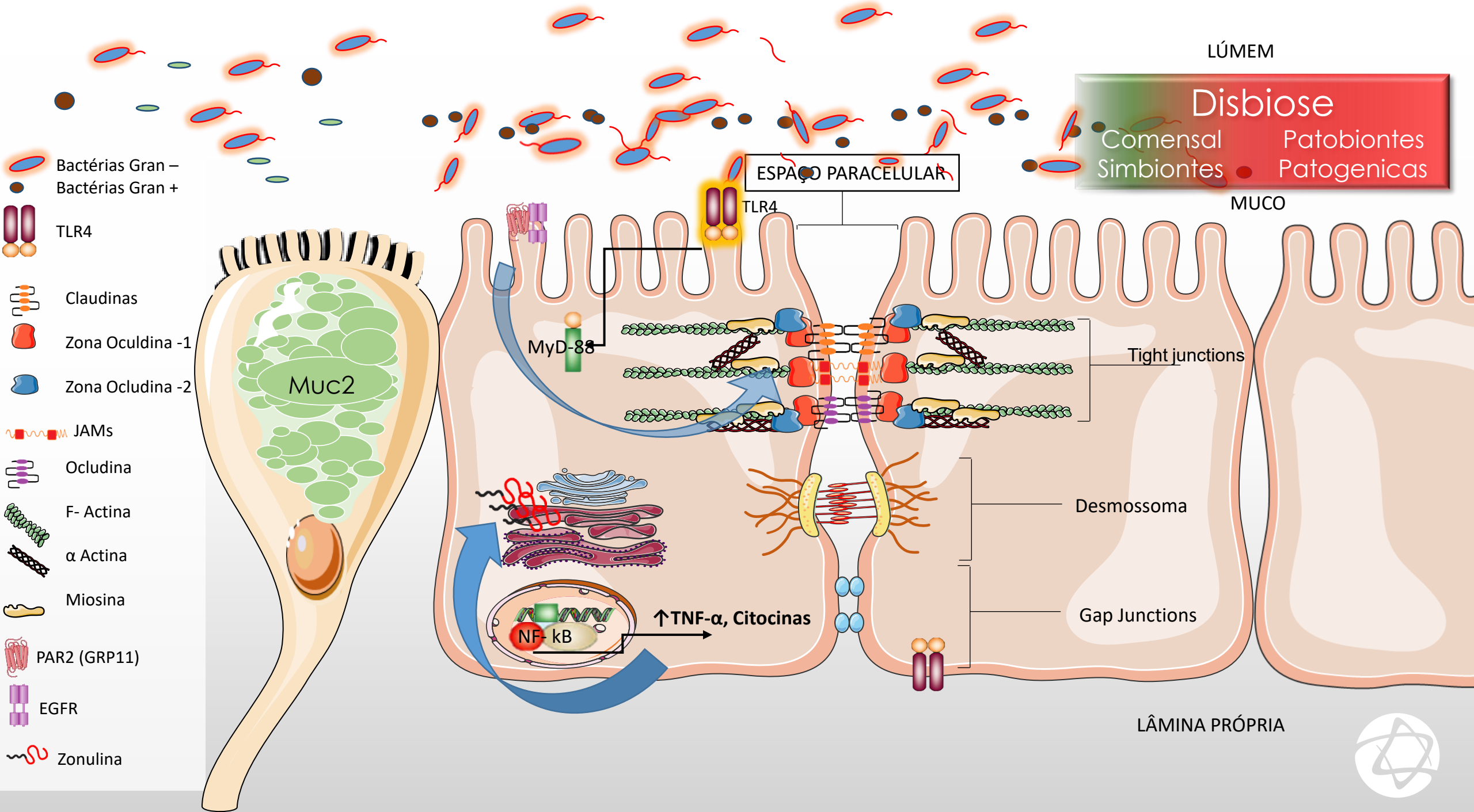


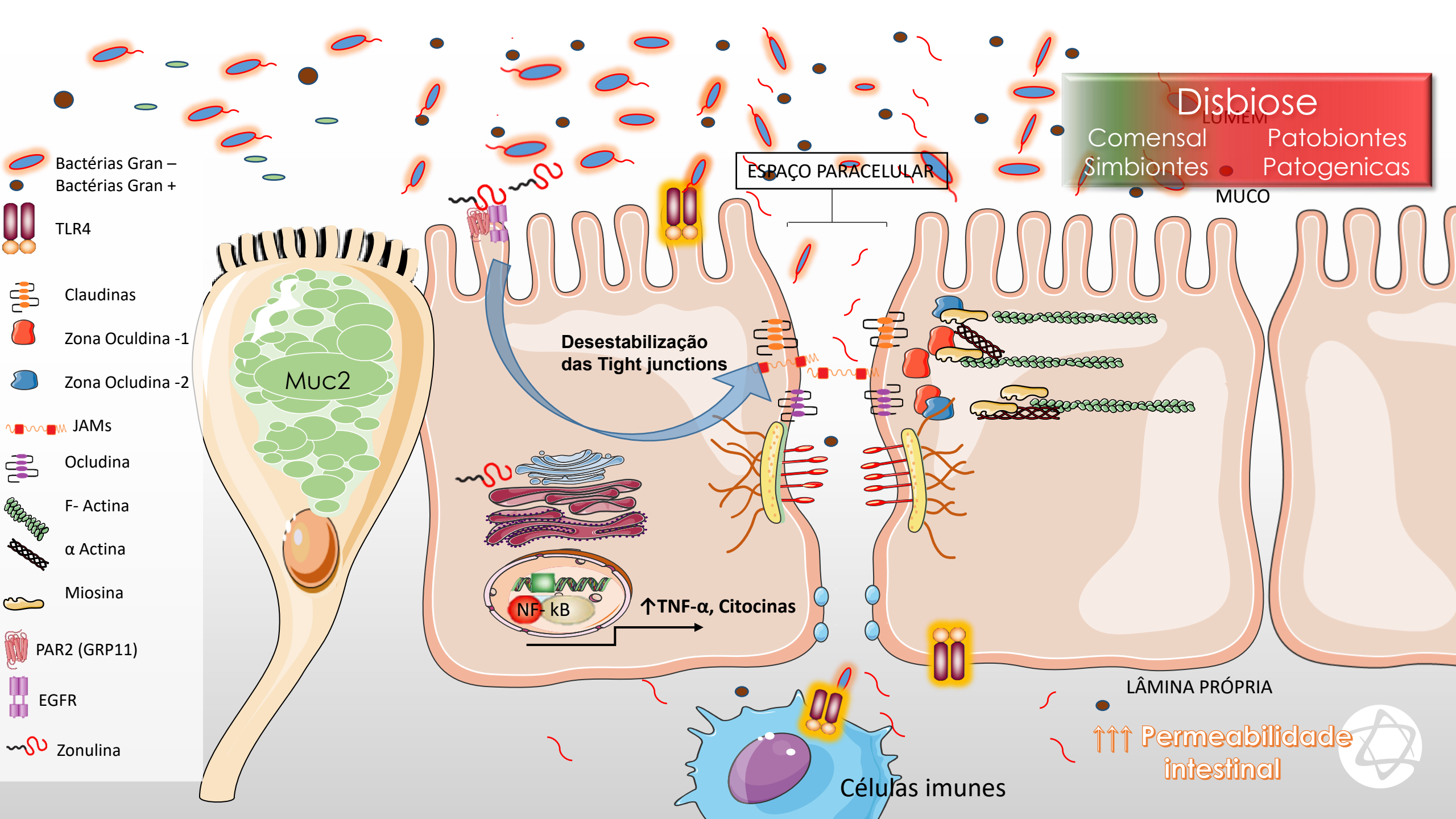
Funcionamento das “*tight-junctions*” e a ação sistêmica do lipopolissacarídeo (LPS)

Epitélio intestinal

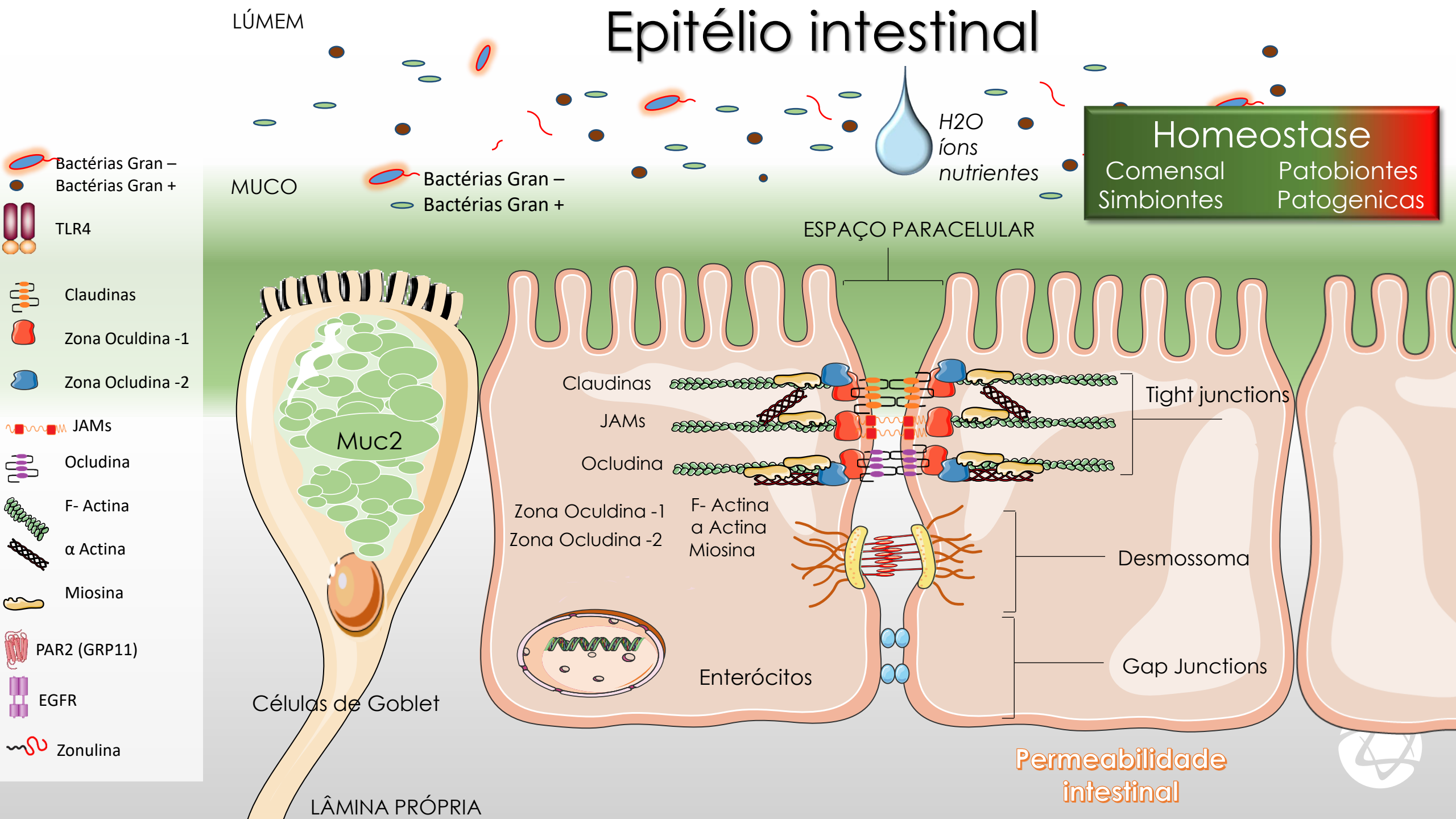








Epitélio intestinal



Homeostase

Comensal
Simbiontes

Patobiontes
Patogenicas

ESPAÇO PARACELULAR

Tight junctions

Desmosoma

Gap Junctions

Permeabilidade
intestinal

LÚMEM

MUCO

Bactérias Gran -
Bactérias Gran +

H₂O
íons
nutrientes

Bactérias Gran -
Bactérias Gran +

TLR4

Claudinas

Zona Ocludina -1

Zona Ocludina -2

JAMs

Ocludina

F- Actina

a Actina

Miosina

PAR2 (GRP11)

EGFR

Zonulina

Muc2

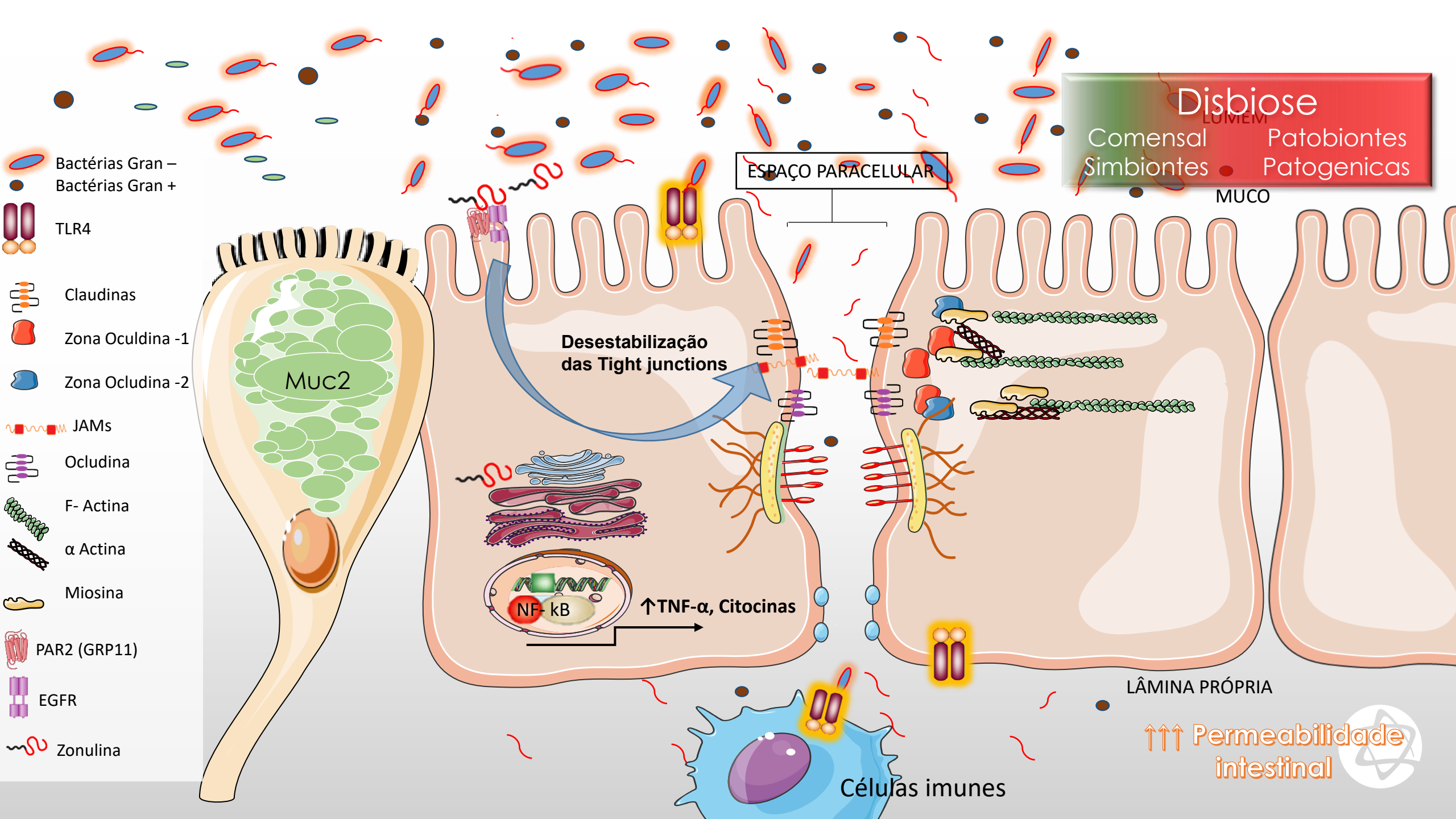
Células de Goblet

LÂMINA PRÓPRIA

Enterócitos

Zona Ocludina -1
Zona Ocludina -2

F- Actina
a Actina
Miosina



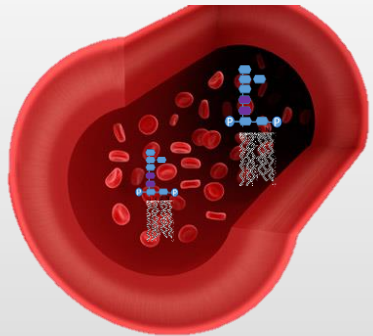
Disbiose



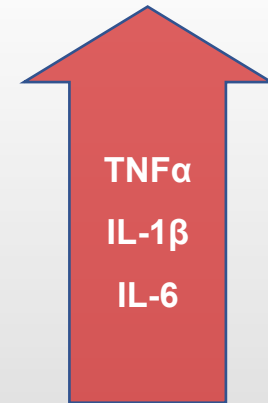
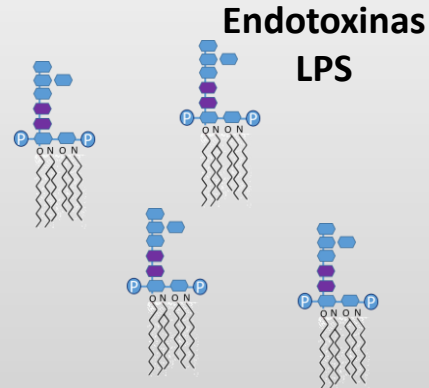
Endotoxemia metabólica



Inflamação crônica de baixo grau



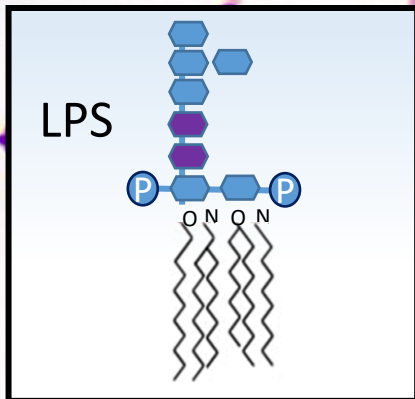
**Endotoxemia
Promovida pela
microbiota intestinal**



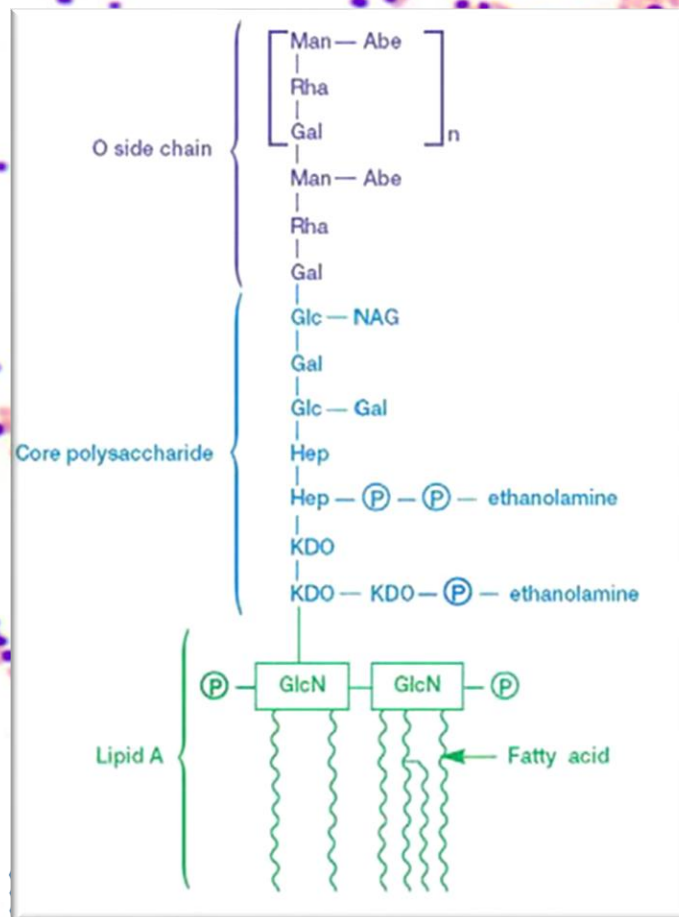
**TNF α
IL-1 β
IL-6**

*Cain et al. Diabetes 2007; 56: 1761–1772.
Kalianan et al., Sci Rep. 2015; 5:11276.*





**Coloração
Gram positiva →**



**Coloração
Gram negativa**

Lipídeo A - responsável
pela ação antigênica do
LPS

AG Saturados

- láurico (C12:0)
- mirístico (C14:0)
- palmítico (C16:0)

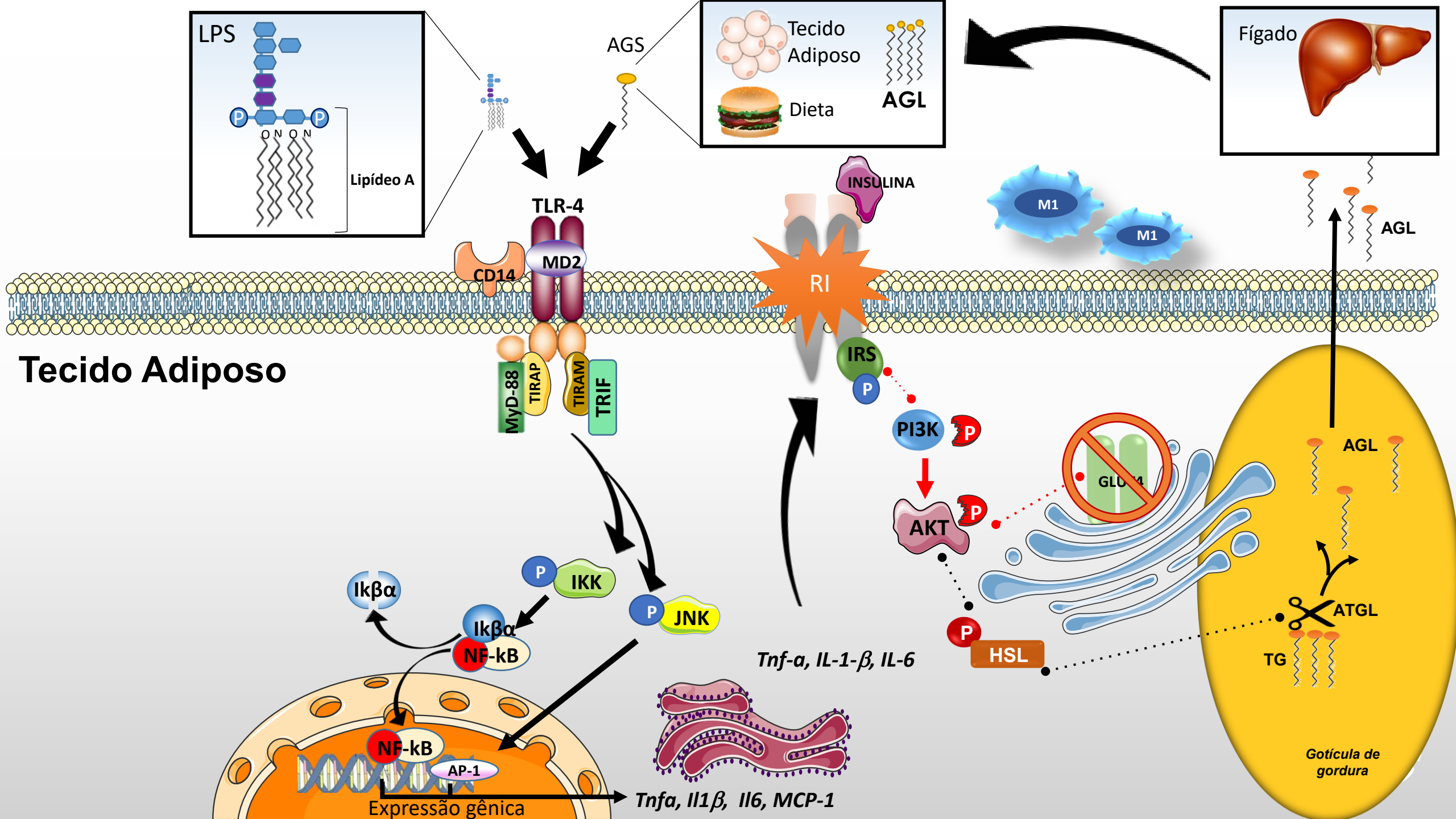
O LPS endógeno é produzido continuamente no intestino pela morte de bactérias Gram-negativas e transportado fisiologicamente para capilares intestinais

Raetz, C. R. *Biochemistry of endotoxins*. Annu Ver. Biochem. 1990: 59, 129-170.

Raetz CR, Whitfield C. *Lipopolysaccharide endotoxins*. Annu Rev Biochem. 2002;71:635-700.

Neal et al., *J Immunol*. 2006 Mar, 1;176(5):3070-9

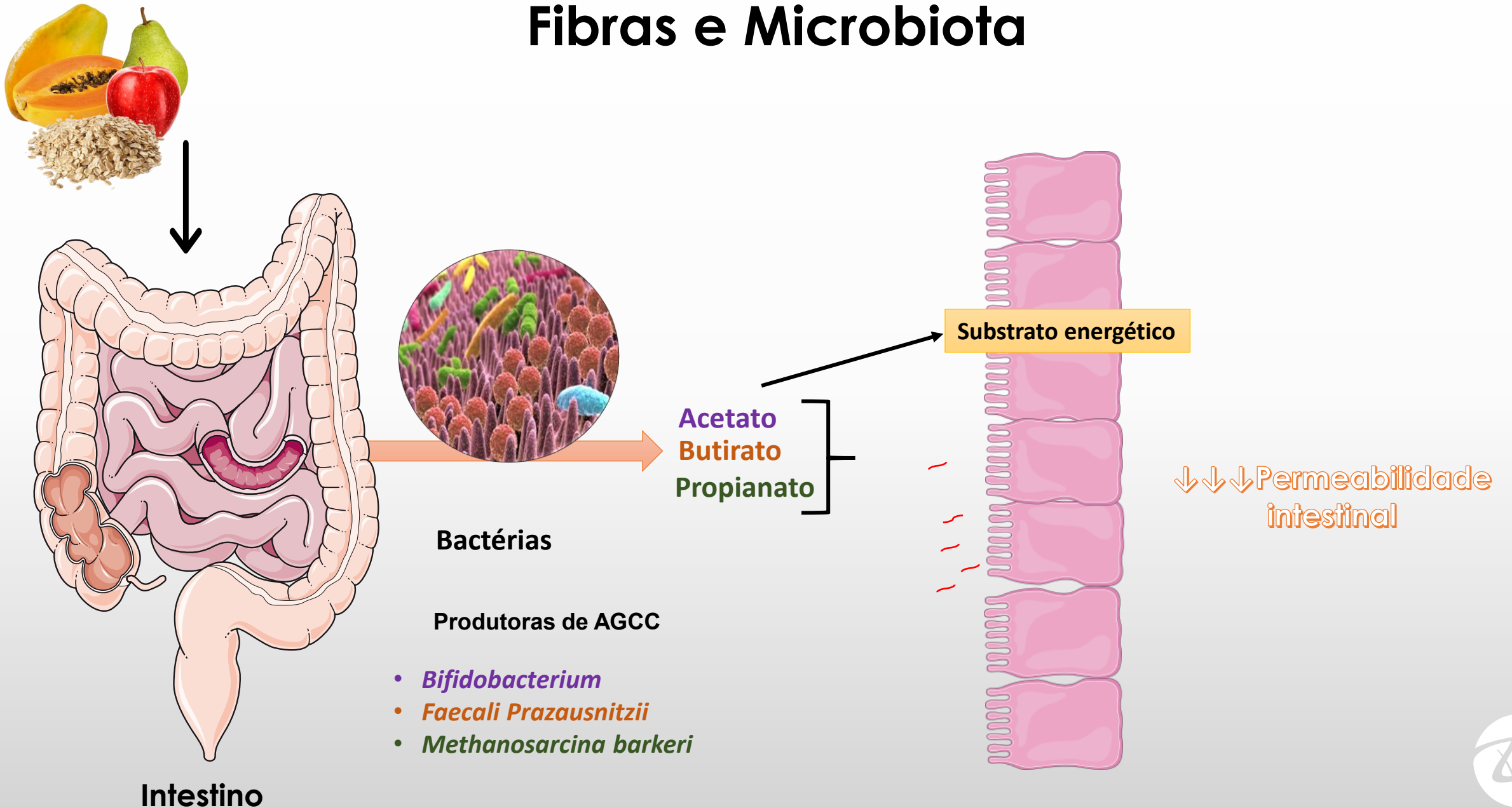




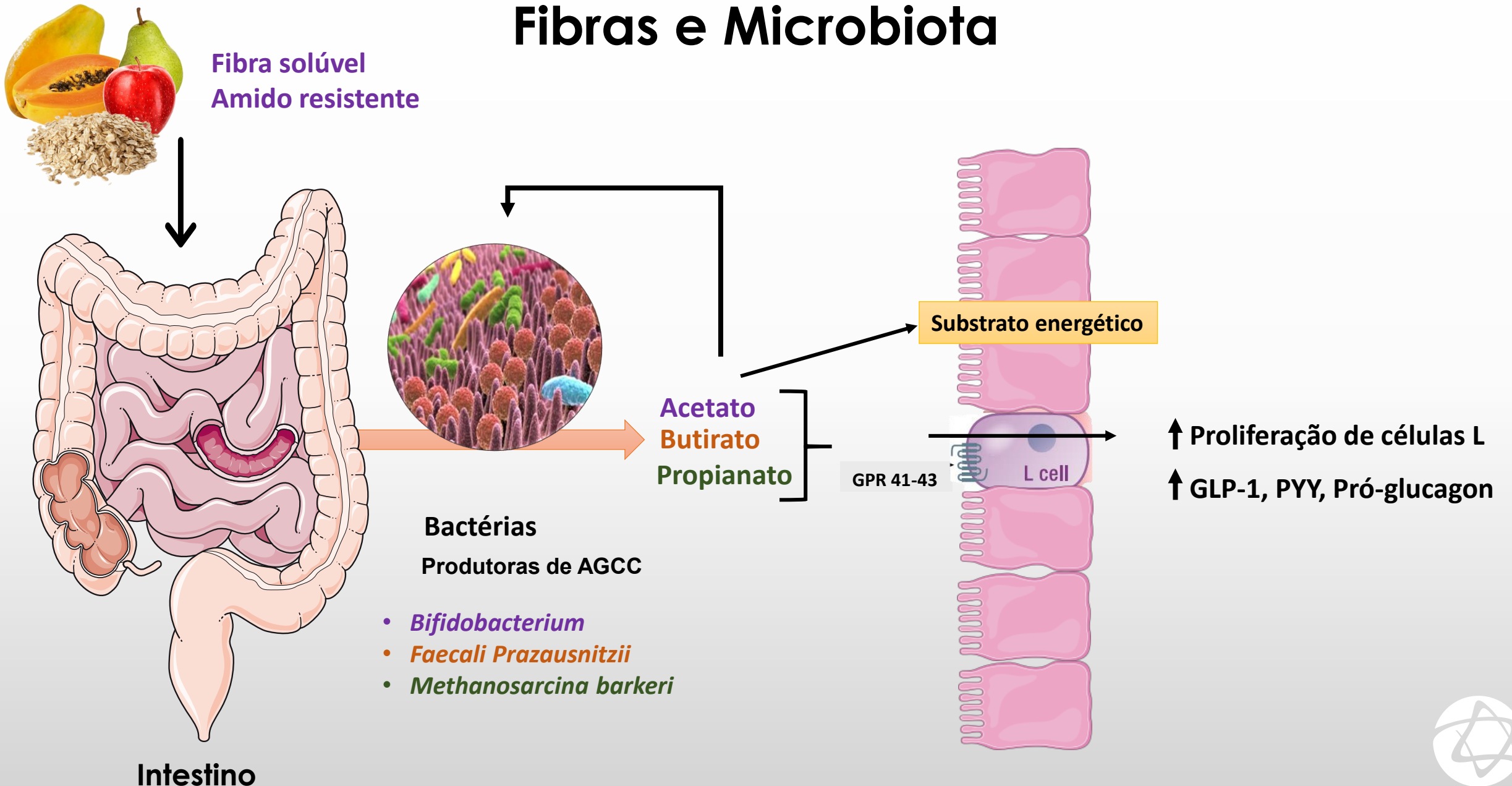


Fibra alimentar e microbiota intestinal: Os AGCC e sua ação sistêmica

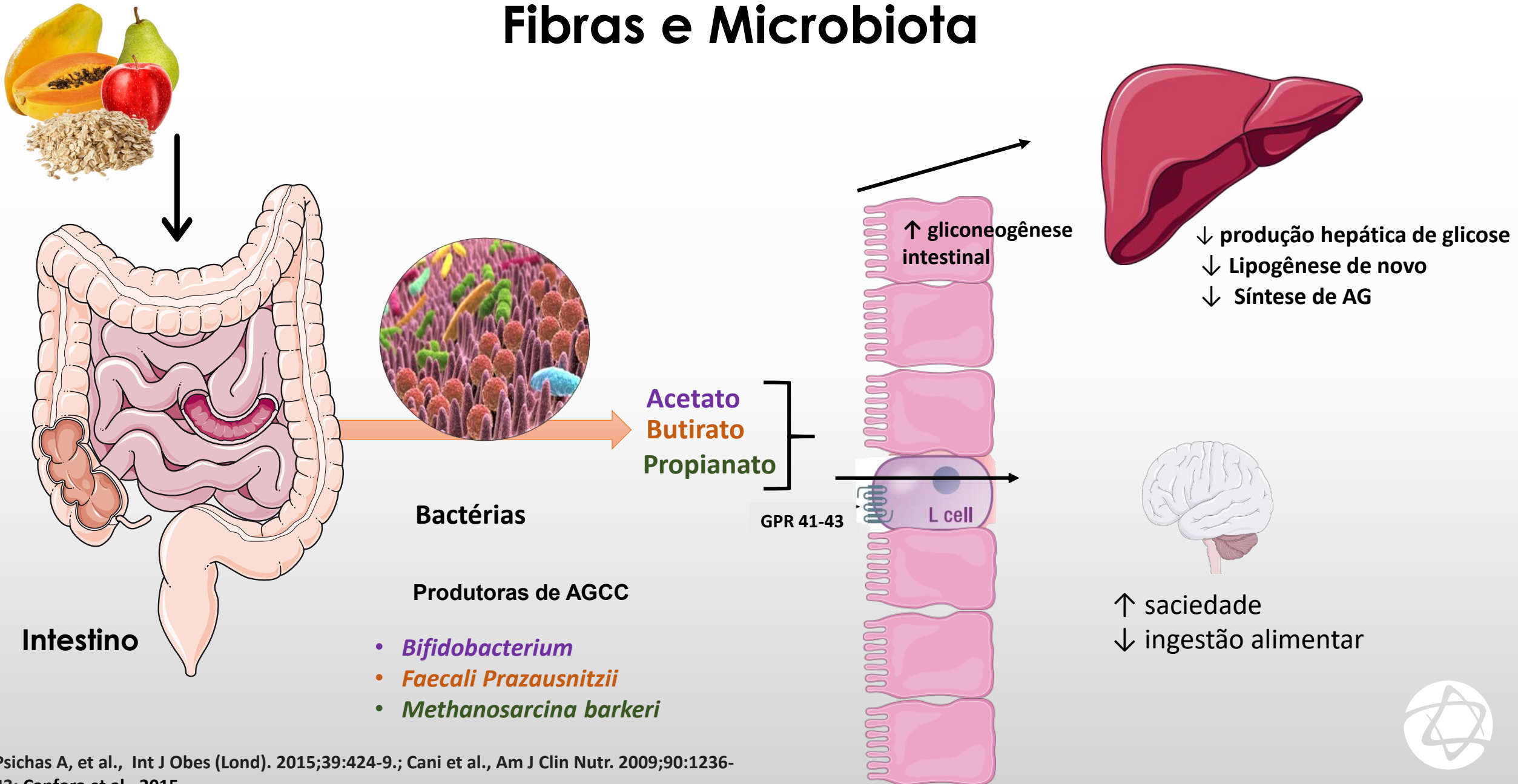
Fibras e Microbiota

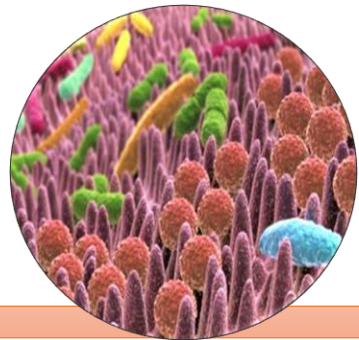


Fibras e Microbiota



Fibras e Microbiota





Produtoras de AGCC

- *Bifidobacterium*
- *Faecali Praausnitzii*
- *Methanosarcina barkeri*

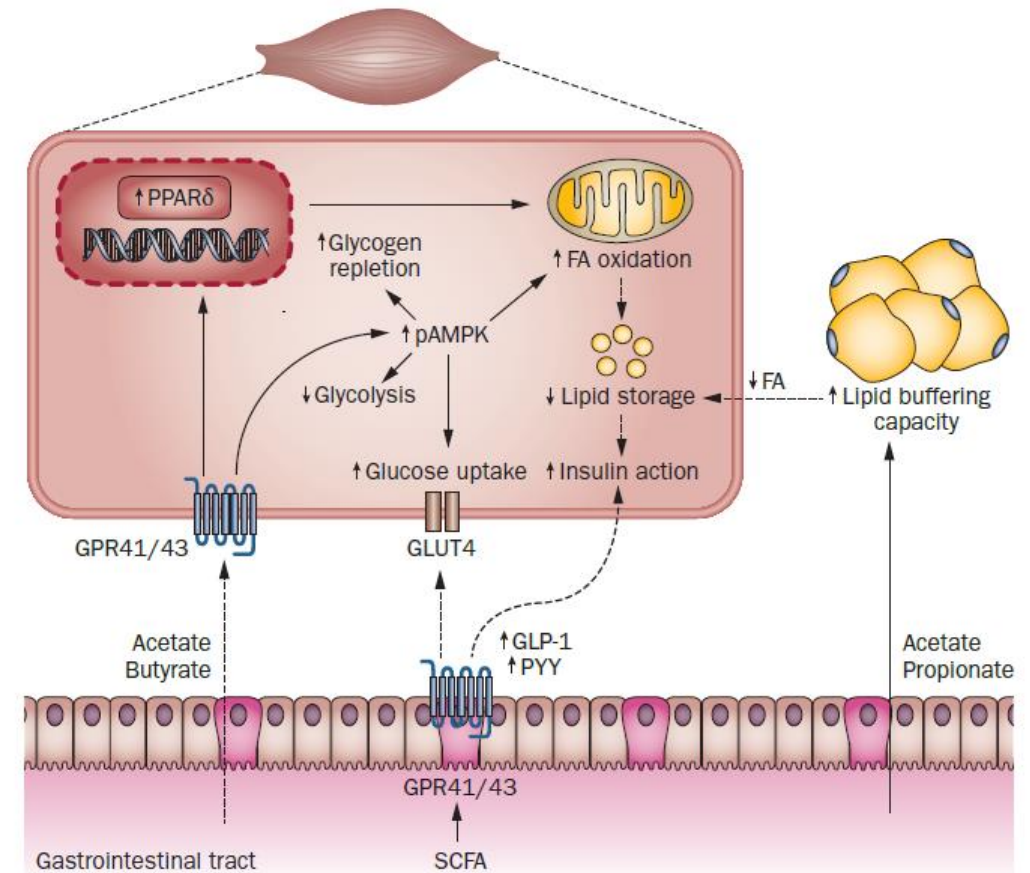
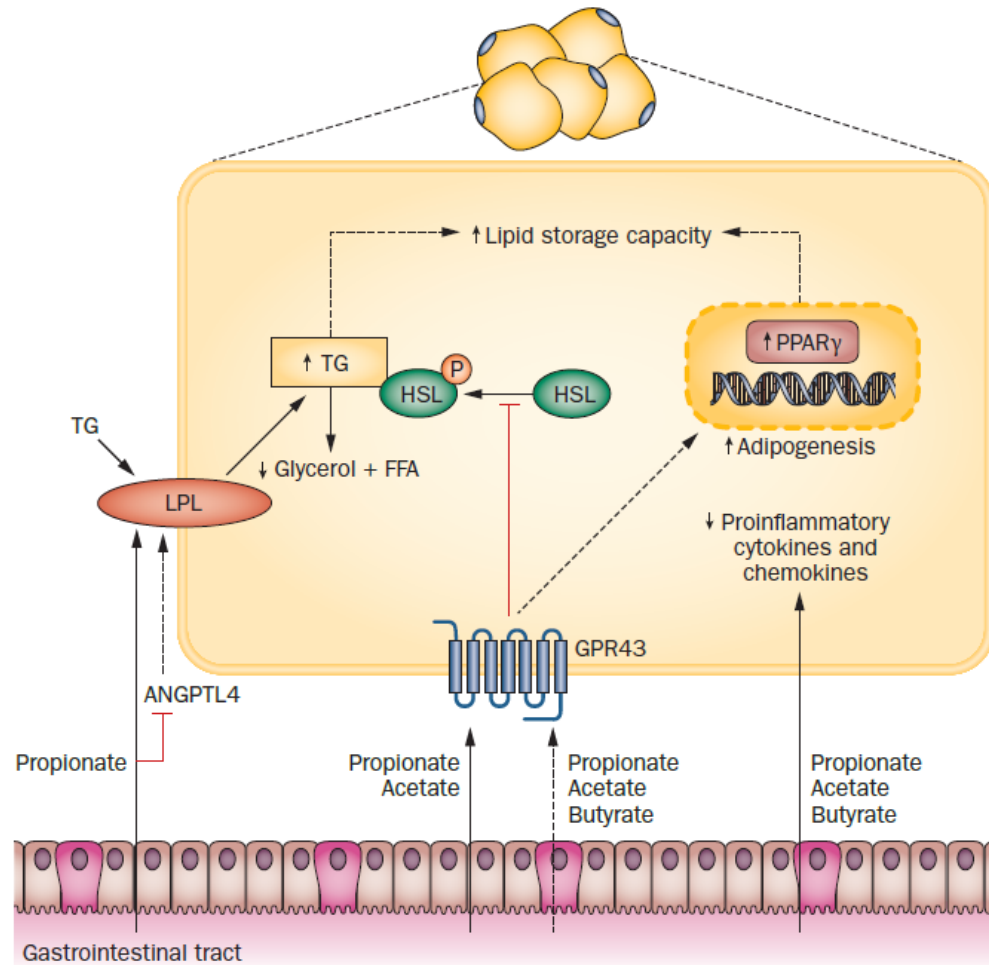
Acetato
Butirato
Propionato

GPR 41-43



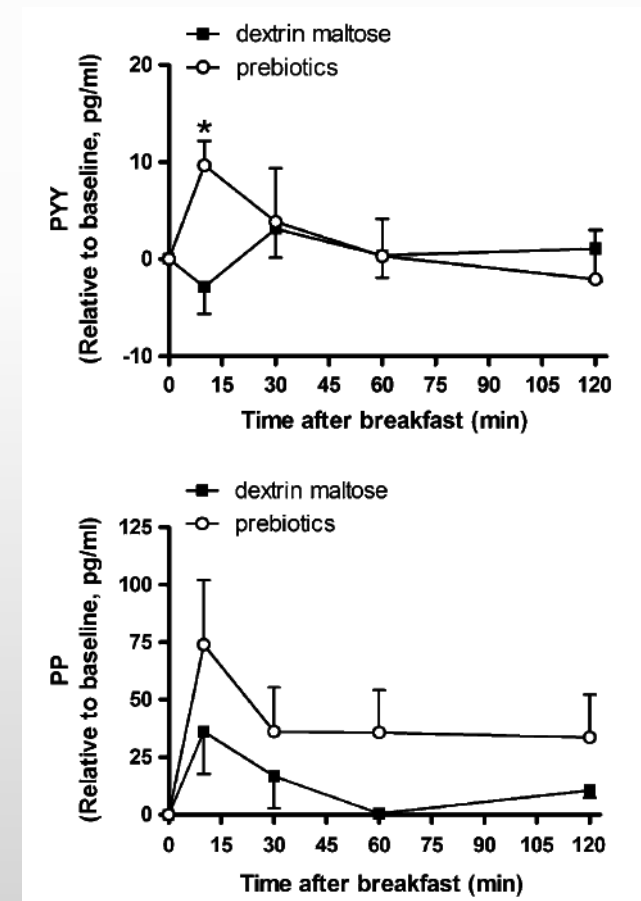
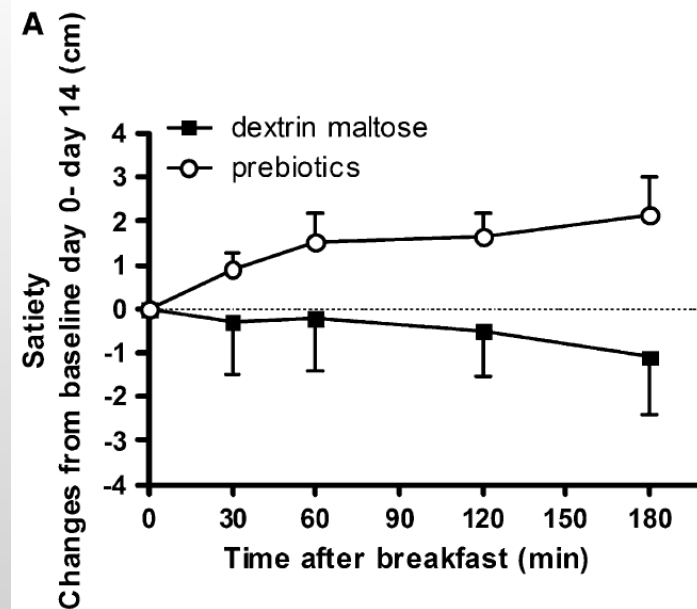
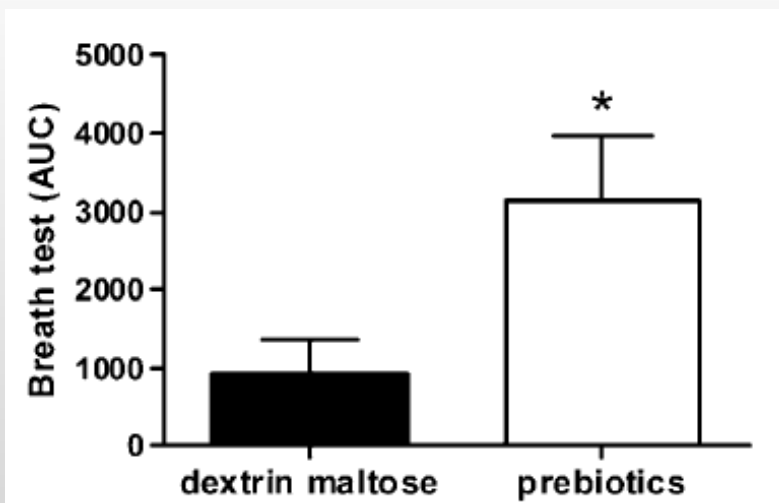
- ↑ captação de AGL
- ↓ lipólise
- ↓ AGL no plasma
- ↓ inflamação
- ↑ sensibilidade insulina

Ação sistêmica dos AGCC



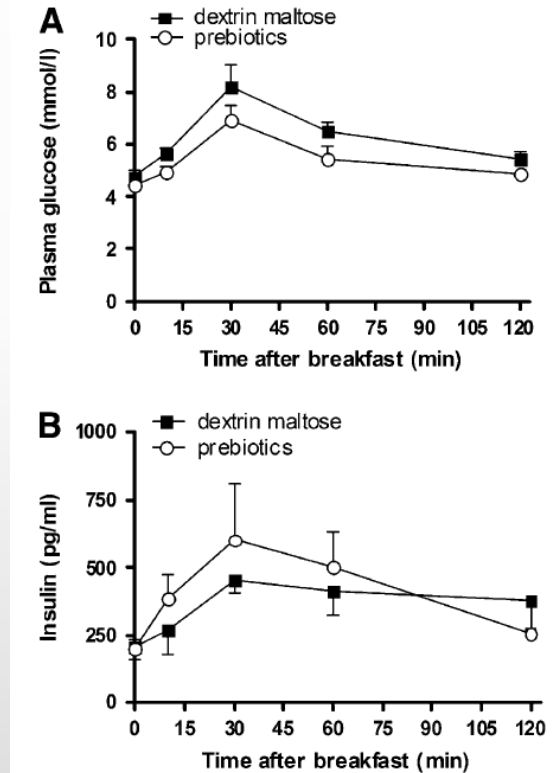
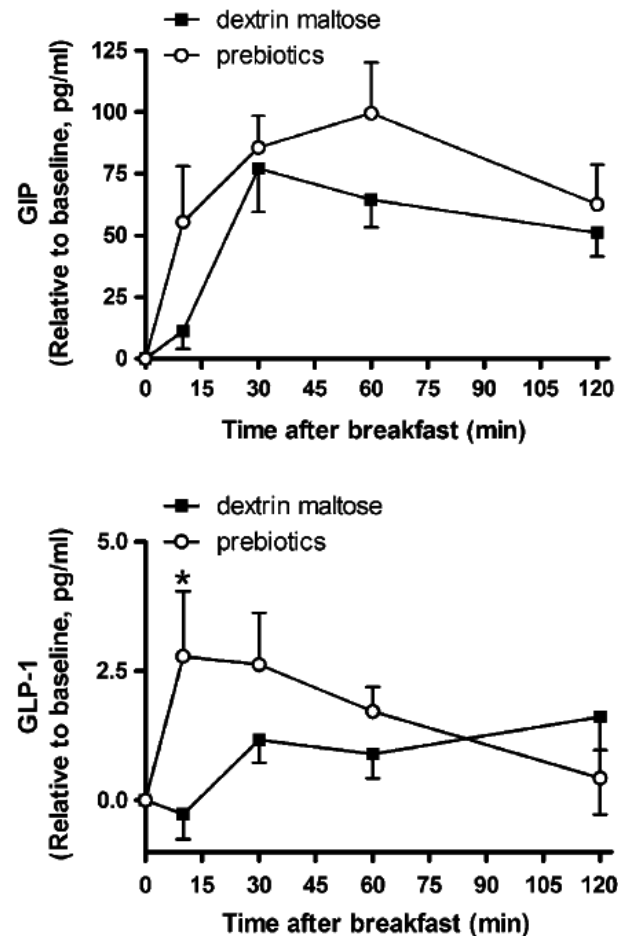
Consumo de fibras induz saciedade, eleva produção de incretinas e melhora resposta glicêmica pós-prandial

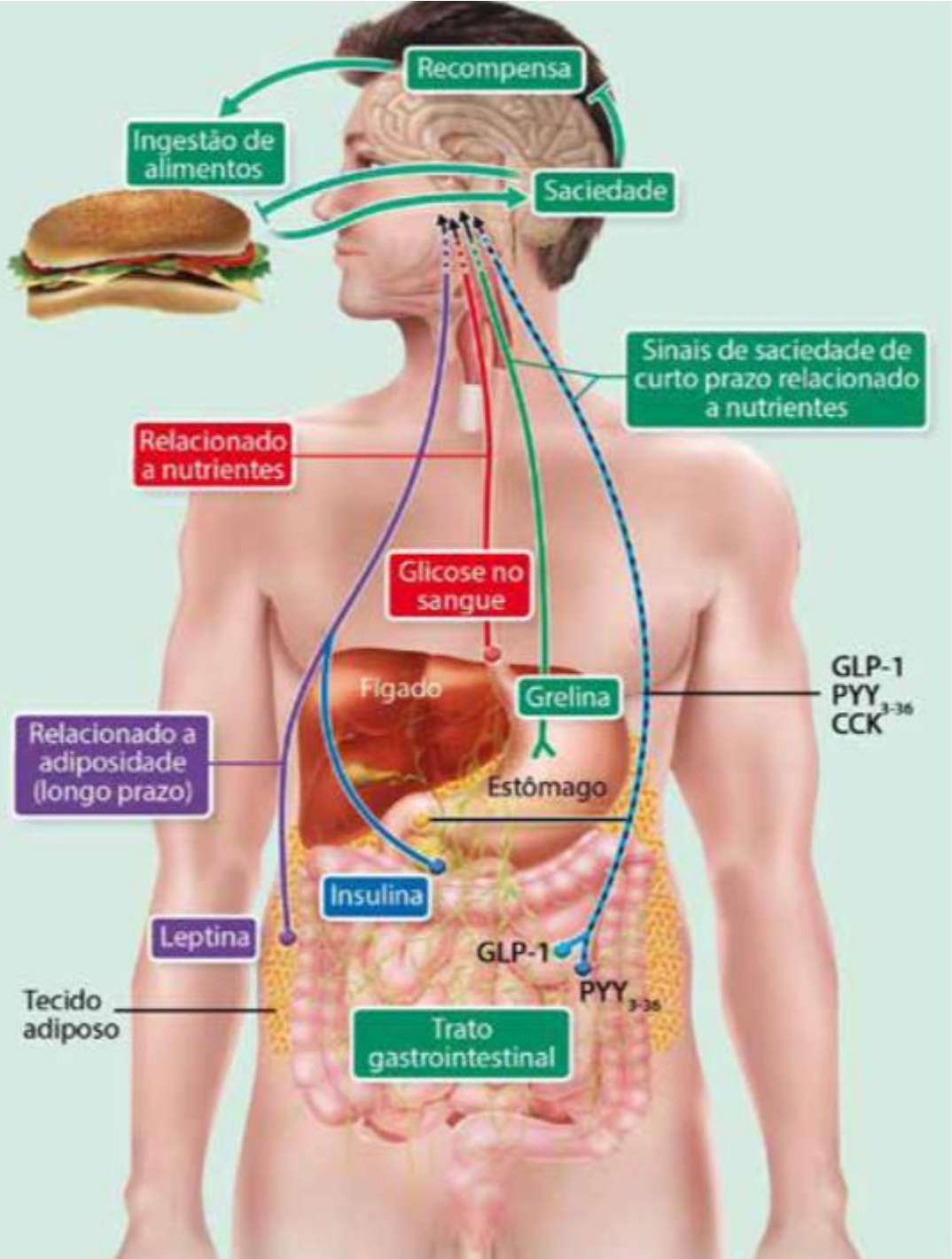
- 10 indivíduos saudáveis, peso normal.
- Estudo duplo-cego, paralelo, placebo x controle
- Prebiótico (fibra solúvel) x placebo – 2 semanas
- 2 tomadas (8g cada) → CM e Jantar



Consumo de fibras induz saciedade, eleva produção de incretinas e melhora resposta glicêmica pós-prandial

- Incretina → estimula secreção de insulina por cel- β
- Melhora da resposta glicêmica

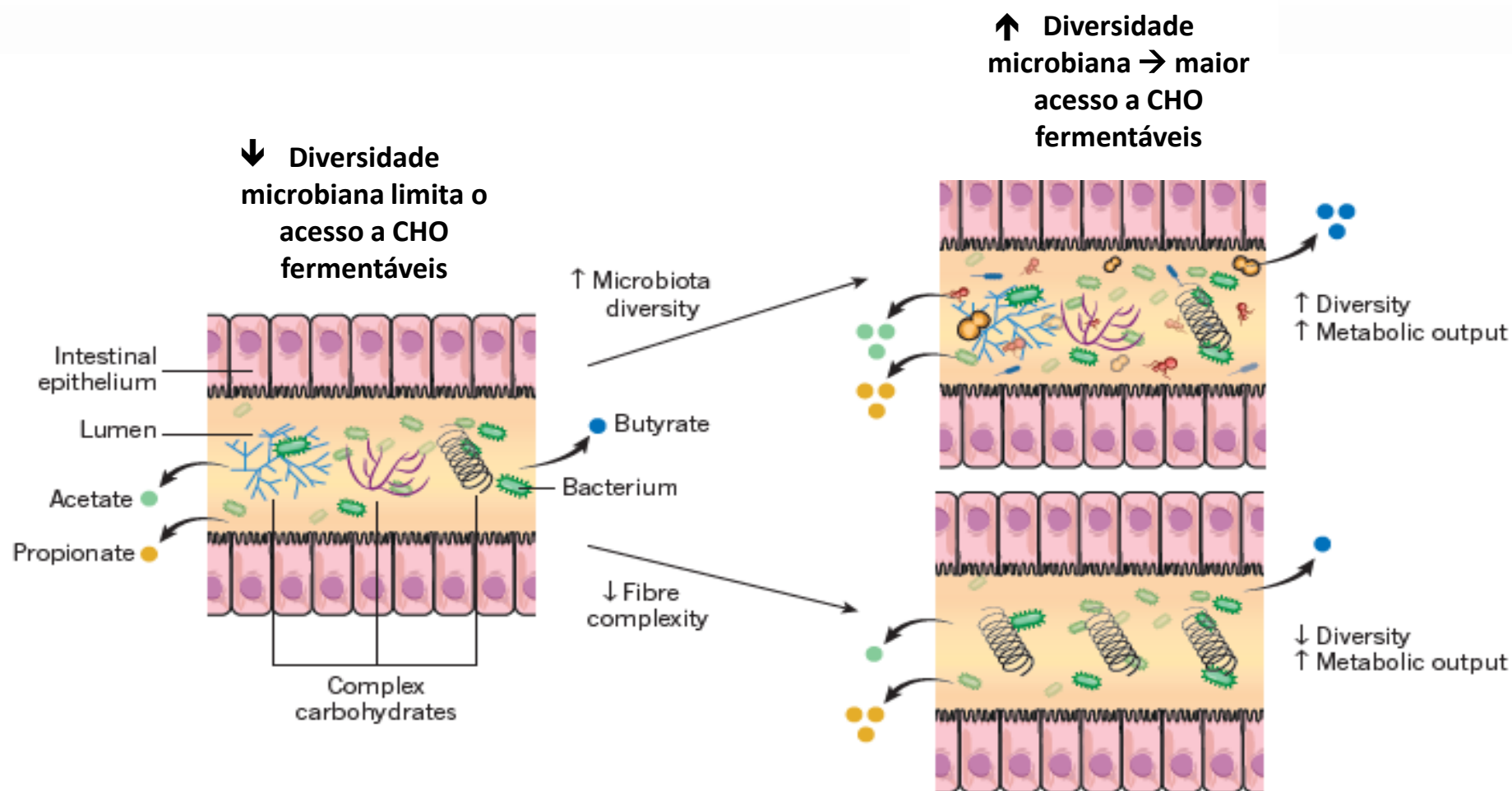




Regulação neuroendócrina do consumo alimentar



Interação entre DIETA x MICROBIOTA determina a produção de AGCC



Causas

- Doenças cardiovasculares
- Diabetes tipo 2
- Neoplasias
- Outras causas

Número de mortes global atribuídas à dieta



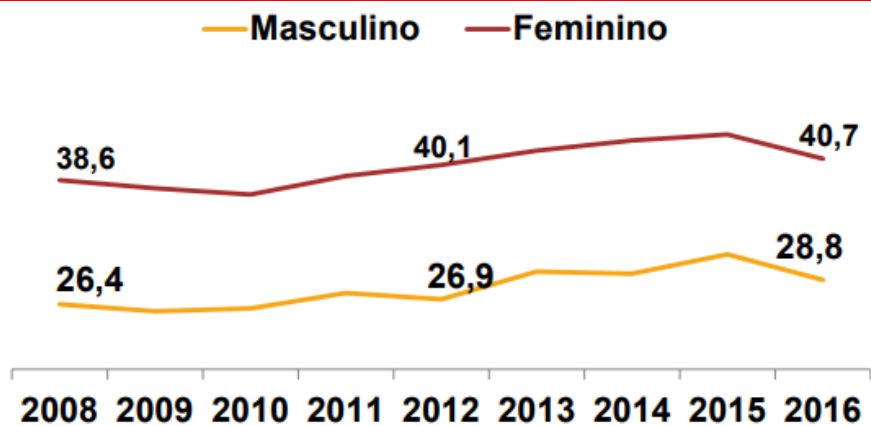
Consumo de Fibras no Brasil

CONSUMO REGULAR DE FRUTAS E HORTALIÇAS

Cresce o consumo regular de frutas e hortaliças.

Em 2008 era **33,0%**, e 2016, **35,2%**

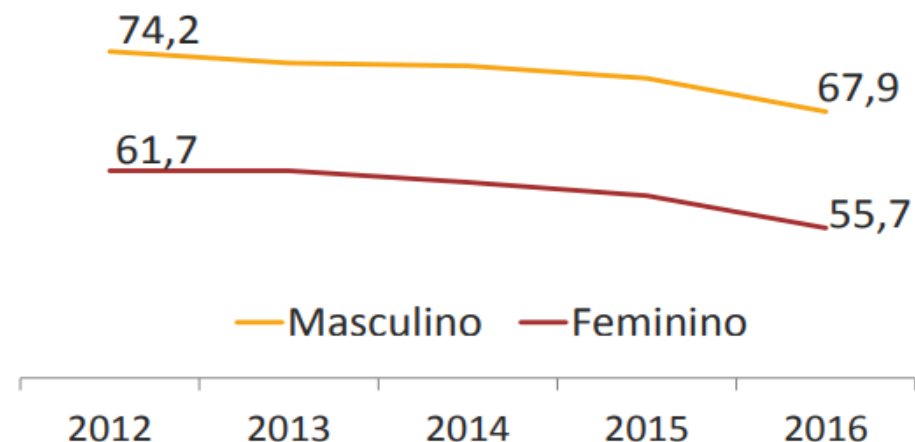
Em 2016, **apenas 1 entre 3 adultos** consomem frutas e hortaliças em 5 dias da semana



CONSUMO REGULAR DE FEIJÃO

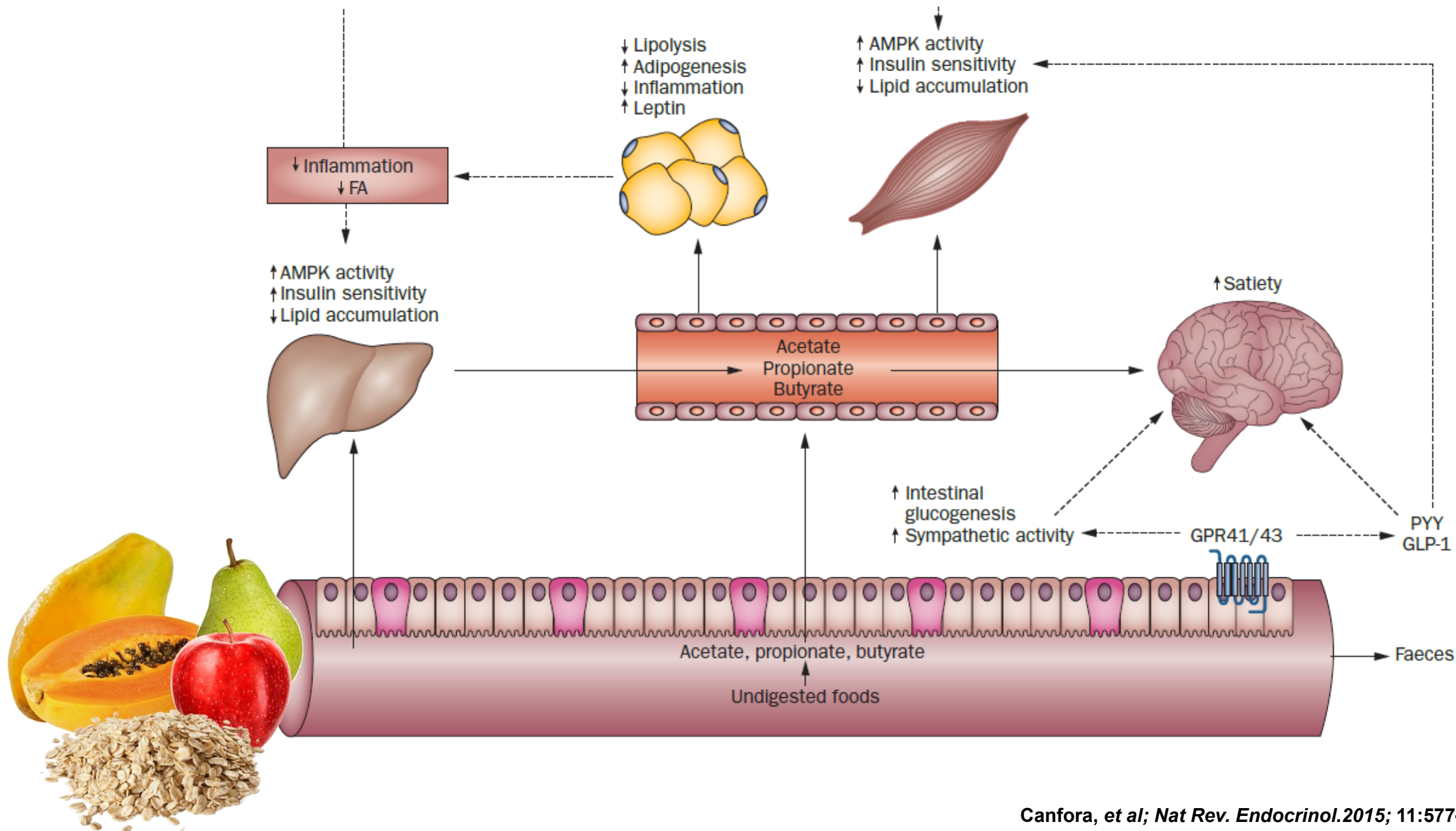
Cai consumo regular de feijão

Saiu de **67,5%** em 2012 para **61,3%** em 2016



* Consumo em 5 ou mais dias da semana.





Eubiose
Dieta rica em Fibras
AGCC



Integridade da barreira intestinal

↓ inflamação
intestinal

↑ produção e secreção
de mucina 2

↑ expressão de TJ

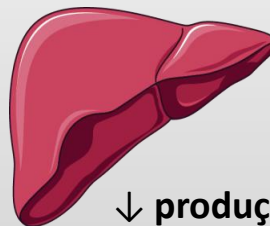
↓ Ativação TLR4 –
secreção de citocinas
pró-inflamatórias

↑ camada de muco

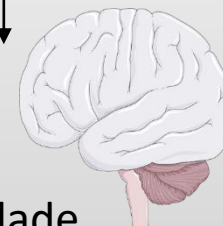
↓ permeabilidade
intestinal



↑ captação de AGL
↓ AGL no plasma
↓ inflamação
↑ sensibilidade insulina



↓ produção hepática de glicose
↓ Lipogênese de novo
↓ Síntese de AG



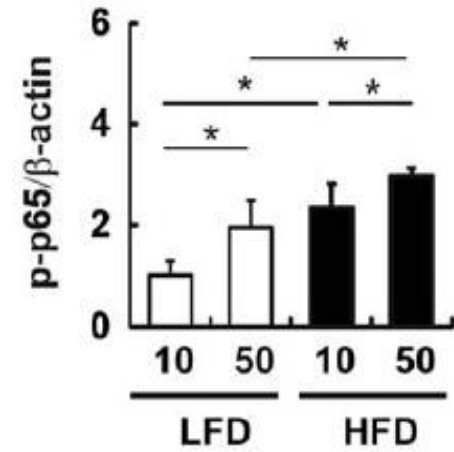
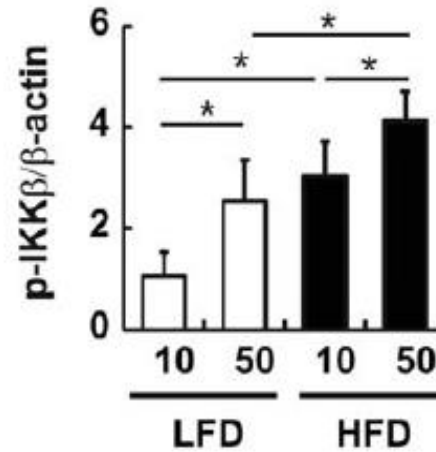
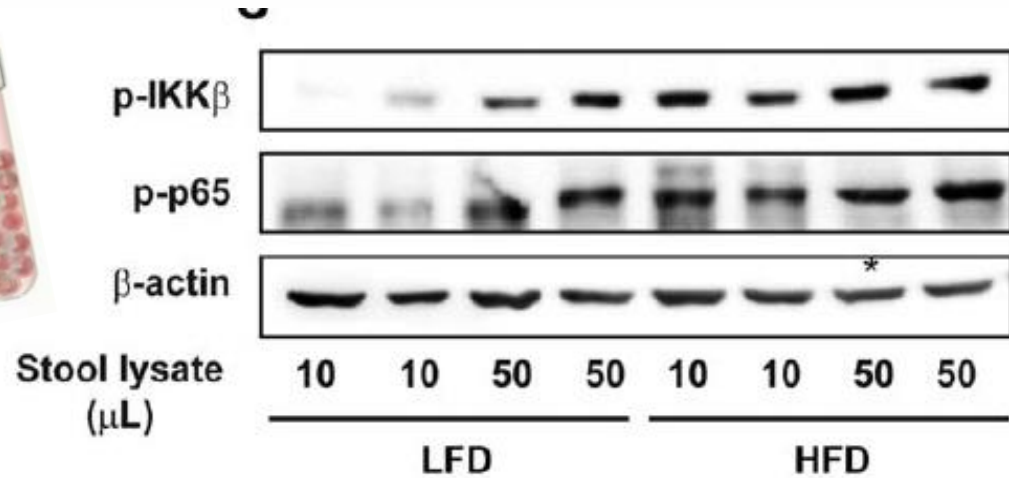
↑ saciedade
↓ ingestão alimentar



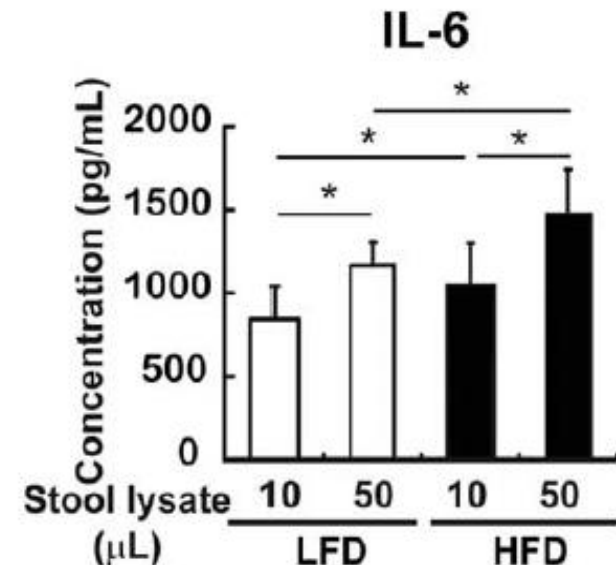
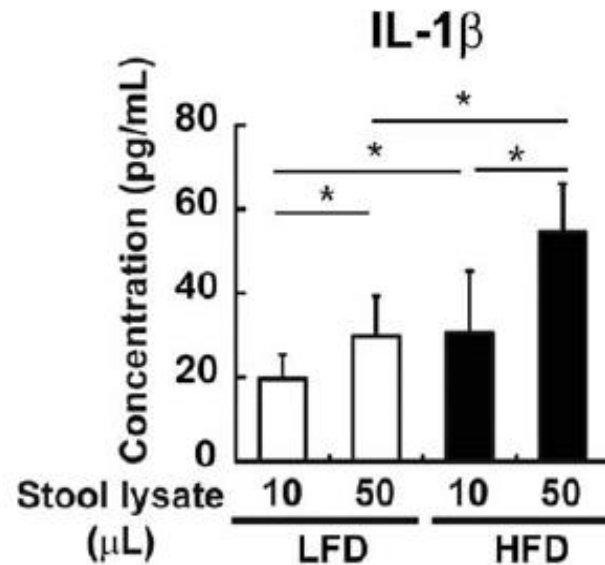
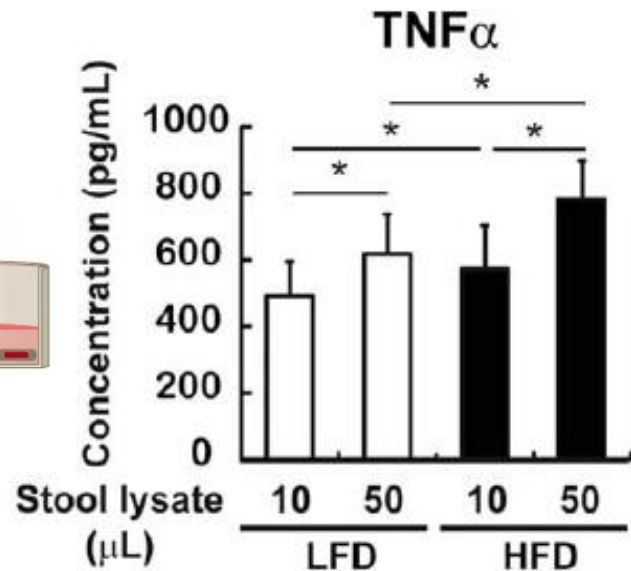


Efeito das gorduras na microbiota intestinal e permeabilidade

Dieta rica em gordura altera a composição da microbiota intestinal de camundongos



D



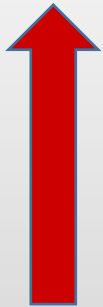
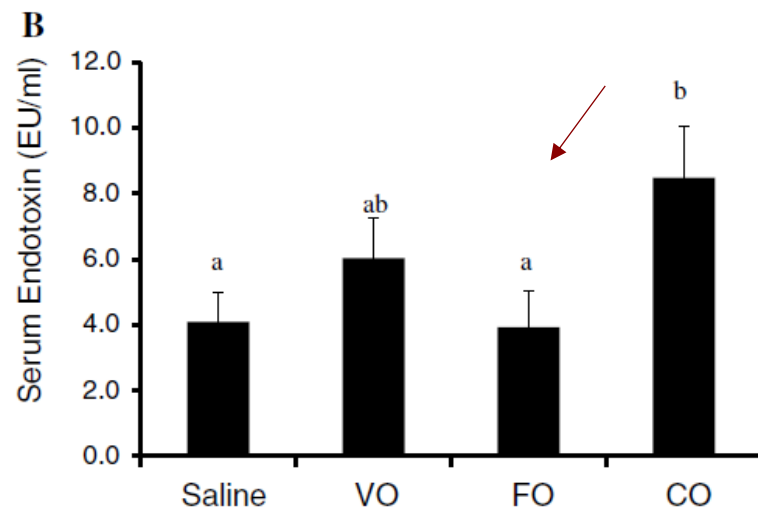
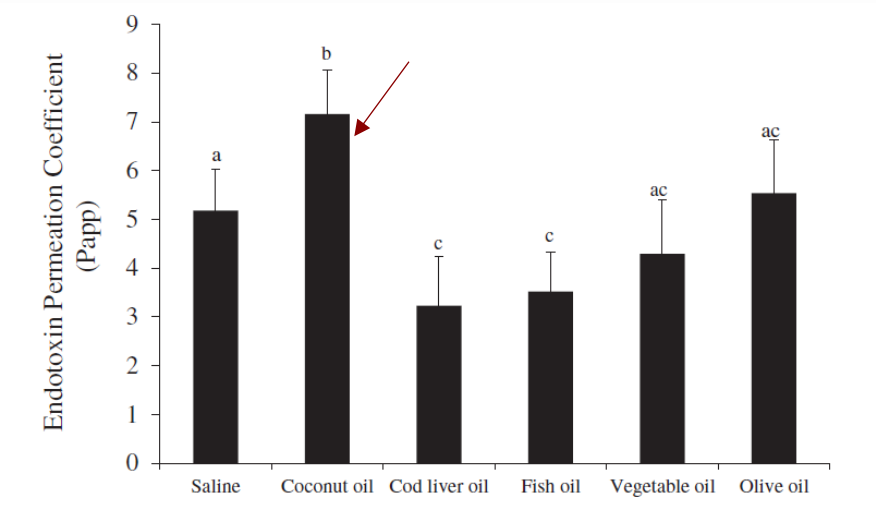
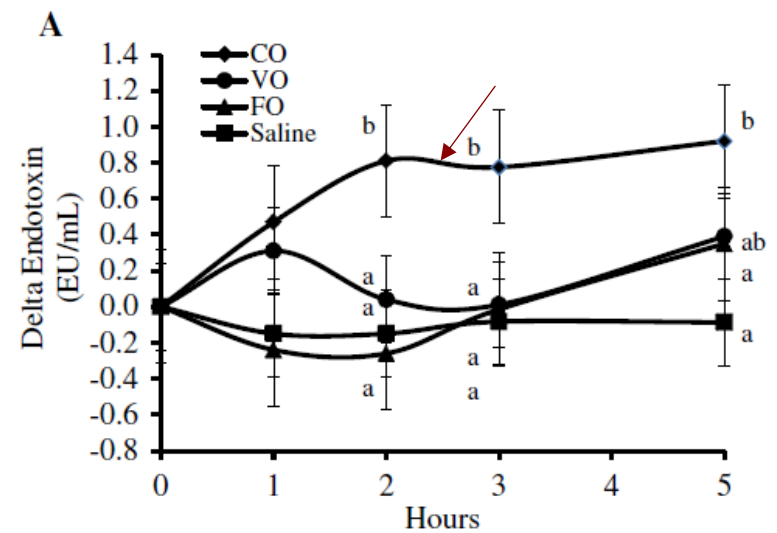
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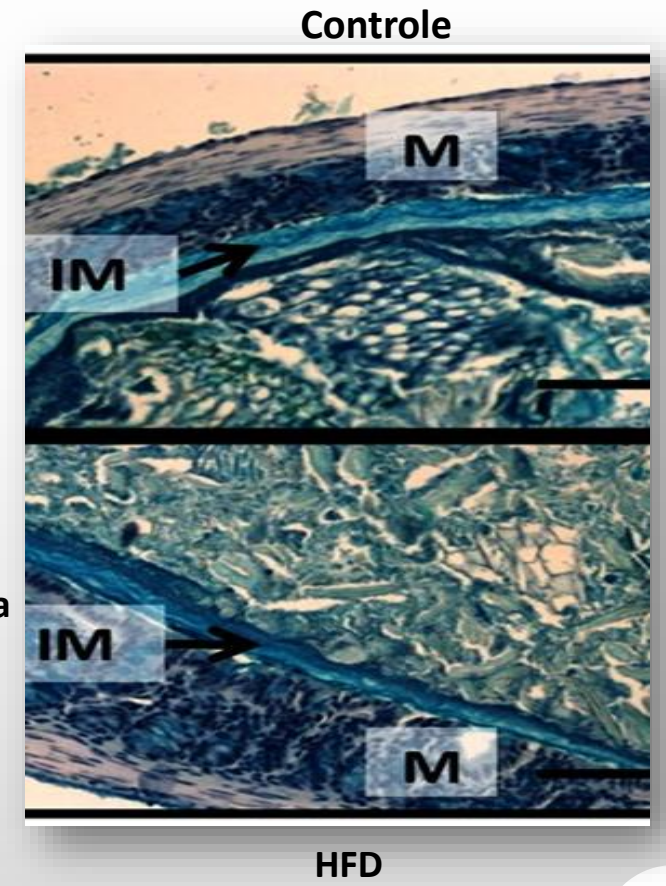
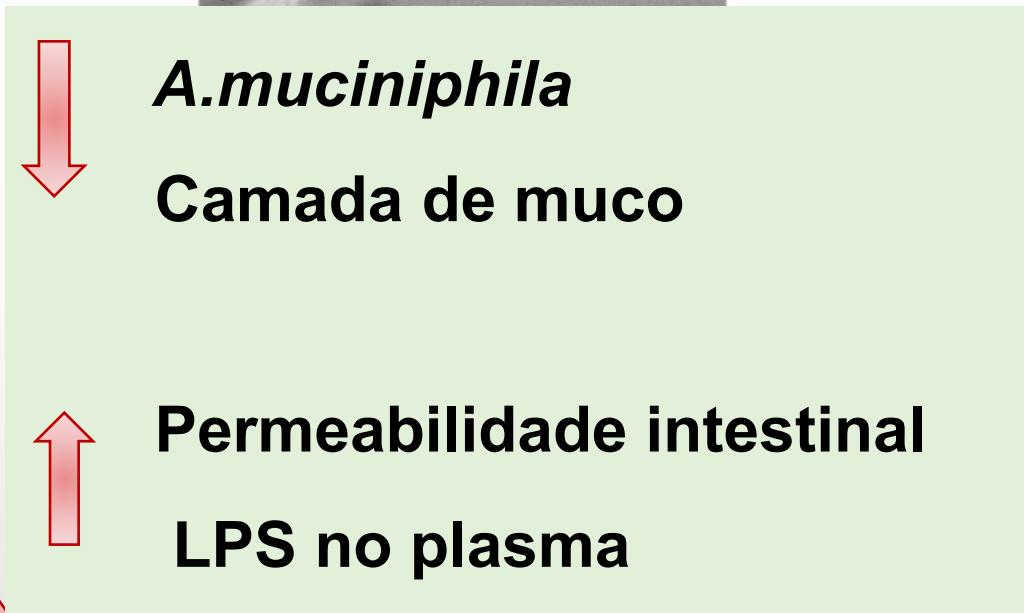
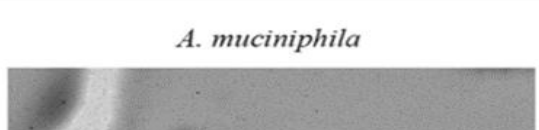
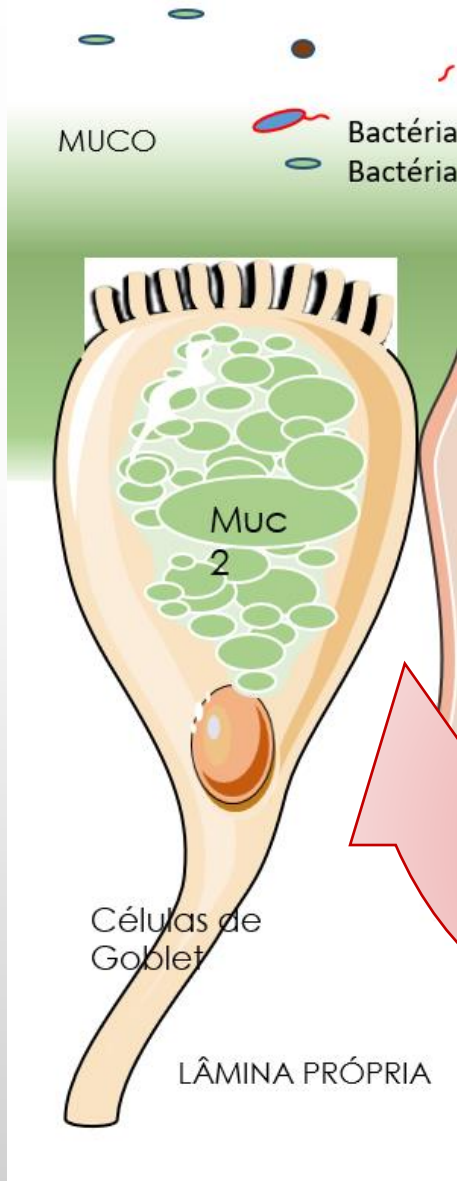
Dietary oil composition differentially modulates intestinal endotoxin transport and postprandial endotoxemia



- Endotoxemia pós prandial
- Permeabilidade da barreira intestinal



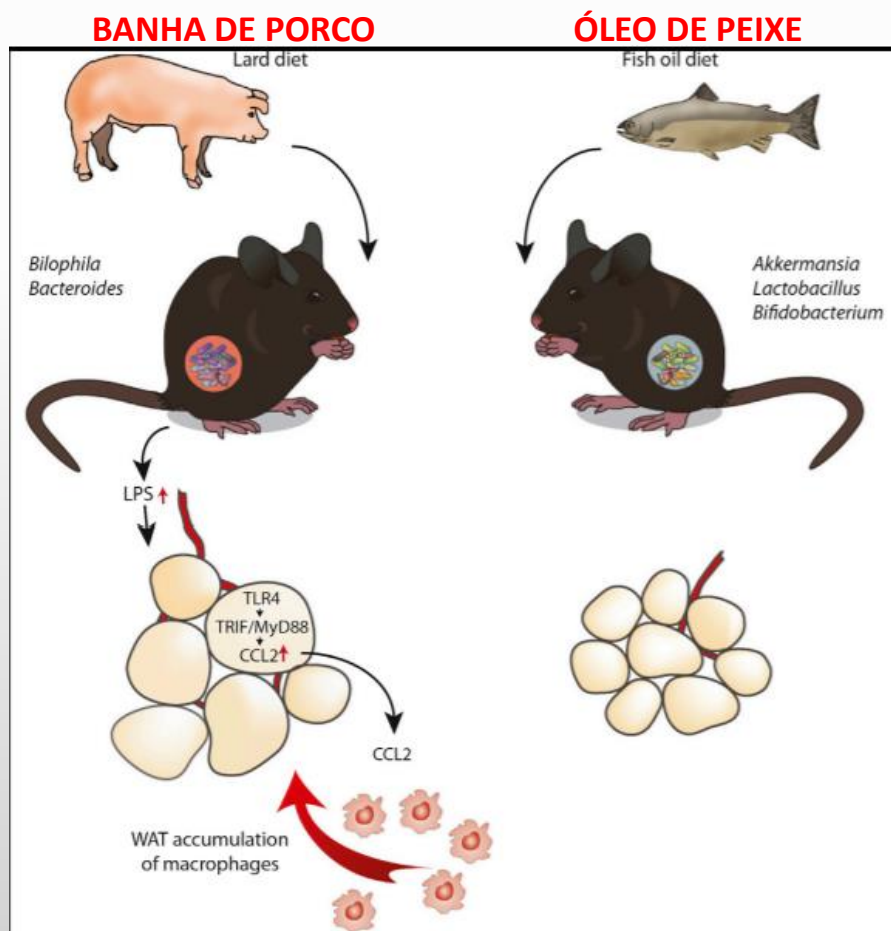
Efeito da dieta rica em gordura sobre a espessura da camada de muco do epitélio intestinal



Cell Metabolism

Article

Crosstalk between Gut Microbiota and Dietary Lipids Aggravates WAT Inflammation through TLR Signaling



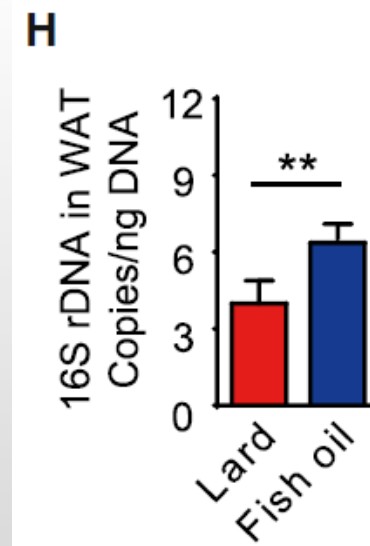
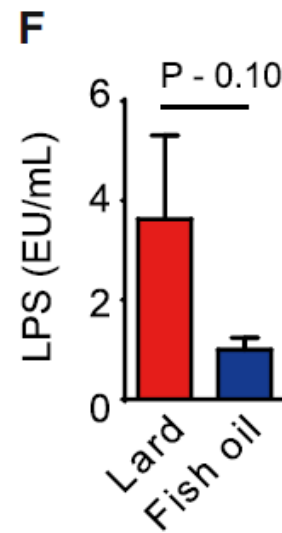
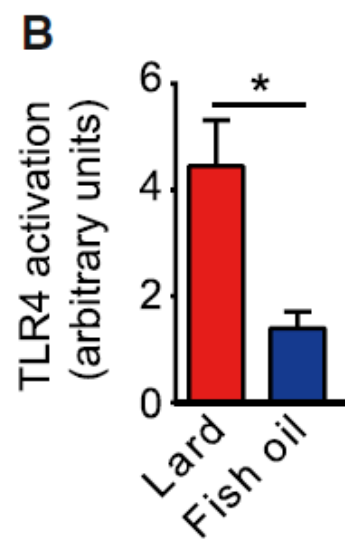
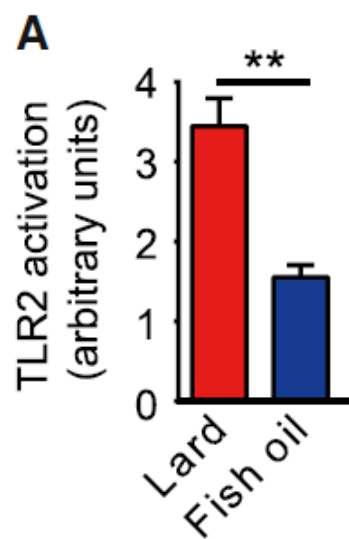
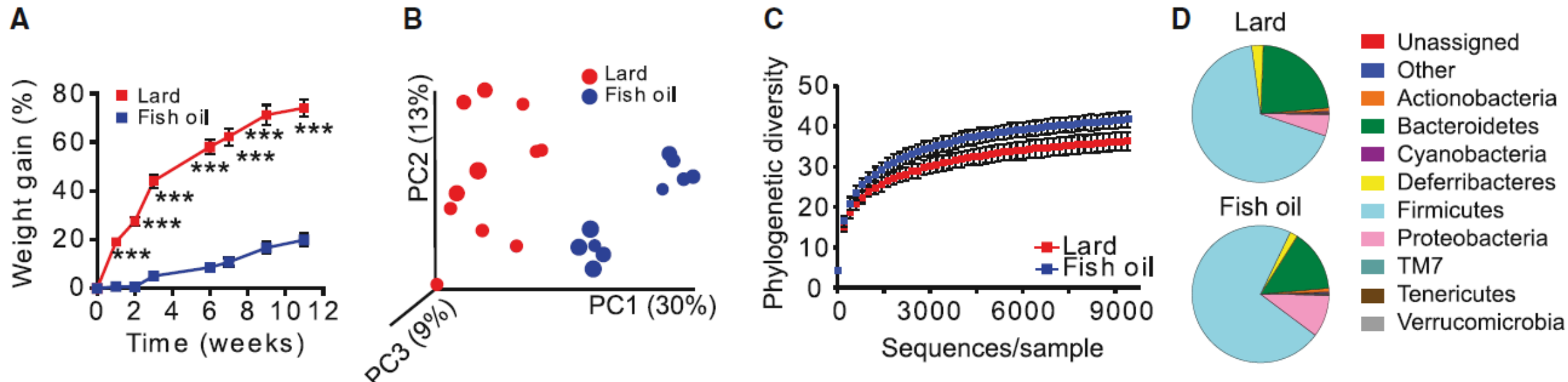
Gordura saturada aumentou:

- Ganho de peso e eficiência alimentar
- Ativação de TLR4
- Inflamação no adiposo

Reduziu

- Diversidade bacteriana
- Sensibilidade a insulina



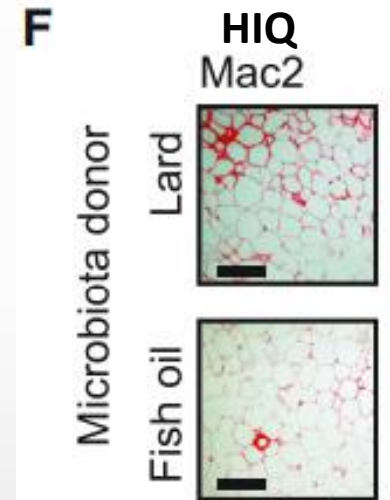
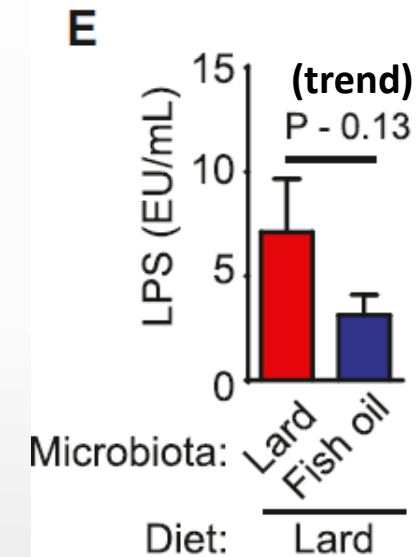
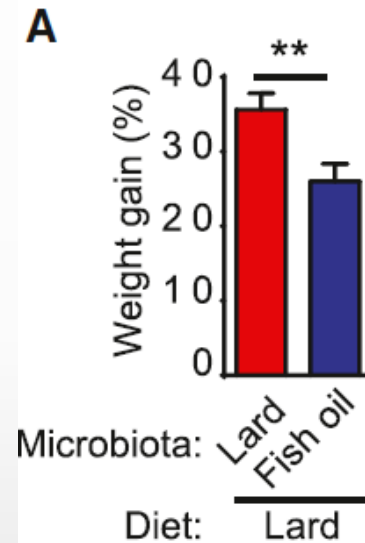
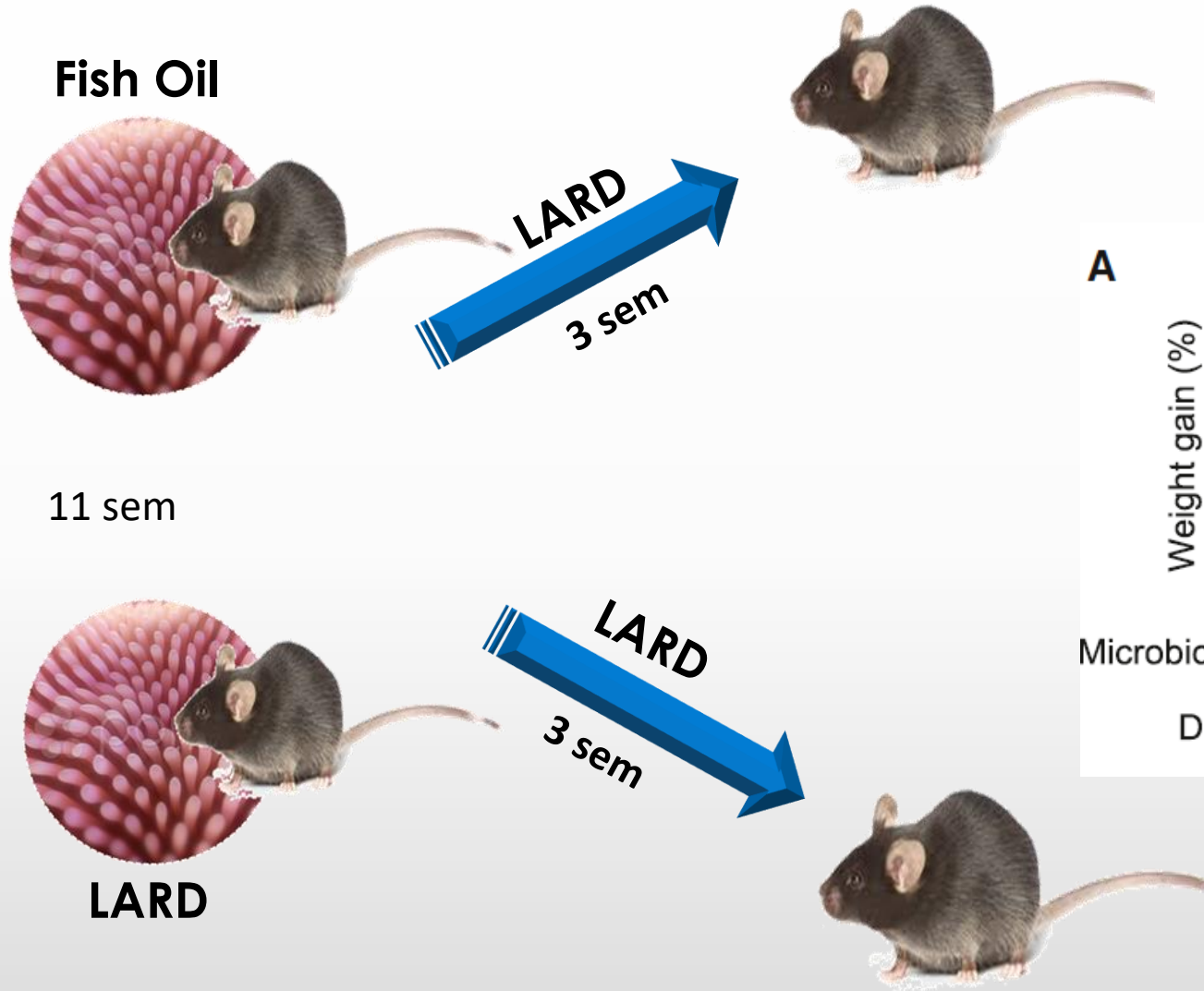


- Soro dos animais LARD promoveu ativação de TLR2 e TLR4



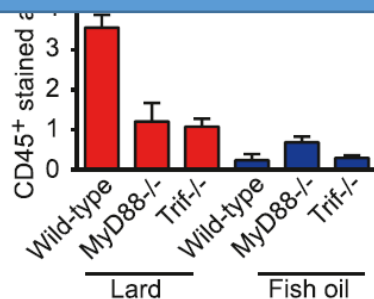
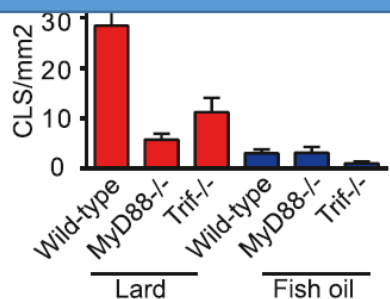
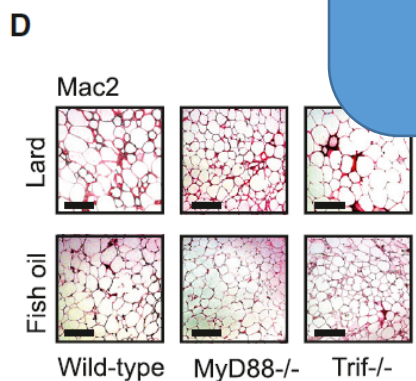
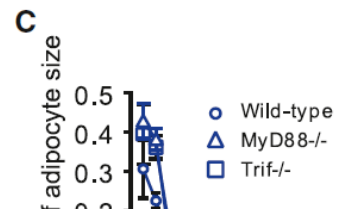
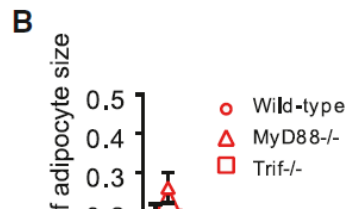
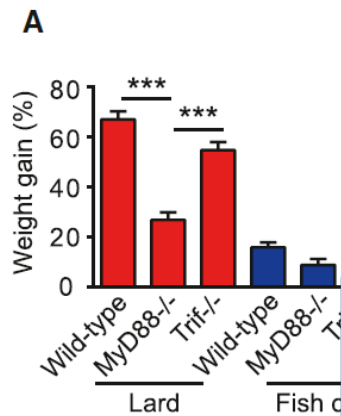
Cell Metabolism

Crosstalk between Gut Microbiota and Dietary Lipids Aggravates WAT Inflammation through TLR Signaling

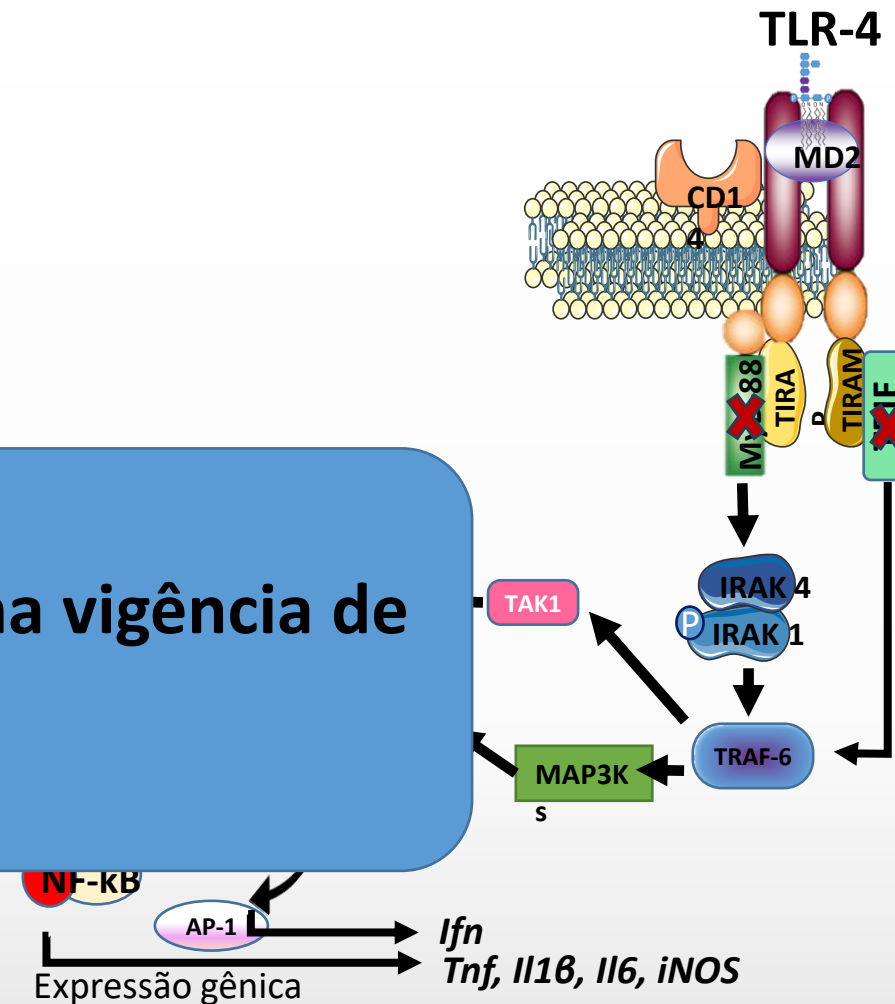


Microbiota proveniente de Fish Oil:

- Resistiu ao ganho de peso
- Menor inflamação WAT

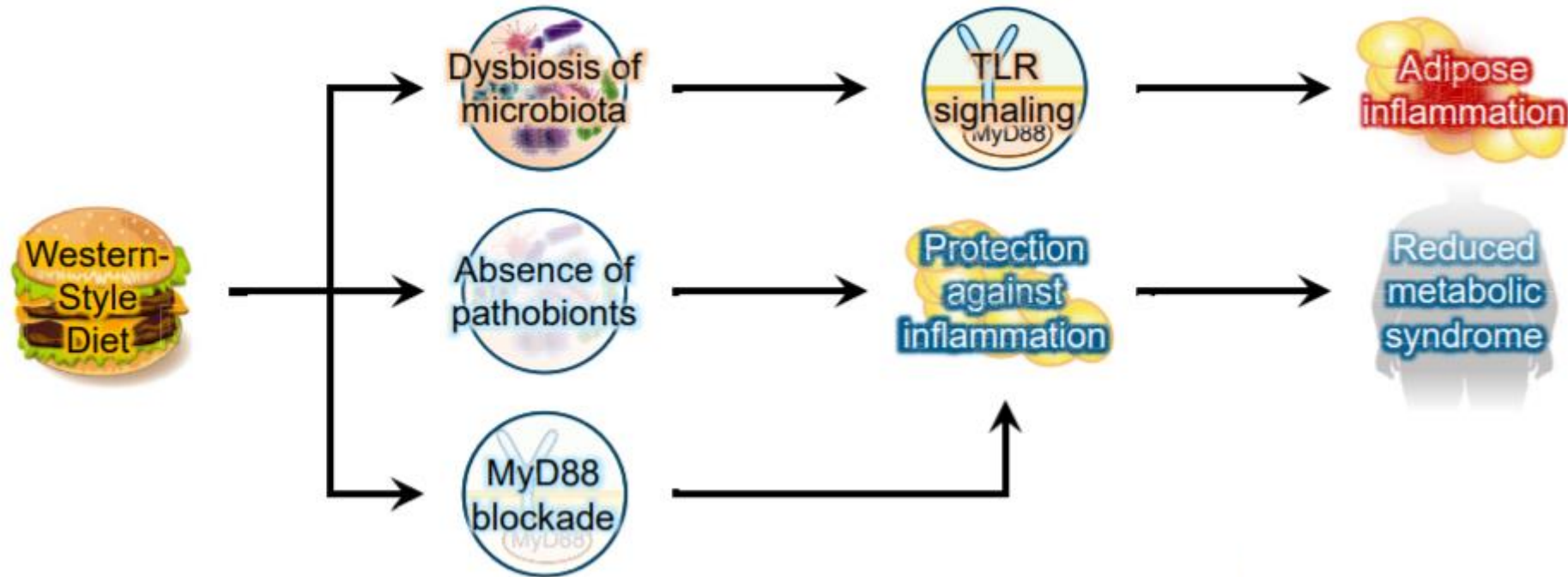


Microbiota → agravo da inflamação na vigência de dieta rica em saturados

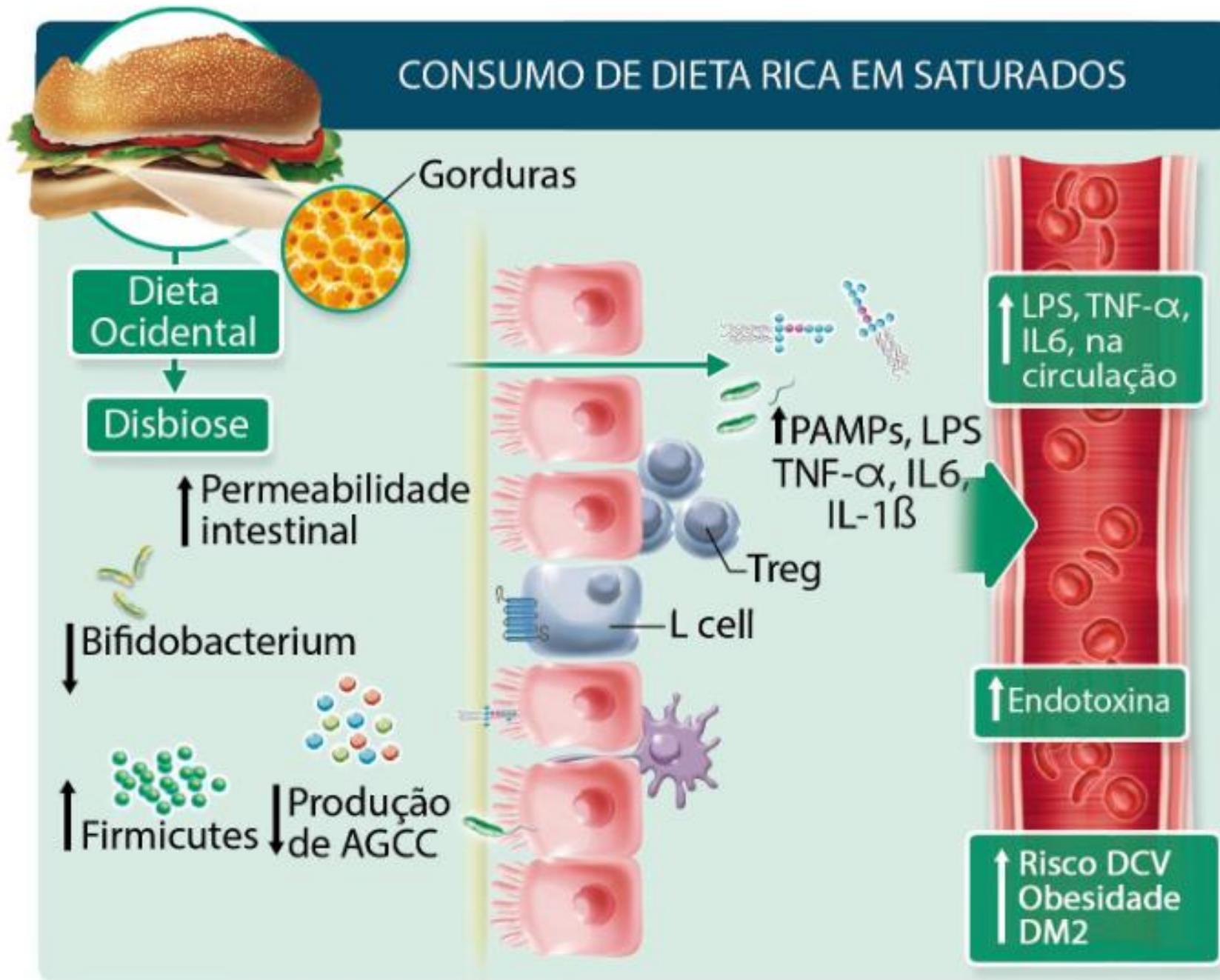


ORIGINAL RESEARCH

“Western Diet”-Induced Adipose Inflammation Requires a Complex Gut Microbiota

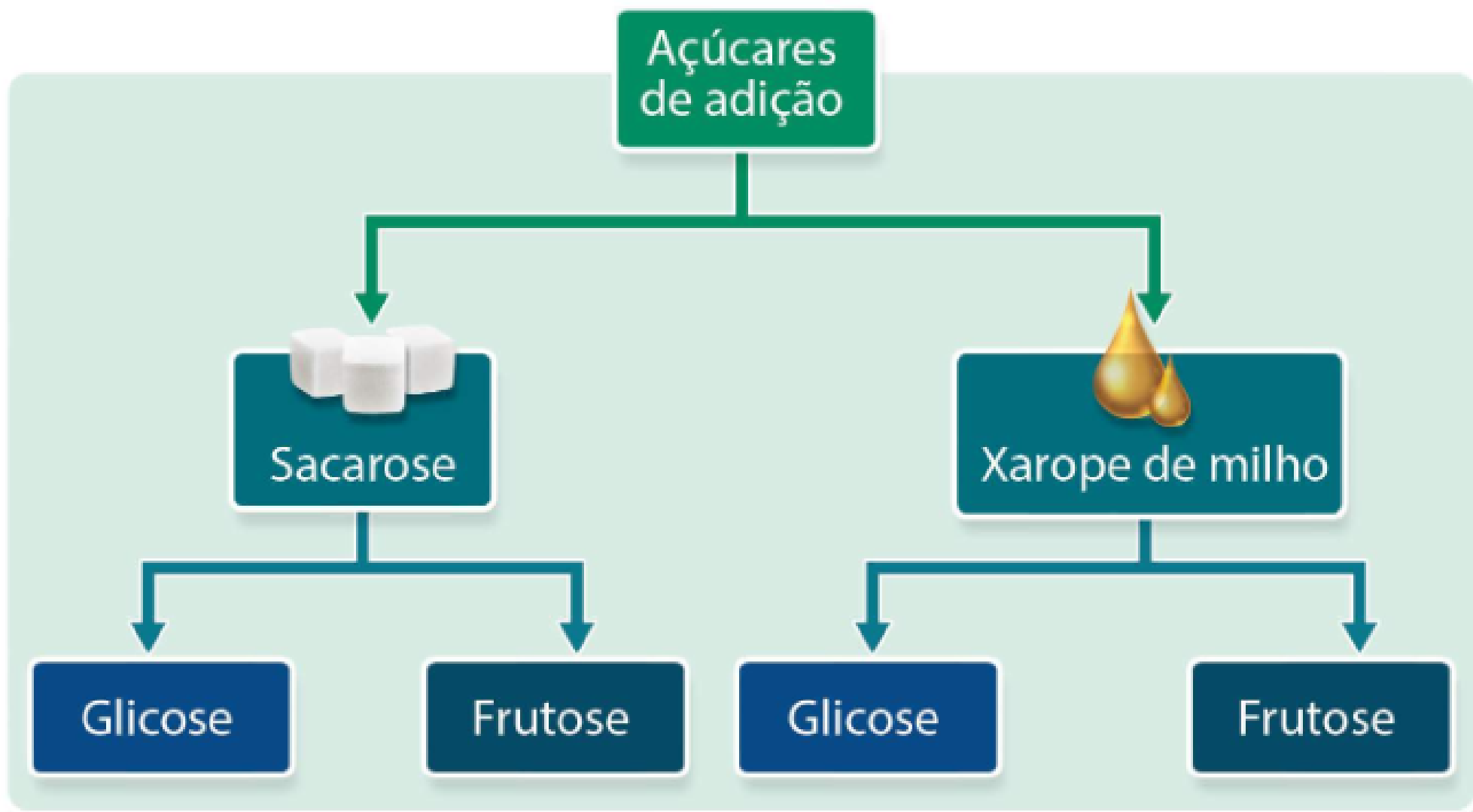


CONSUMO DE DIETA RICA EM SATURADOS





Ação dos açúcares na modulação da microbiota e permeabilidade intestinal



Impacts of Fructose on Intestinal Barrier Function, Inflammation and Microbiota in a Piglet Model

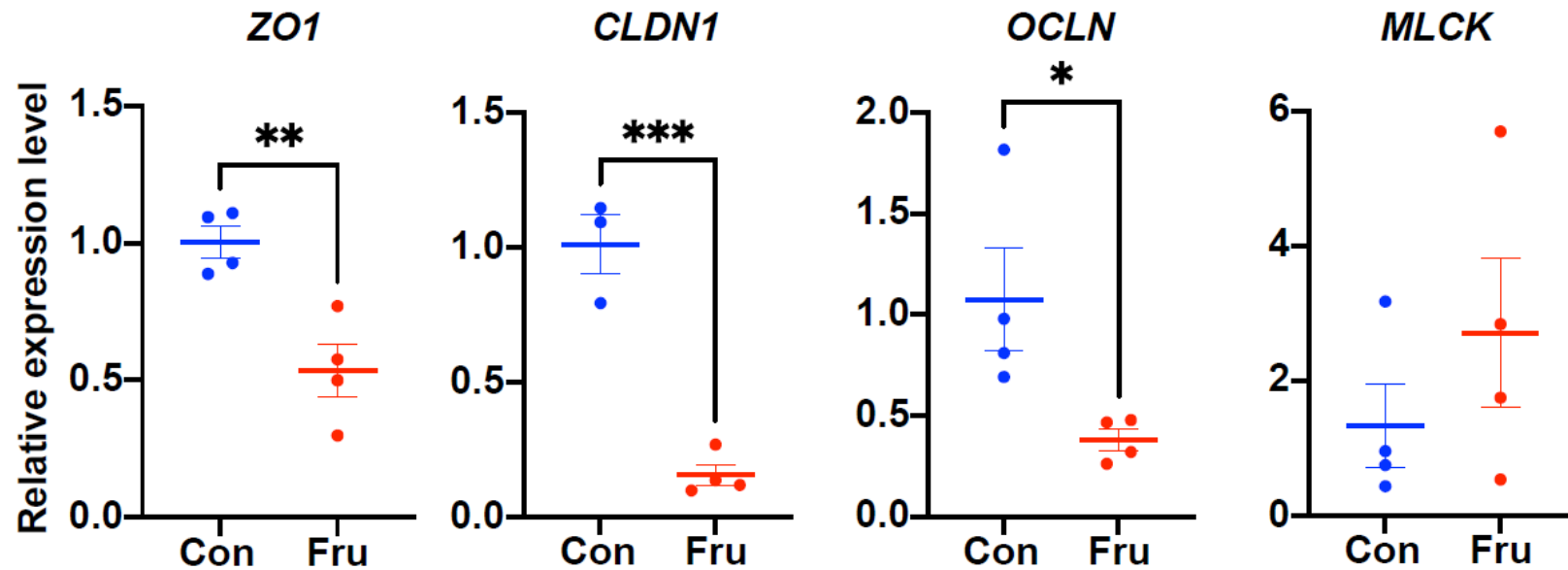
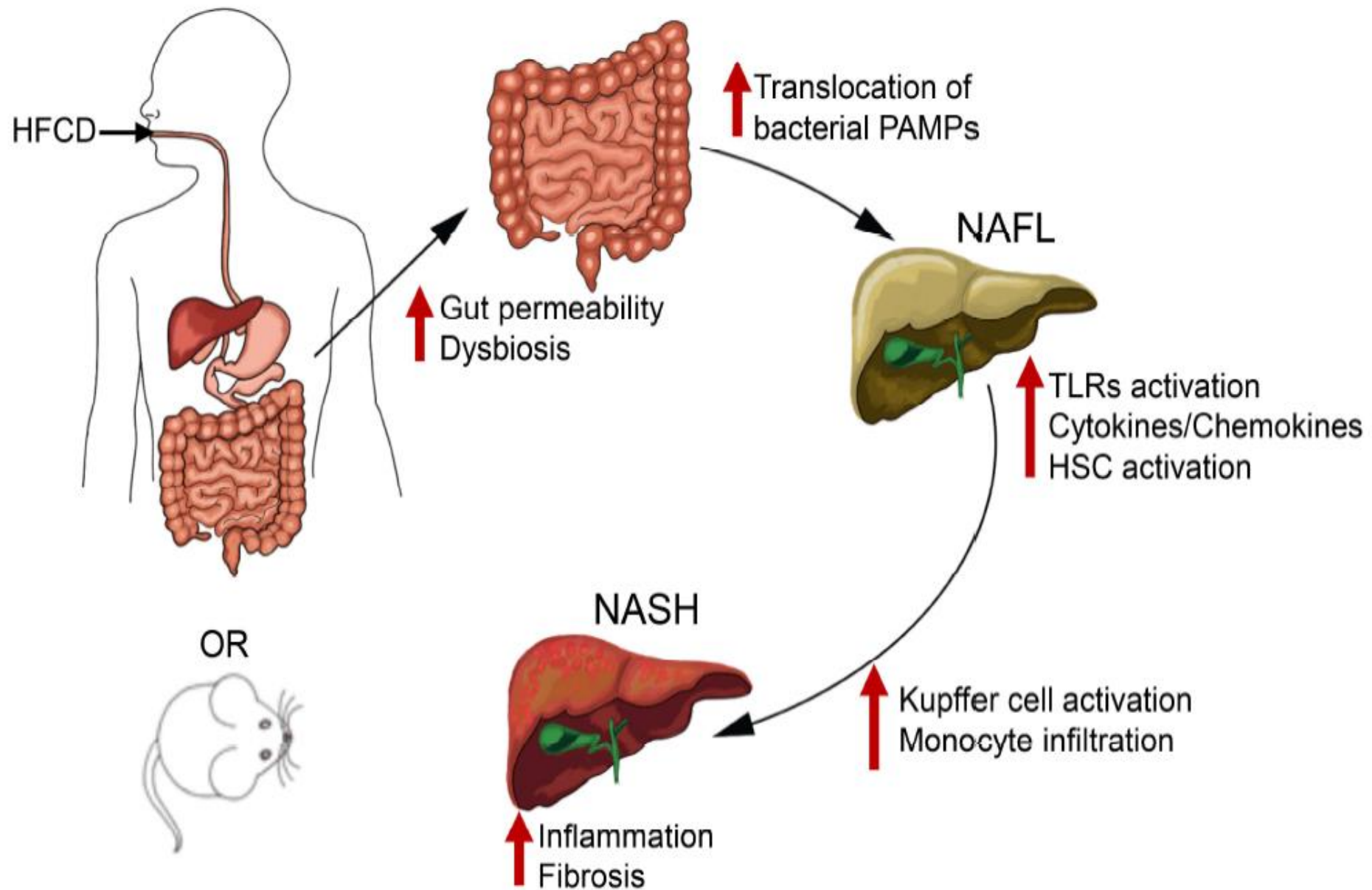
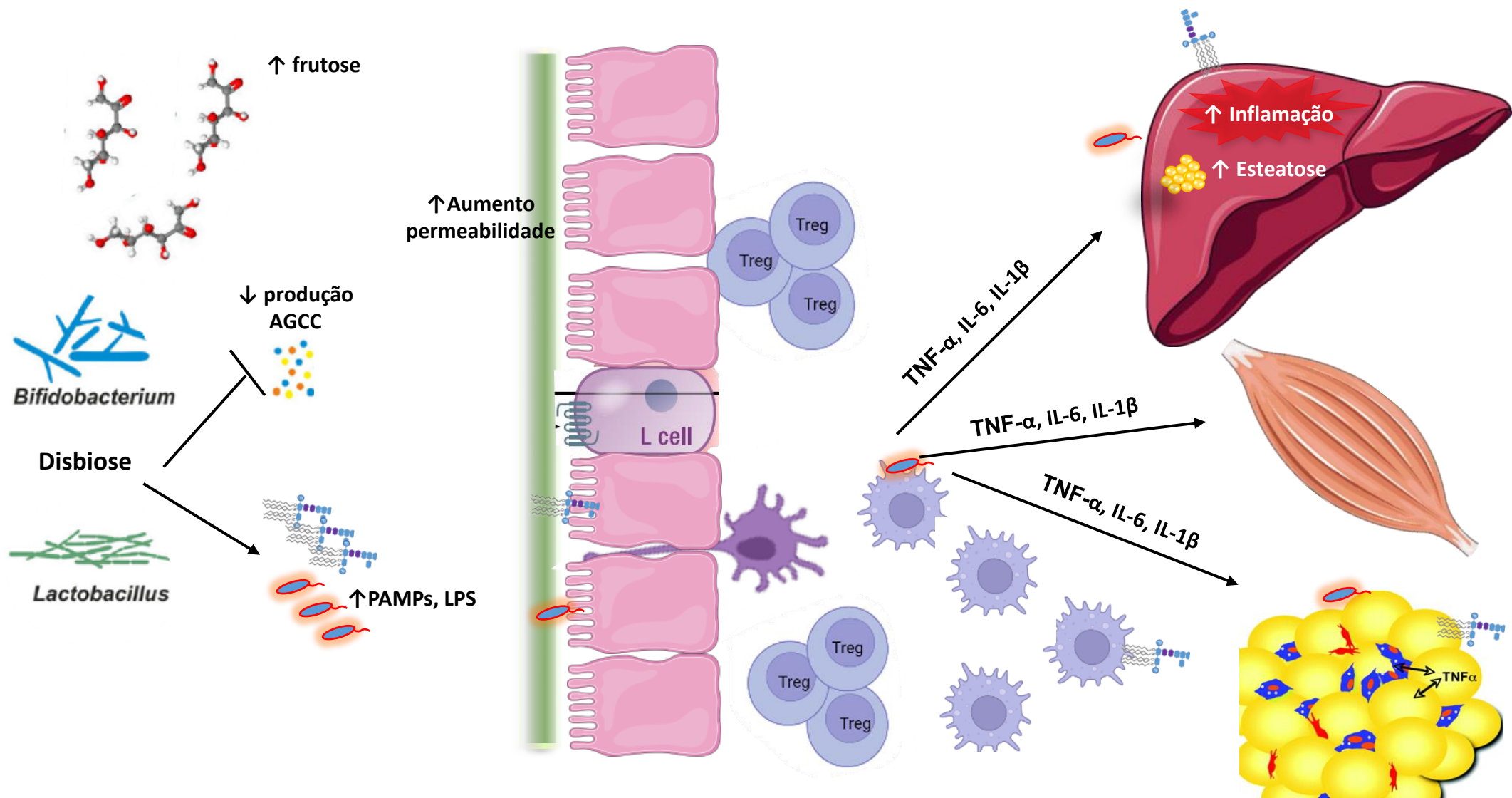
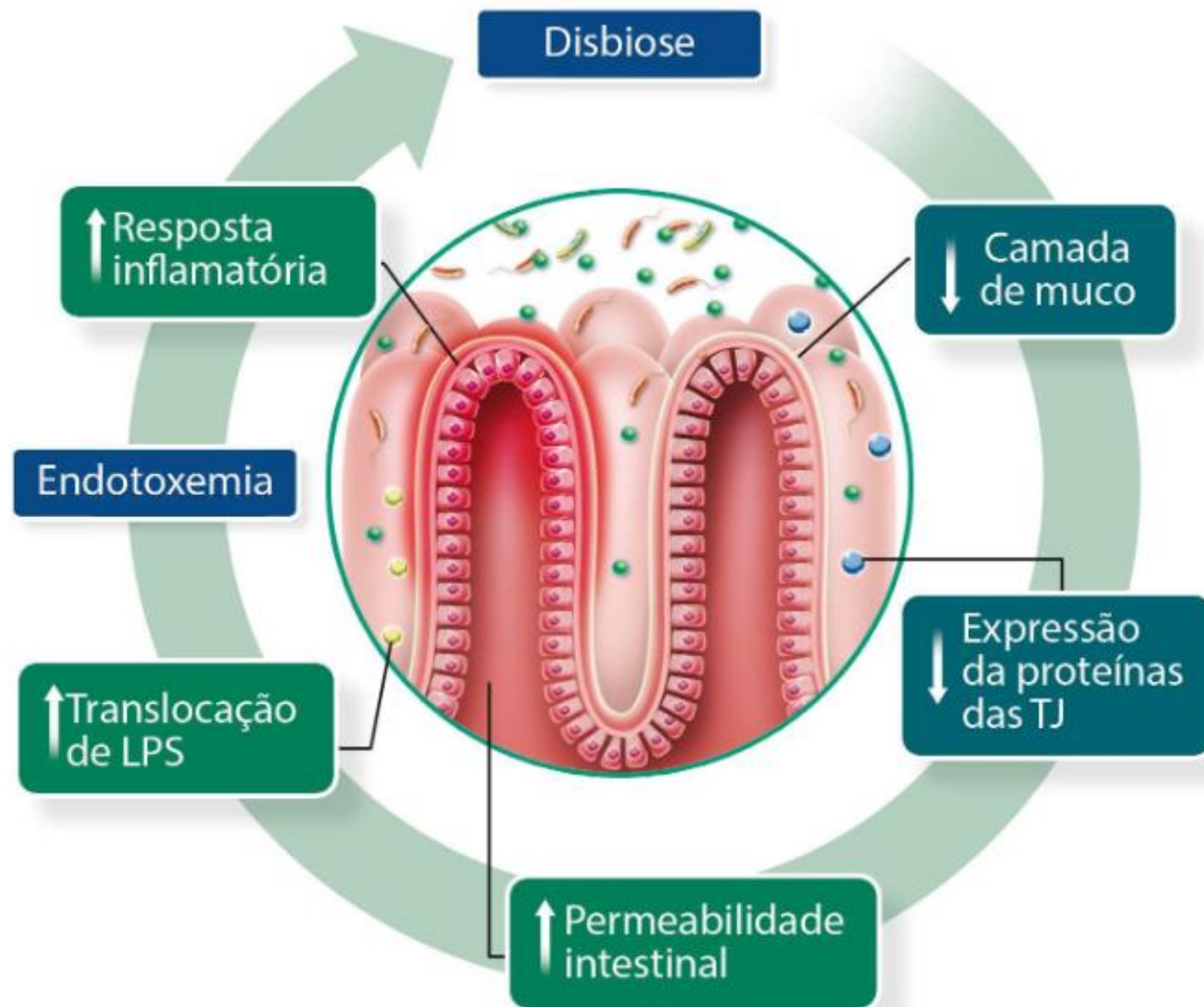


Figure 1. The ileal barrier function of weaned piglets fed a basic or 0.2% fructose-supplemented diet. The relative mRNA expression levels of tight junction (ZO1, CLDN1, OCLN) and MLCK genes were shown here; $n = 4$. Con, the control group; Fru, the 0.2% fructose-supplemented group. *, $p \leq 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$.



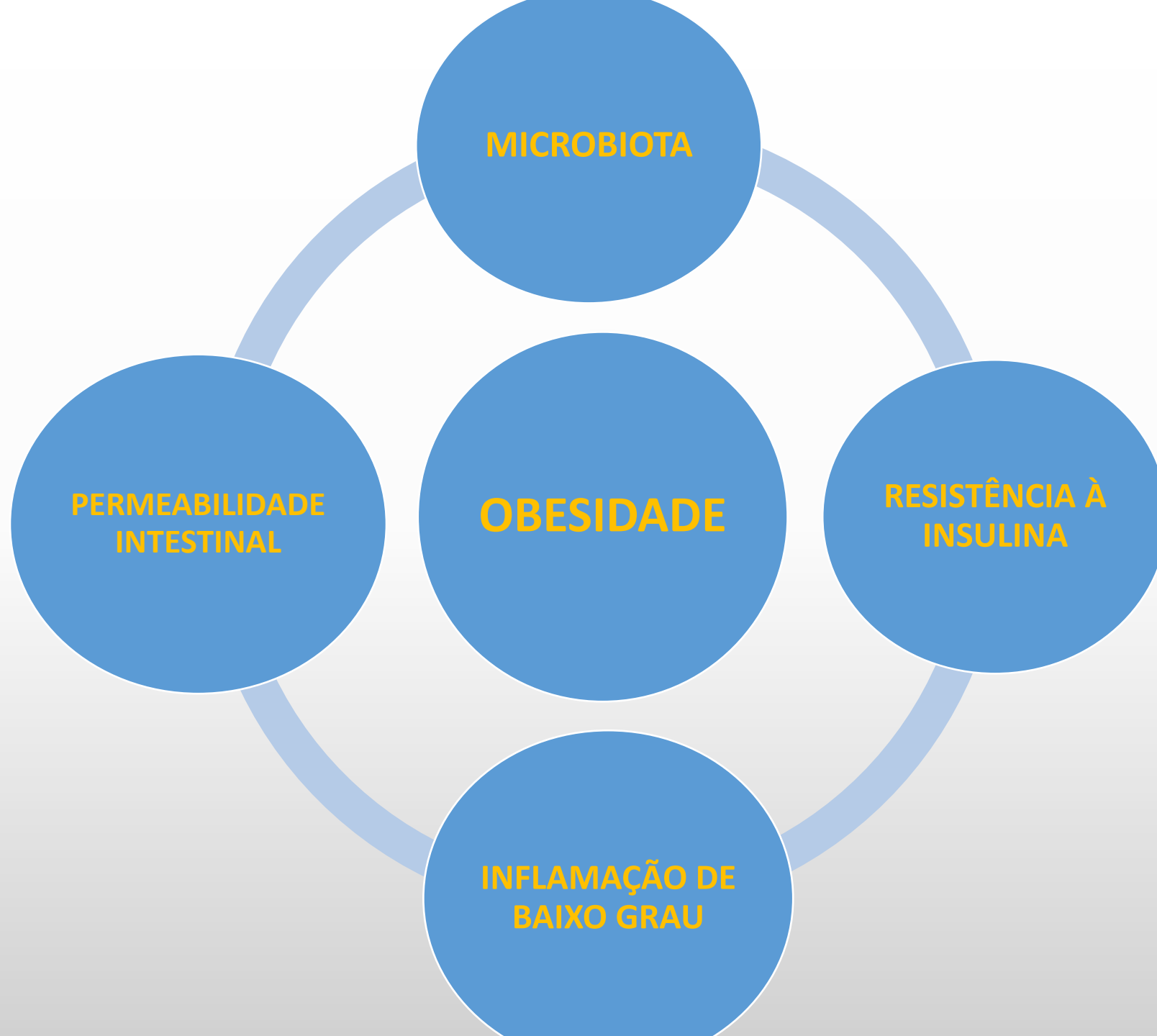








Microbiota e Obesidade



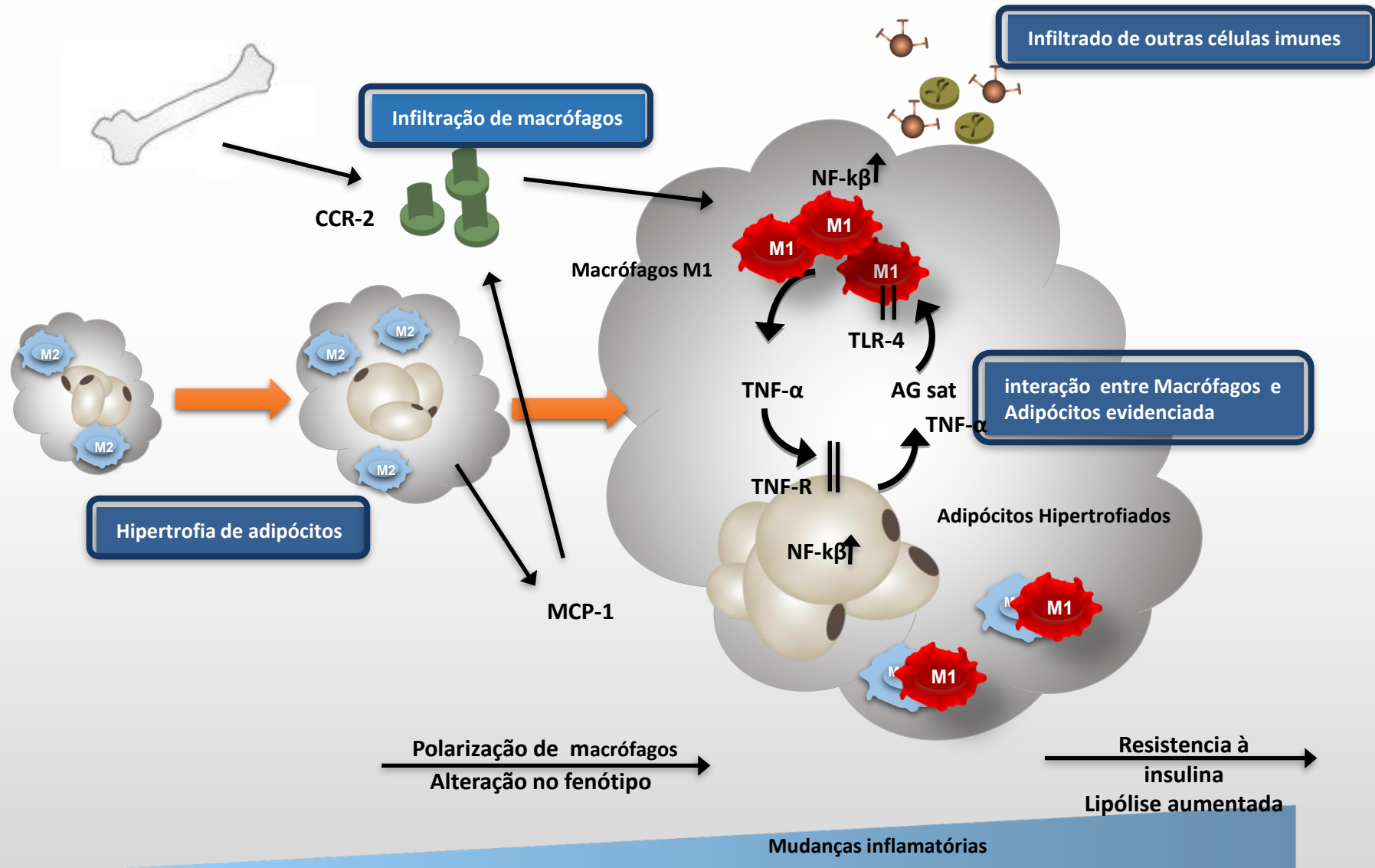
Desenvolvimento da obesidade é lento e progressivo

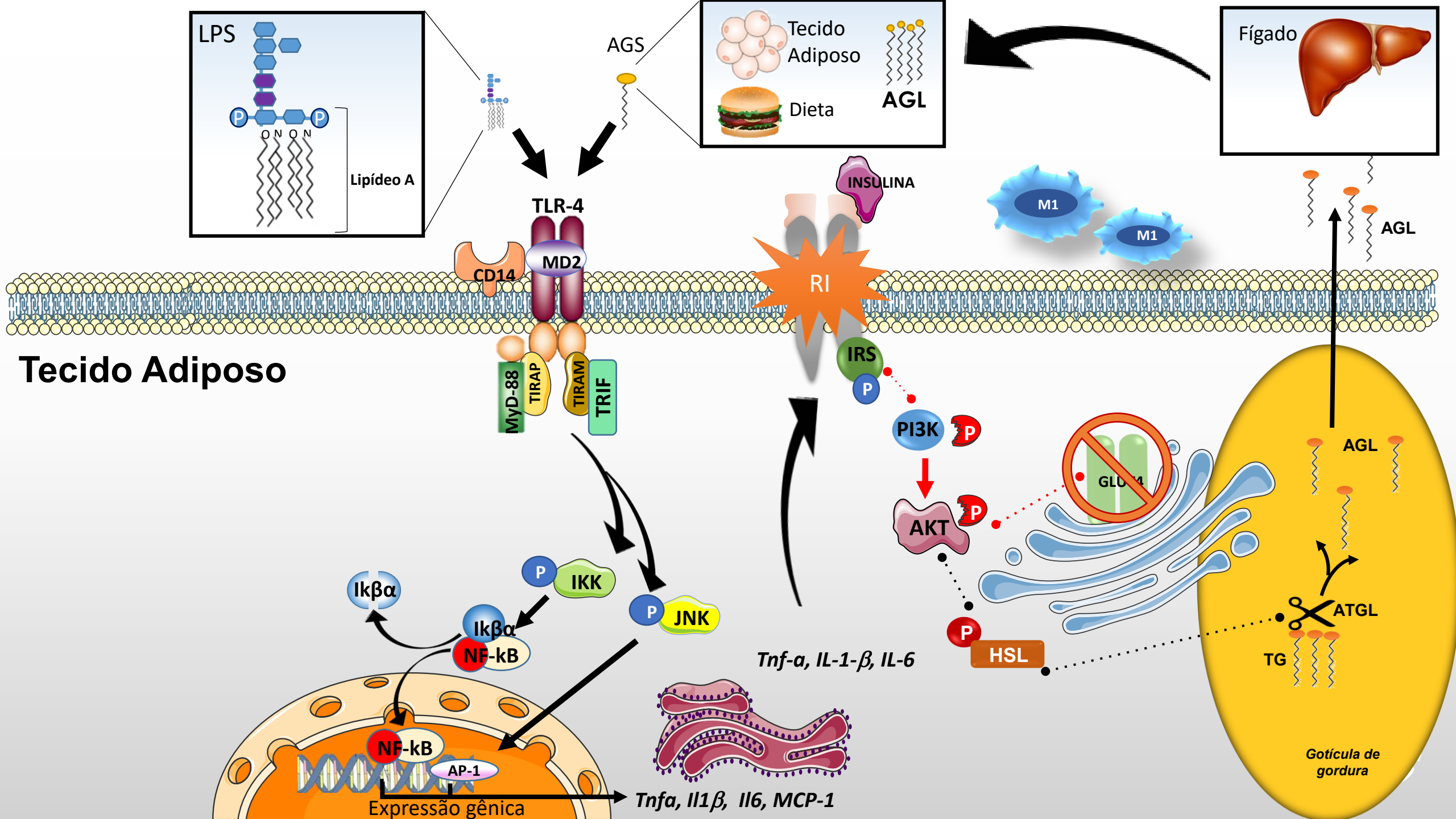
Forte associação hábito alimentar:

- ✓ **Alimentos processados, ricos em gorduras e "Fast Foods"**
- ✓ **Baixo consumo de frutas, vegetais e alimentos integrais**
- ✓ **Alto consumo de álcool (7Kcal por grama)**
- ✓ **Alimentar-se fora de casa (maior teor de gordura, sódio e açúcar)**
- ✓ **Porções maiores de alimentos**
- ✓ **Alimentação reconfortante; Hábitos familiares inadequados**



Obesidade e alterações metabólicas

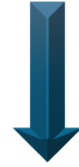






Dieta

Obesidade



**Alteração da composição da
microbiota intestinal**

- **A microbiota de animais obesos inoculada em animais magros induz obesidade**

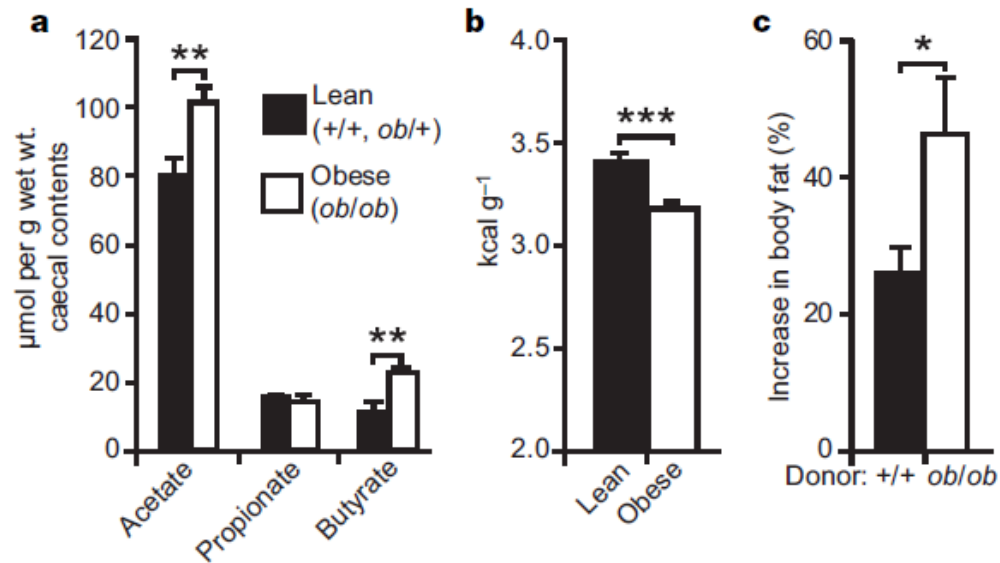
Turnbaugh et al., Nature. 2006;444:1027-31.

- **Indivíduos obesos apresentam menor diversidade bacteriana**



An obesity-associated gut microbiome with increased capacity for energy harvest

Peter J. Turnbaugh¹, Ruth E. Ley¹, Michael A. Mahowald¹, Vincent Magrini², Elaine R. Mardis^{1,2} & Jeffrey I. Gordon¹



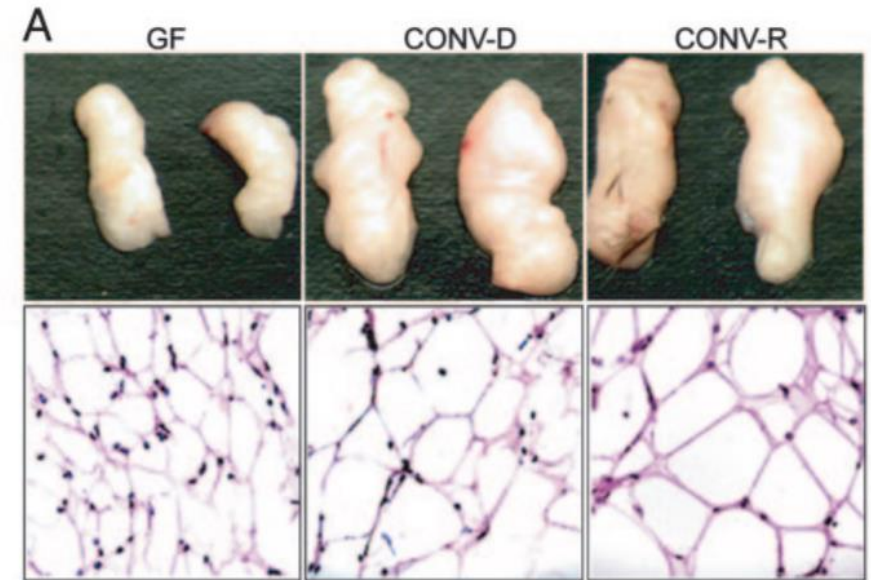
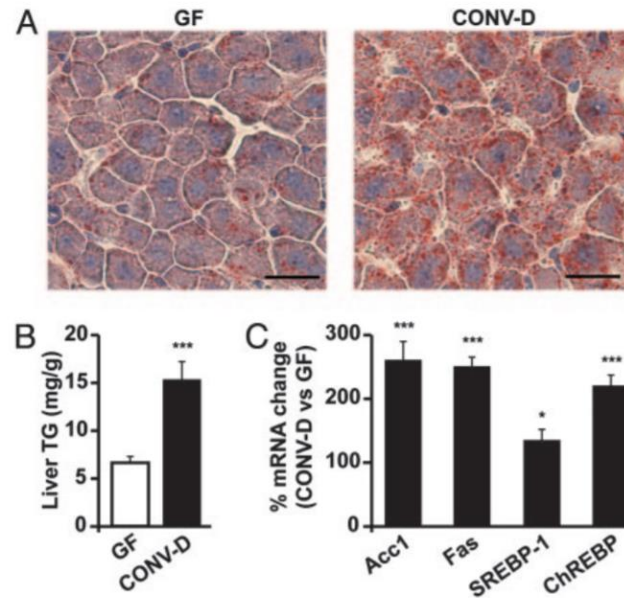
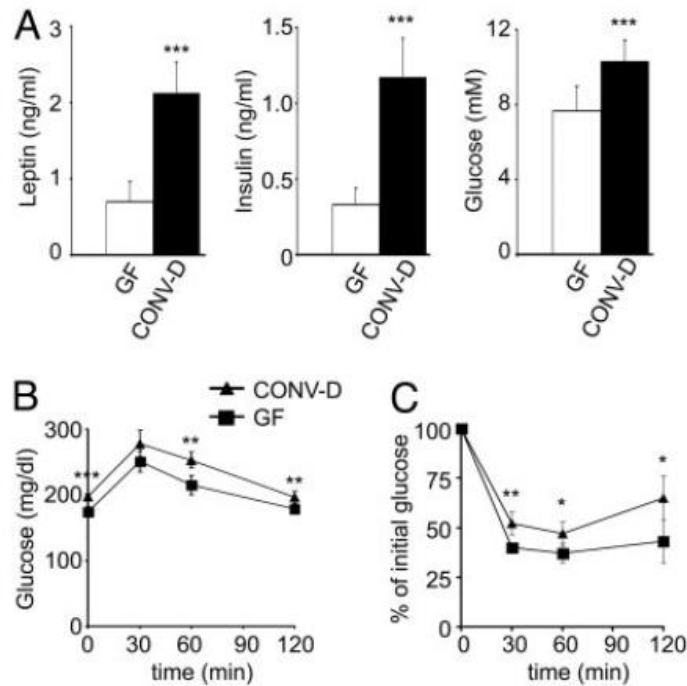
- ✓ Microbiota Ob/Ob: maior capacidade de extrair energia da dieta para o hospedeiro
- ✓ Capacidade foi transferida com transplante da microbiota



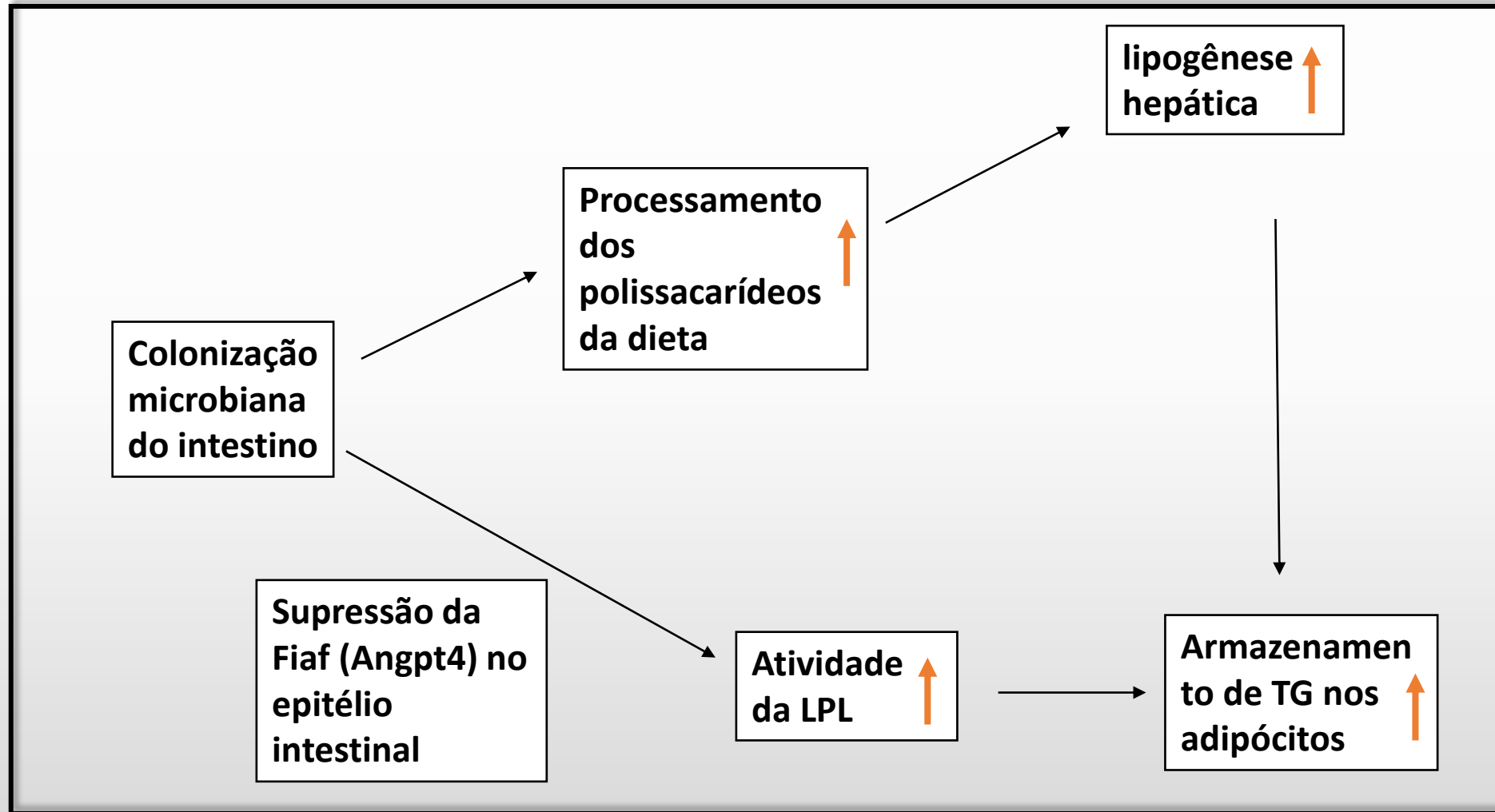


The gut microbiota as an environmental factor that regulates fat storage

Fredrik Bäckhed^{*†‡}, Hao Ding^{§¶}, Ting Wang^{||}, Lora V. Hooper^{†**}, Gou Young Koh^{††}, Andras Nagy^{§‡‡}, Clay F. Semenkovich^{§§}, and Jeffrey I. Gordon^{*†¶¶}



The gut microbiota as an environmental factor that regulates fat storage





Probióticos e Prebióticos

PROBIÓTICOS

- São microrganismos associados a ações benéficas na saúde do hospedeiro.
- As bactérias mais comuns pertencem ao grupos *Lactobacillus* e *Bifidobacterium*,



PREBIÓTICOS

- São alimentos não digeríveis, que estimulam o crescimento e atividade metabólica de certas cepas bacterianas, conferindo benefícios para a saúde do hospedeiro.
- EX: Fibra solúvel (Inulina e transgalacto-oligossacarídeos), amido resistente.



O que dizem as diretrizes para o tratamento de OBESIDADE:

Não existem evidências científicas suficientes para recomendação e indicação clínica da suplementação de probióticos, bem como para a realização de testes genéticos de microbiota na prática clínica para o tratamento da obesidade.

Classe de recomendação III e nível de evidência A





ALBERT EINSTEIN
INSTITUTO ISRAELITA DE
ENSINO E PESQUISA

CENTRO DE EDUCAÇÃO EM SAÚDE
ABRAM SZAJMAN